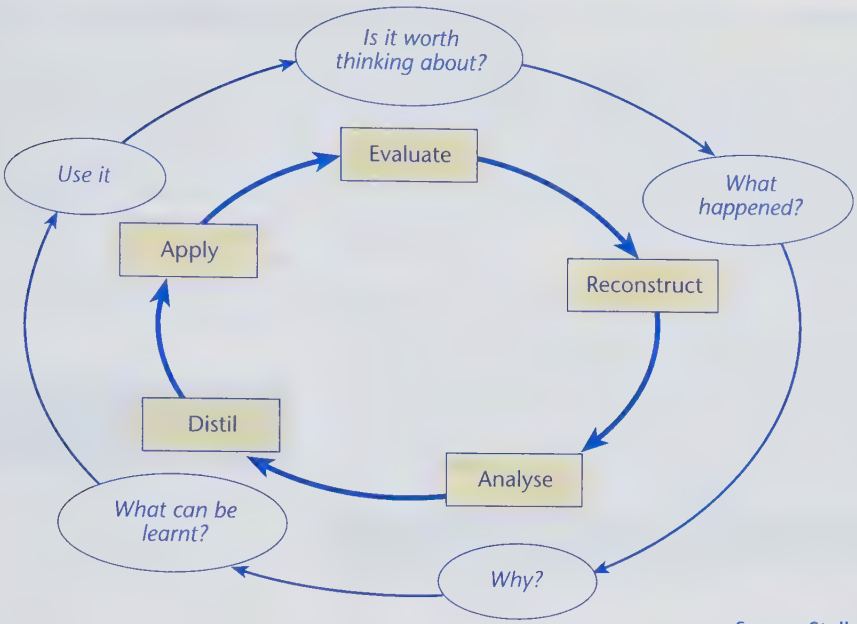
The Core Model for critical reflection



Source: Stella Cottrell (2010).

Core Model: Stages and prompts

The Core Model for critical reflection consists of five stages, with question-based prompts to support each of these (Cottrell, 2015). You can adapt, select from or delete these stages. You can add further stages if you feel that a particular prompt or set of prompts merits separate attention, either for your own purpose or because these are especially important in your discipline. The five stages are described below.

1 Evaluate

The kind of critical reflection described in this chapter is, as we have seen, intensive, time-consuming and potentially emotionally demanding; you need to select a focus. Given the demands that it makes upon you, it is worth spending time critically reflecting upon your choice and the direction that you are taking. As this is so important, the first stage of your reflection is a critically evaluative one, examining your purpose and focus and deciding whether these will really yield the best results.

Examples of prompts useful for this stage

- Does this provide enough challenge and material – or have I gone for too easy an option?
- Is this too challenging? Am I too angry or affected by this to address it at this time or as part of an academic assignment?
- Am I ever likely to be in a similar situation again? If not, what will I gain from focusing on it now?

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

2 Reconstruct

Each time we go over an event in a reflective way, the account is likely to vary. That is why this stage is referred to as 'reconstruction'. This is typical phase 1 reflective activity. We are putting together, again, the events, feelings, details as we now recall them. Going over an event in detail can bring up unexpected reminders or draw attention to items that we may not have noticed previously.

Input from others about the event, or perspectives gained from our reading about the subject, may lead us to reconstruct the event in a slightly different way with different aspects coming to the forefront or gaining a new emphasis or significance. If we write these down and return to them, the differences between one account and another may strike us as significant.

Examples of prompts useful for this stage

- What was I aiming to achieve through my actions, what I said, or what I didn't do or say?
- Did this work out as I expected?
- How did I feel when X happened, and how did I respond?

3 Analyse

Examine your reflective materials, your thoughts or your notes, from different angles; interrogate your material; answer your prompt questions; compare your emerging findings with those expected from your reading or training; question whether you are really confronting your own motivations and taking on board the difficult messages; take a walk; talk to a peer about what you are finding; jot down new thoughts; sleep on them; sit quietly with your reflections; go back over everything again. In other words, work with your material in many ways, so as to encourage fresh insights to emerge.

Examples of prompts useful for this stage

- Which actions or omissions were most significant in the way events unfolded?
- What were the consequences of unexpected or difficult feelings?
- What interpretations or theories help to make sense of what happened?
- Were there deeper roots to this than were apparent at the time?

4 Distil

If your analysis helped to make sense of your experience, then it should be a relatively easy next step to draw out the lessons learnt. Your analysis may have generated many thoughts, of different levels of significance. Draw out the most relevant and synthesise these into a workable set of key ideas or conclusions.

Examples of prompts useful for this stage

- What worked well that I could use again in the future?
- What were the trigger points? How would I manage those differently?
- How does my way of approaching matters or thinking about them help or hinder the desired outcome?

5 Apply

If you have really arrived at new understandings and drawn lessons for yourself, then the logical next step is to consider how you will make use of that learning before you forget it. This may mean that you need to bring others on board so that they understand why you want to do things differently, and are prepared to support you.

Examples of prompts useful for this stage

- To what kinds of situations will I apply these insights?
- What support will I need?
- Who else needs to be on board? How will I persuade them of the benefits?

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Applying reflection to professional practice

Why reflect on practice?

Critical reflection has long been used in health-related, caring and teaching professions, and its value is recognised increasingly in other areas such as management and business. There are a number of reasons for this, such as:

- it anchors theory in meaningful, concrete experience, helping to bring it alive;
- it gives recognition to learning gained in non-academic contexts;
- it provides a bridge between practical experience and academic study;
- it helps develop understanding of difficult work situations, improving professional practice.

The workplace as a resource

Work contexts offer rich resources for meaningful study. However, there are ethical considerations that you must take on board.

Ensure that you retain confidentiality and anonymity with respect to workplace information. You may be able to use examples if names and details are removed in such a way that the workplace and individuals cannot be identified, or if you gain written permission.

You will need permission to use workplace documentation if it is not already in the public arena. It is likely that your programme will ask you to consider the ethical implications of the focus you have selected and the material you will be using.

Reflection on the knowledge base

Your workplace practice is likely to be based, at least in part, on theories, research or established professional wisdom. You can build your own expertise and understanding by drawing on these as part of your critical reflection.

- **Select a relevant theory** or piece of research that relates to your work context.
- **Identify what it would mean in your context**. What, in summary, do the theory or research findings propose? If this theory or research applied to your work context, what would you expect to happen? What would you expect to see, hear, feel, experience, understand better, etc.?
- **Check whether it applies**. In practice, do the theory or research findings apply to your context as expected? How do they throw light on your own experiences? In what ways do they help – or not help – you to make sense of your professional practice?
- **Identify probable cause**. If the theory or research findings do not seem to apply as expected, how do they differ? Why is that likely to be the case? For example, are you working with a different client group than that used for the initial research? Are other conditions different? Do any other theories throw light on your findings?
- **Support for change**. Do your reflections suggest that any change is needed in your workplace – such as a review of procedures or a change in expectations? How do research findings or established theories support your case?

Challenge your own previous practice

Use your critical reflections to challenge your previous thinking, assumptions and actions. Re-examine these in the light of what you have read. If you find that this, in itself, raises difficult issues, reflect on where these difficulties lie, how they have arisen, and how you will address them. Talk to your tutor or your workplace supervisor about any complex issues or concerns.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Following procedure

In clinical practice and workplace settings, there are usually set procedures and customary working practices that you are required to follow. These have an important function in ensuring that everyone knows what to do, that teams can work together, and that health, safety, security, quality assurance, statutory obligations and other requirements are met.

When procedure isn't enough

However, from time to time, situations arise that make it difficult or inappropriate to follow procedures exactly. This requires you to make a judgement call – sometimes on the spot. Many poor outcomes in the workplace arise because people don't have sufficient understanding of the relation between theory and practice. This means they are unclear:

- when to deviate from procedure;
- what to do in situations where no procedure yet exists;
- how to read a situation so as to make an appropriate decision for action.



John was a stickler for following protocol to the letter.

Making good judgement calls

If you have a good grasp of the knowledge base and working context that underpin everyday practice, you are better able to make a good call when needed. You are more likely to:

- understand why the normal procedure was established;
- understand the assumptions that inform normal procedure;
- be able to identify whether these apply in the current context;
- judge whether the procedure itself can be adapted or over-ruled;
- evaluate what aspects of the normal procedures still need to be met;
- make a decision that meets the requirements of the current situation.

Reflecting on non-routine action

The following prompts can assist you in reflecting on how you act when routine responses may not be appropriate.

- What is the usual course of action in this kind of instance?
- What is the rationale for that routine procedure (i.e. what are the reasons for doing things that way)?
- What was different in this instance such that it was difficult or inappropriate to follow usual procedures?
- Which aspects of procedure had to be adhered to? Why were these essential?
- Which aspects could be adapted or over-ruled? Why is that the case?
- Did you stick to the routine or adapt it? Why was that?
- Did it work well? What were the immediate effects? What were the consequences?
- What, if anything, would you do if this situation arose again?

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Good and bad critical reflection

From what you have read already, you will have started to formulate a sense of what constitutes good reflection. The following table highlights the differences between good and poor critical reflection.

Aspect	Good critical reflection	Poor critical reflection
1 Experience	Draws on personal, group or workplace experience as a means of testing out theory or learning new lessons, looking at experiences with a critical eye.	Assumes 'experience' is an end in itself; that one's own experience is typical of others' without good evidence that this is so; that experience equates to 'insight', without bringing a critical eye to it.
2 Personal responsibility	The best critical reflection demonstrates integrity both in focusing on one's personal role, such as the assumptions brought to a situation or actions taken or omitted, and in taking responsibility for the consequences of these.	Poor critical reflection finds ways of deflecting blame onto other people or the context itself for the way events unfolded. Alternatively, personal responsibility is addressed in a superficial way, so that the relation of action to consequence is not considered in depth. It is also poor practice to assume personal responsibility and guilt for matters that are not simply down to oneself.
3 Focus	Selects a focus such as a particular time-period, set of events, specific kinds of incident or examples of interactions.	Rambles around or covers too many dimensions so it isn't clear what is the focus of the reflection.
4 Scale	The focus is broad enough to offer challenge and meaningful insights but can be reasonably explored in the timescale and word limit.	Is either too narrow to provide the insights needed or too broad to look at things in any depth.
5 Direction	However it may start, good critical reflection begins to take direction as the person starts to identify, and then focus on, selected themes for closer attention.	Rambles or hops about rather than finding a direction.
6 Depth	Delves below the surface: it picks up on initial thoughts and insights, analysing these further with the aim of gaining deeper insights or broader applications.	Is superficial. It doesn't demonstrate any interest in burrowing beneath the surface to understand more.
7 Challenge	Usually tackles a difficult area or enters difficult terrain, such as matters that are personally difficult, or issues that are complex and don't lend themselves to easy answers.	Tends to stay within safe territory or deals with difficult issues in a superficial way or doesn't seem to take the person forward in their understanding.
8 Theory	Draws on relevant theoretical standpoints, research, or established professional practice in ways that demonstrate how these have helped understanding. Where relevant, it relates the particular incident to broader social and political issues.	Draws only on the person's own ideas, experiences and anecdotes, or makes superficial passing references to theory and research.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Good and bad critical reflection (continued)

Aspect	Good critical reflection	Poor critical reflection
9 **Criticality**	Brings a searching, critical eye to the focus of the reflection, to emerging insights, and to any theories or sources of information. This criticality is used to take the person forward in their understanding of the core emerging issues, such as by challenging their own ideas and actions, or showing how their experience supports or challenges existing knowledge.	Is preoccupied mainly with describing situations, content or events. May include critical analysis but this doesn't seem to be used in a way that really develops an understanding of the core emerging issues.
10 **Insight**	The reflection takes the person forward in their understanding, such that they can make more sense of their situation, work or study, manage better within it, do things differently, apply understandings to new contexts, etc.	The reflection gives little indication that the person has moved forward in their understanding of the context or issue, or self-knowledge.
11 **End-points** (extrapolated conclusions)	The process of reflection may take the person in many different directions. However, by the end, they have stood back, drawn out the key messages of what they have learnt, and summarised these as conclusions or recommendations.	The reflection reads more as description of a process or rambling free association. The lessons learned are not drawn out clearly as conclusions or recommendations.
12 **Audience** (if reflection is to be shared)	If this is to be used in academic, work or public contexts, the writing up of reflection demonstrates a sound understanding of ethical considerations and stylistic or academic conventions that may apply. Issues of confidentiality have been addressed appropriately.	The reflection is submitted or made public without all due care being taken to ensure that confidentiality and other data protection issues are addressed; little thought is given to how to make the reflection manageable for others to read.



Reflection: The quality of your critical reflection



Using the table above, consider your own critical reflection.

- What do you think are your areas of strength in producing good critical reflection?
- Which aspects do you need to improve?

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Logs, blog, journals

It is likely that your initial reflections will be written primarily as phase 1 reflection in a journal or blog. If you are asked to submit these as part of an assignment, then read through them following the guidelines in the Audience sections of pages 193 and 208. Amend them accordingly, deleting or blocking out material that you decide not to submit. If your initial reflections are very lengthy, it may be advisable to edit these, or to highlight the salient parts, to help your reader manage the bulk and identify the most relevant sections.

Reflective summary or essay

Typically, you will be asked to produce a critical reflective summary, essay or report. This tends to carry the most weight in an assignment and merits time and attention so as to do justice to what you have learnt. It is important to bear in mind the following.

Academic conventions

Ensure that you apply the usual academic conventions. See pages 147–69.

Referring to theory and research

You may have provided an outline of your background research as part of a project report or made reference to different theoretical perspectives within your phase 1 reflection. However, it is essential that you also relate theory to practice within your reflective summary or essay, bringing a critical eye to these, and making clear how they influenced your reflection and practice.

Examples: Making use of theory

Example 1: Business student
When interviewing these participants on their employment history, details of their personal lives emerged. I hadn't expected to become preoccupied by my own anxiety. Smith (1992) feels that for sensitive topics, there should be an experienced co-leader present 'to adequately monitor the group's comfort level'. I had thought this to be an extreme position for business settings but these interviews made me think again. My difficulty with Smith's suggestion is that having someone present with only that role might be off-putting to the group – and make them feel spied upon. One alternative might be to use two interviewers …

Example 2: Education student
A number of authors draw links between feeding and teaching (Coren, 1997; Williams, 1997). Salzberger-Wittenberg (1983) notes that the processes involved in learning and digestion are similar, i.e. taking in, absorbing and producing. … Within education, 'spoon-feeding' is often used as a metaphor for poor practice, encouraging shallow learning and 'regurgitation'. My reflections on my teaching are that the process is considerably more complex than I had believed. I shall demonstrate that …

Common errors to avoid

When relating theory and practice:

- Avoid putting all references to theory in the opening paragraphs and then forgetting it. If you summarise the research at the outset, make sure that you refer back to it at relevant points throughout the essay.
- Avoid referring to theory without directly linking it to your reflections.
- Avoid reproducing at length the content of theory, such as *Freud said X. He also said Y. He also said Z*. Rather, be specific about how the theory or research is useful or limiting in explaining why something happens.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Summary

This chapter has looked at critical thinking from a different angle, turning the spotlight onto your own actions. Critical reflection of this kind is used increasingly within both academic and professional contexts.

Critical reflection is challenging, requiring you to analyse your own behaviours and assumptions, relating these to broader perspectives from theory and research and, if relevant, professional practice, and doing this with integrity, even when the material itself may be taxing or reveal your actions in a less than perfect light. In addition, initial reflections need to be sifted, the key lessons extracted and related back to theory and practice, and all this presented to others using the conventions of academic writing. Along the way, it is likely that the material will be personally challenging, probably evoking difficult feelings and emotions which you will need to manage well for yourself and others.

The achievement of good critical reflection can feel elusive. Motivation can be an issue, and it can feel difficult to balance the different aspects in a coherent way. This chapter provides ways of addressing these difficulties. It structures the activity into distinct phases and stages, gives examples so that you can see what is expected, and provides step-by-step methods and practical tools that you can use and adapt for your own critical reflection.

It is not surprising that, if you are successful in bringing together all of the above, you are likely to gain enormously. Those who engage deeply in this process generally find that, within a relatively short time, they move from initial, superficial observations to deep, thoughtful conclusions about themselves, their learning, their professional roles, ways of changing things for the better, and even their way of being in the world.

Information about the sources

Borton, T. (1970) *Reach, Touch and Teach* (London: Hutchinson).

Boud, D., Keogh, R. and Walker, D. (eds) (1985) *Reflection: Turning Experience into Learning* (London: Routledge).

Coren, A. (1997) *A Psychodynamic Approach to Education* (London: Sheldon Press).

Cottrell, S. (2010) *Skills for Success: The Personal Development Planning Handbook*, second edition (Basingstoke: Palgrave Macmillan).

Cottrell, S. (2015) *Skills for Success: Personal Development and Employability*, third edition (London: Palgrave).

Driscoll, J. (1994) 'Reflective Practice for Practice'. *Senior Nurse*, **13** (7), 47–50.

Gibbs, G. (1988) *Learning by Doing: A Guide to Reading and Learning Methods* (Oxford: Further Education Unit, Oxford Polytechnic).

Johns, C. and Freshwater, D. (1998) *Transforming Nursing through Reflective Practice* (Oxford: Blackwell Scientific).

Salzberger-Wittenberg, I., Williams, G. and Osborne, E. (1983) *The Emotional Experience of Learning and Teaching* (London: Karnac Books).

Smith, L. (1992) 'Ethical Issues in Interviewing'. *Journal of Advanced Nursing*, **17**, 98–103.

Williams, G. (1997) *Internal Landscapes and Foreign Bodies: Eating Disorders and Other Pathologies* (London: Tavistock Clinic Series, Duckworth).

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Chapter 13

Critical thinking for your future career and employability

Learning outcomes

- To know how to apply critical thinking to your own professional and career development
- To be able to apply critical thinking skills to the process of applying for jobs
- To recognise the value of critical thinking to employers
- To understand how to articulate your critical thinking abilities to employers
- To think critically about your own performance, to help with future job applications

Critical thinking for professional life

Critical thinking skills are highly valued in the workplace, nationally and internationally, and can make the essential difference to the development of your career and success in the job market.

That means it is well worth putting time aside to bring a critically analytical and reflective approach to your consideration of the world of work and your place within it. You can apply your critical abilities to key questions such as:

- Am I considering the issues that I should in order to further my longer-term life, career or work-related aims?
- What will my career pathway look like?
- What kind of jobs should I apply for?
- Are my job applications as good as they could be?
- What am I learning from my experiences of applying for jobs that will help me in the future?



© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

The criticality of work decisions

Most people need to refresh their job-hunting and application skills when looking for work, as trends in the labour market change over time. It is also useful to step back from time to time and think critically about your life and work decisions 'in the round', as you, too, may have changed.

The decisions you make about work will, for many years, affect almost all of your waking life:

- how each day is spent and where;
- the kinds of people you get to meet;
- the conversations you will have;
- the life opportunities that open or close;
- what you wear, eat, see, hear, and feel;
- how much time you have for yourself;
- your health, well-being and happiness.

That is why it is worth giving time to thinking through how your decisions about work and your career will affect other aspects of your life – and the role that you want work to play in your life.

What matters to me most?

Considering what you want from life in general terms might seem rather abstract, especially if you need a job soon and feel you have few choices. However, you can improve your short-term decision-making by thinking critically about what you want from life in the longer term, the roadmap to achieving it, and how your current decisions could help or hinder in achieving your goals.

Ultimately, decisions about your career path will be particular to you, your circumstances and your considerations of such issues as family, friends, relationships, lifestyle, working life and so on. For example, if being creative, or engaging in sport, community work or politics are important to you, then check that jobs you apply for are compatible with furthering those ambitions. If you will have family responsibilities, will you be able to meet these in jobs that require long days, irregular hours or business trips abroad?

Understand the commitment



Whilst a particular job may suit you in the short term, it is unwise to commit to a career until you research what it really means on a day-by-day basis. There isn't much point becoming excited about a job that sounds prestigious, or has a good salary, if you loathe the people, working hours, tasks and commitments that come with it.



Reflection: Lifestyle and values?

- What do you value most in how you spend time? What kinds of jobs would enable that?
- In which sorts of jobs are you most likely to meet and work with the kinds of people you like?
- What other values are essential to you that should inform your career decisions?

© Stellla Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

What is my next step?

This is an important first question. Many graduates take a year or more experimenting with jobs, locations and life-styles. Some take a gap year to travel, develop their creative practice, or work as interns or volunteers. Such an approach can be helpful if you don't have a clear idea yet about what you really want to do.

However, even if you have a fixed idea about how your future career will pan out, it is still useful to pause and reflect critically about your future. Is that career route still best for you? Is it still feasible? Does it inspire you? If your reasoning supports your hypothesis about your career, and there is evidence to back that up, that is good to know. If your critical reflection demonstrates some uncertainties, then it can be useful to explore these before committing further time and expense.



Reflection: Next steps?

- What are the options open to you for how you spend the first year after graduating?
- Are you more interested in getting straight into a graduate career route? Or taking a break before following a career route? Or developing your own business? Or engaging in further study?

If you are not sure of the impact of such choices on your career development, speak to a careers adviser at your university or college. They will be able to help you to explore your options.

Ultimate role

If you do have a clear idea about the kind of job or role that you wish to achieve at the peak of your career, then that can be helpful in giving a sense of direction. In this case, speak to a careers adviser at your university about the routes that you can take that are most likely to make that a reality.



Reflection: Plan towards ideal job roles

- What kind of profession or senior role would you like to aim for?
- What qualifications would you need for this? How long do these take to achieve?
- What kinds of interim jobs and roles would take you to where you want to be? How long would it be expected that you would spend in each?
- What kind of timeframe is needed to achieve your goals?

Which sector and profession?

If you have studied for a qualification that leads into a particular professional area, you may wish to continue on that track. However, if this feels an unwelcome obligation, apply your thinking skills to analysing the job market for other jobs that reward the attributes you have developed.

Conversely, if you want to work in a particular sector and your degree doesn't seem to lead to the obvious careers in that field, there may well be many other good jobs in that sector that need your skills. Most businesses, whether in finance, arts, care or engineering, need a range of skills: technical, management, IT, creative, marketing and so on. Around half of graduate jobs are open to graduates irrespective of degree subject.

Activity **Labour market research**

Find out:

- The best websites for jobs that interest you.
- The number of those jobs available, where they are located, and typical starting salaries.
- What skills and attributes are needed for these jobs.

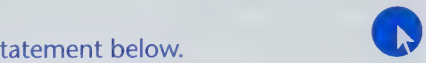
© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Self-evaluation: Thinking critically about your career path

It can be a challenge to know where to start thinking about a career pathway, whether or not you already have a job. Everyone has a different starting place. You may also feel quite differently about your career options as you progress through your course or once you try out a new job in a particular field. Self-evaluations can provide a useful starting point.

Critically evaluating statements, giving thought to your responses, can draw attention to aspects you might have overlooked in the past. Reflecting about your ratings can assist you in drawing conclusions about your career path and actions you should take.

The following self-evaluation provides a starting place for such critical reflection, enabling you to analyse your own career awareness in terms of key components.



- Read each statement below.
- Rate yourself on a scale from 0–4 where 0 is 'not true at all' and 4 is 'very true'.
- Jot down your thoughts as you go – such as actions you can take, things to investigate, or matters you could follow up further with a careers expert.

A	Career and lifestyle choices	Rating ☹ 0 1 2 3 4 ☺	Your thoughts and observations on your ratings
1	I have thought through, in detail, what kind of job would best suit me.		
2	I have a good sense of the professional path that I want to follow.		
3	I have a good sense of the kind of life and lifestyle I want, and jobs that fit with that.		
4	I am aware of the kind of demands that go with the jobs that interest me.		
5	I have a good sense of the kinds of jobs available that match what I am looking for.		
6	I am clear about the full range of job choices open to any graduate.		
7	I am clear about the jobs open to graduates only with my kind of degree.		
8	I am clear whether I prefer to work for an employer *or* have my own business.		

B	Qualifications	Rating ☹ 0 1 2 3 4 ☺	Your thoughts and observations on your ratings
1	I am clear what qualifications are required for the kind of career I want.		
2	I am currently studying the right course for the kinds of jobs or career I want.		
3	I am aware of any further qualifications I may need to study for to reach my goal.		
4	I am aware of the requirements of professional bodies that are relevant for careers that interest me.		

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Self-evaluation: Thinking critically about your career pat

C	Experience of employment	Rating ☹ 0 1 2 3 4 ☺	Your thoughts and observations on your ratings
1	I am aware that I am more likely to get a job interview if I have had a job before.		
2	I am aware that I am more likely to get a job interview if I am in work already.		
3	I have a good track record of work experience already.		
4	I have gained a good understanding of the demands of the 'real life' workplace.		
5	I have experience of work relevant to the career or jobs that interest me.		

D	Skills	Rating ☹ 0 1 2 3 4 ☺	Your thoughts and observations on your ratings
1	I have a good portfolio of skills and experience to draw upon.		
2	I am clear how to articulate my current skills in ways that resonate with employers.		
3	I am aware of the skills that employers are looking for in areas that interest me.		
4	I have strong written communication skills.		
5	I have addressed any weaknesses I had in my use of grammar and punctuation.		
6	I have the skills necessary to run my own business.		

E	Experience of applying for work	Rating ☹ 0 1 2 3 4 ☺	Your thoughts and observations on your ratings
1	I know where to look for jobs that interest me.		
2	I have had success in applying for jobs.		
3	I have learnt good lessons from the process of applying for jobs.		
4	I know how to make a sensible selection of jobs to apply for.		
5	I know how to make a strong written application.		
6	I know how to do well at assessment centres.		
7	I know how to do well at interview.		

© Stellla Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Thinking critically about your career: Taking action

One important dimension of thinking critically about your career is acknowledging the power of your own actions – or 'agency'. This will affect your options and the outcomes of your career. That is especially so at key moments such as when at university, at graduation, or at a cross-roads in your professional life. Use the table below to identify, with a ✓, the actions you consider would be useful for you at this time.

Careers Preparation 'To Do' list	Top priorities from my Careers Preparation 'To Do' list
❒ See a Careers Adviser at my college or university.	1.
❒ Use a specialist text to assist my career development in more depth. See page 228.	
❒ Check that I have the right qualifications for the career path I want to follow.	2.
❒ Browse job websites to get a sense of the jobs that are available and what attributes are requested.	
❒ Find out more about what employers say about graduate skills. See page 219–23.	3.
❒ Gain any work experience (if I haven't had a job recently).	
❒ Gain work experience in a relevant field.	
❒ See if there are jobs available on campus.	4.
❒ Find out more about setting up a business of my own.	
❒ Find out how to grow and develop my current business.	5.
❒ See if there is an enterprise module that I could study as part of my course.	
❒ Find out about prizes and schemes for promoting enterprise where I am studying.	**Others (list here)**
If in employment now	
❒ Make more effective use of the staff appraisal or review scheme where I work now.	
❒ Consider going for promotion in my workplace.	
❒ Speak to my employer about opportunities for development in my workplace.	
❒ Make better use of my employer's training schemes.	
❒ Express an interest in work that would widen my experience, outside of my usual job.	
❒ Look for ways of showing initiative and taking on responsibility.	

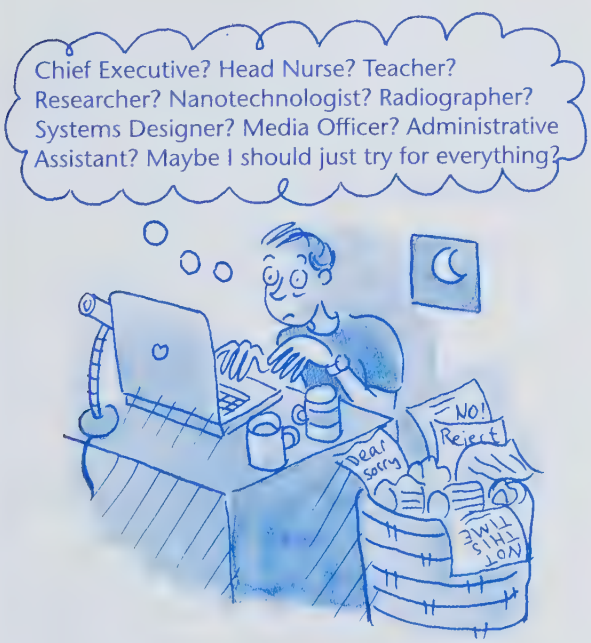
© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Applying critical thinking when looking for a job

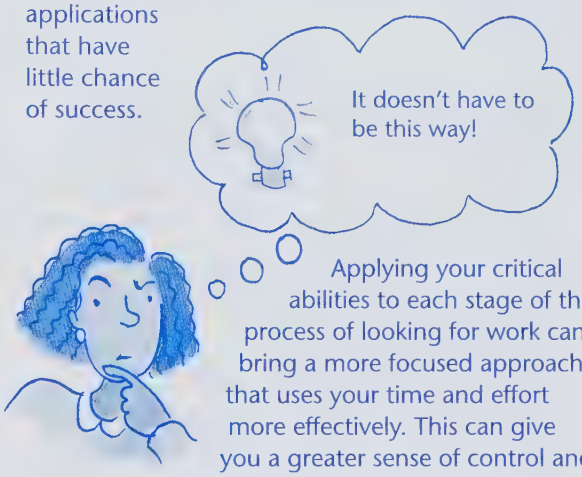
It may be that you find, apply for, and gain the ideal job with your first application. That would be great, but it isn't usually the case, especially if you are ambitious about the kinds of jobs you go for.



Failed applications can deplete your motivation and even undermine your sense of self-worth.



Typically at the point where you are applying for a job, there are many pressures at play. These might be life changes, a need to earn a living or gain a higher income, pressure to prove yourself to others, move home and so on. At such times, it is all too easy to get caught up in the process, applying for many jobs, using time and energy on applications that have little chance of success.



Applying your critical abilities to each stage of the process of looking for work can bring a more focused approach that uses your time and effort more effectively. This can give you a greater sense of control and increase your chances of success.

Apply your critical thinking abilities

- Evaluate your best job options on the basis of the evidence, selecting the right job vacancies on which to spend your time.
- Use the clues: examine the 'person specification' and other employer information closely to identify what the employer is looking for; use that information in order to make a strong application.
- Read critically: interpret the person specification sensibly, in the context of other information provided about the level of responsibility that goes with the role.
- Select the best evidence about your own attributes to match the person specification.
- Check that you have demonstrated why you should be considered for the job, based on the person specification.

More detail about what this means, in practice, is provided on the next page. If you have already made job applications, consider whether you used your critical faculties to best effect at each stage of the process.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

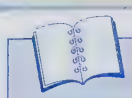
Critical consideration of the 'best fit' jobs for you

What am I looking for?

Before launching into job applications, consider critically what is essential to you in the job you will accept. Just as when you conduct information searches for academic study, you are more likely to find what you want if you know what you are looking for.

What kind of job best suits me?

- *Working for myself or for others.* Am I more suited to running my own business? Do I want to work for a large organisation? A small company? The private, voluntary or public sector?
- *Type of job.* Am I looking for a graduate job that makes use of my degree? Or a starter job to develop experience, expertise and skills? A part-time job to bring in money whilst I pursue my studies? Promotion in my current profession?
- *Roles and sectors.* What kind of roles am I aiming at this time around? What wouldn't I consider? Are there any work sectors that I prefer – or would not consider?
- *Geographically.* Where would I be prepared to work – and where not? How much travel or commuting am I prepared to accept?
- *Financially.* What salary am I aiming for? What is realistic? What is the lowest salary I will accept?
- *Work culture.* What kind of work environment am I looking for? What workplace values are important to me – or would I find unacceptable? How far do things such as driving up profits, customer service, professionalism, teamwork, or social responsibility matter to me? Excite me? Drive me to do better?
- *Lifestyle.* What kinds of hours am I prepared to work? Am I prepared to work evenings? Weekends? To commute daily? Do I want to travel in my work?



Reflection: The right kind of job?

- Jot down your answers in response to the questions opposite. Which factors matter most to you?
- Consider what your answers suggest about the kind of job that would work for you.

Use time effectively before applying

Apply close critical reading skills to all the information available about a job you are considering. Look at the 'evidence' in full, and reflectively, before deciding whether to apply. More information about this is provided on page 219.

Before applying

Before spending time on an application, consider whether you can say 'yes' truthfully to all of the following.

- ❒ Do I really know enough about this job?
- ❒ Is this really the kind of work I am looking for?
- ❒ Does it suit my minimum requirements?
- ❒ Can I see myself in this job role?
- ❒ Am I eligible for this job: do I meet the person specification?
- ❒ Could I really do the work required?
- ❒ At interview, would I be able to sound convincing that I want, and can do, this work?
- ❒ Would I really accept the job if offered it?
- ❒ Will I learn much through this application?
- ❒ Is it worth spending time on this application?

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Use the clues: Information provided by employers

Get into the employer mindset

When putting together your application, first put yourself in the employer's shoes. Deploy your critical thinking skills to consider:

- What kind of person do they want to employ?
- What will they expect of the successful job candidate once employed?
- How will they use their own critical abilities to select the job candidates they want?
- How would a busy employer go about 'weeding out' the least suitable job applicants in order to spend more time considering those applicants that look like a good match to what they need?
- What does all this tell you about what to include and emphasise in your application?

What the employer wants: The clues

Employers are busy people. They don't want to waste their own time nor that of applicants. They make information available so that potential applicants can select themselves in, or out, of the application process. They expect applicants to look very carefully and critically at:

- the job advertisement;
- the job description;
- the person specification;
- background information.

Typically, an employer will want to see that you:

- are capable of finding the information available;
- understand what it means;
- can interpret it in the light of the job role;
- use information effectively and consistently.

Job advertisement

The brief detail in the initial advertisement can signal such things as:

- the reason why the job has arisen now;
- the aims of the organization;
- the level of seniority associated with the role;
- the scale and scope of the job;
- the personal attributes and characteristics that are most important to the role.

Job description (JD)

This tells you what the successful job applicant will have to do. Read it critically to decide whether:

- this is the right level of job for you now – or too senior or junior than would be suitable for you at this point;
- you would be able to 'hit the ground running' from day one and deliver on all, or almost all, of the substantial responsibilities. Employers don't, usually, want to hear that someone could do the job if they were trained to do it. They prefer someone who is 'job-ready'.

Person specification (PS)

The person 'spec' is where you need to focus the most when writing your application as this usually constitutes the list of criteria that employers use to select candidates. Deploy your critical reading skills to analyse exactly what is meant by each of these in the context of this job.

- Do you meet all items designated as 'essential'? If you can make a sensible case about how you meet all of these, you have a reasonable chance of progressing to the next round of the process. Ultimately, the candidate that best meets these attributes will get the job.
- Consider carefully how you meet any attributes designated as 'desirable'. These are generally used to decide between candidates when many applicants meet the 'essential' criteria.

Clues in the 'background material'

- *The application pack*: what message does it convey about the company and the role?
- *The employer's website:* what can you find out about company values, mission, structure, strategies, plans, financial reports?
- *Public information:* there may be useful information about the employer on the internet, such as social network sites, and that held by government bodies or the Charity Commission.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Where do job applicants go wrong?

Understanding where others go wrong can provide a useful steer on where to apply your powers of critical analysis when applying for a job yourself.

1. Not using the evidence available



a) They don't seem to have read the material provided and thought critically about what is needed for the particular role.
b) They contact employers for information already available in the pack or on the website.
c) At interview, they ask questions for which they could have found out the answers themselves.
d) Their application or answers reveal they have misunderstood what the job is really about.

2. Not following instructions!



a) They don't follow, exactly, the instructions about how to complete the application forms.
b) They either don't send documents required, or provide documents that are not needed.
c) They submit applications after the deadline.
d) They don't arrive in good time at assessment centres or for interviews.
e) They don't answer, in a precise way, the questions asked at interview.

3. Taking 'short-cuts' unhelpful to me as the employer



a) They go about things in ways that inconvenience the employer, adding to the time it takes them to process the application.
b) They refer the employer to a CV, additional document, or an earlier paragraph, rather than providing information in the location requested.
c) They ask employers to make special arrangements such as changing interview dates or using Skype, rather than changing personal arrangements.
d) They merge responses to several aspects of the person specification into one answer. This can make it harder for the selectors to allocate points, either because items get overlooked or because of the way information is presented on the employer's recruitment system.

4. Making empty statements



a) They paraphrase the person specification as a statement of fact without providing examples. For example, if the employer asks for 'excellent people skills', they write such things as: 'My people skills are excellent' or 'I am told I have good people skills' or 'I have always had to use good people skills in my work.' That doesn't provide the evidence needed.
b) They make broad statements about the world, their beliefs, or values, in response to questions that call for 'good knowledge and understanding' of the sector or job – rather than giving examples of how they would have developed this through experience at work.

5. Vagueness about details



a) They omit detail that would enable employers to know they really meet all essential criteria.
b) They refer to what 'we …' or 'the team …' achieved in such a way that the employer can't gain a sense of their personal contribution.
c) They omit detail that would enable the employer to gain a sufficient sense of the level of seniority and personal responsibility for achievements.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

6. Stating one thing; demonstrating the opposite



a) They claim good communication skills but do not communicate convincingly that they are a good match for the post.
b) They state they have good communication skills or good 'attention to detail', but their application is poorly phrased, contains errors, or doesn't appear well proof-read. This undermines their credibility and can be taken as evidence that they don't meet the criterion.

7. Lack of attention to detail



a) They do not address every aspect of the person specification (so lose points).
b) They omit information requested, such as dates, salaries, qualifications, references, etc.
c) They submit an incomplete work history – leaving gaps open to the employer's imagination.
d) They make silly mistakes such as spelling errors, or getting the company name wrong.

8. Appearing more interested in themselves than the employer



a) They describe their skills and qualities without making it clear to the employer how these are relevant for the job for which they are applying.
b) They talk about longer-term ambitions which are not manifestly in the interest of the business.
c) They appear to be over-interested in holiday entitlements and benefits, asking about this when information is already provided.
d) They apply for a full-time job whilst having their own business as well – so it isn't clear about their commitment to the job.
e) They appear more concerned about getting a job than in this particular employer and role.
f) They state, in interview, that they didn't have time to make a good application and/or to research an aspect of the company.
g) They appear to criticise the employer, such as by saying that it was hard to find information on their website.
h) They appear insensitive to, or ignorant of, the nature of the business and the demands it would make of employees.

9. Not applying for suitable jobs



a) They apply for jobs that are manifestly well above their current grade.
b) They apply for jobs where the salary is much higher than they are paid in their current role (usually an indication of a senior post requiring significant experience).
c) They assume that a job that requires skills and job-based experience is one that they could do simply because they care about the area or have life experience.
d) They apply for jobs that don't really interest them, so that they don't really sound convincing about wanting the job.

10. Not tailoring applications/ responses



a) They recycle information that they have used on a previous application, without adapting this sufficiently to suit each application.
b) They submit applications that contain details that were clearly written for a different job application.
c) They simply send in a standard CV and hope that that will suffice.
d) They write brief answers on application forms and assume that the employer will use the CV to fill in any gaps.
e) They provide incomplete information in interviews, assuming the employer will refer back to the application documentation (which they rarely do).

© Stellla Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Where job applicants go wrong: Examples

Using the errors outlined on pages 220–1 as a guide, identify where each of the candidates below is going wrong. There may be more than one error in each case. What advice would you give them? When you have thought critically about these, see pages 301–2.

Example 1: Serina

It is April, and Serina is applying for a teaching post in a primary school. She will graduate this year with the relevant qualifications. The advert says that interested candidates can attend an open evening to be held on April 20th. The start date for the new job is given as August 14th. Serina and her friends have already booked a holiday overseas for three weeks in September. She contacts the employer to enquire:

- *Will the school consider newly qualified teachers?*
- *Is it possible to visit the school before applying – such as one weekend?*
- *If she gets the job, could she delay starting in the post until October?*

Example 2: Arno

Arno is a recent graduate in Sports Journalism. He has struggled to find a job that suits his circumstances so is widening his net. He applies for a post as deputy-director of student services at a university. The person specification asks for 'significant experience' in managing a service budget, and leading one or more student services, such as Counselling, Disability, Financial Advice, Accommodation, or International Office. Although he hasn't the exact experience requested, he writes a detailed application arguing that:

- *He is a recent student so has a good understanding of what students need.*
- *He ran a student Judo Society so has experience of leading and inspiring others.*
- *He managed the Judo Society budget (£500).*
- *He worked part-time over the summer in the International Office at his university.*

Example 3: Kim

Kim is a final year business management student, and keen to find a job. He has written dozens of applications. One is to Kiaru Holdings for a graduate internship. The fifth item in the person specification asks for 'Ability to work independently and as a member of a team.' Kim's response to this section is:

> *I am told I am easy to get on with and I enjoy working with others. As a student, I have undertaken several projects that called for good team working. I am an avid sports player and a keen member of several societies such as the Radio Society. I believe I would be able to transfer these skills easily for the role at Kairu Holdings and be a good team member.*

Example 4: Lizzy

Lizzy is a Social Policy student applying to Gener8, a large national energy company, based near the capital. The job advert asks candidates to submit an application form online, with a covering letter, without a CV. The person specification includes requirements for a person:

- Educated to degree level in any subject.
- Excellent communication skills.
- Willing to travel and work flexible hours.

In her covering letter, Lizzy's writes:

> *I am excited by this opportunities, especially as I have always wanted to live in the capital. I grew up in a small town but my university is in a large city and I really like city life so would be flexible about moving to the capital. I will be graduating this summer a degree at university level 2.1 in Social Policy. I would love a job with MTZ-Co. I am delighted attach my CV which outlines the skills I gained through study degree and my part-time job as a store supervisor for 18 months and a volunteer care worker 10 months.*

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Employer demand for critical thinking skills

A global demand

Employers all around the world are looking for graduates who are good critical thinkers – who can bring an open mind, be self-reflective, able to manipulate multiple perspectives and challenge conventional ways of thinking skills (Diamond et al., 2011).

In the UK, surveys of employers over several years indicate the ongoing demand for critical thinking skills and for associated abilities, such as problem-solving (CBI, 2009; Lowden et al., 2011).

In 2013, a major survey in the USA found that over 93 per cent of employers placed more value on the ability to think critically, solve problems and communicate effectively than they did on the degree subject. This trend is increasing in the USA. By June 2014, US job advertisements were twice as likely to require critical thinking skills as had been the case in 2009 (Korn, 2014).

In India, recruiting companies have noted gaps in the level of critical thinking abilities needed for jobs in the global workplace (Fischer, 2012).

In China, employers are looking for employees with 'soft skills' such as communications, innovation, and taking the initiative in problem-solving. The lack of such skills has been linked to gaps in critical thinking abilities (Chan et al., 2015). The importance of critical thinking is recognised in the Chinese government's flagship International Scholarship Exchange Curriculum (ISEC), which includes critical thinking as one of its eight core aspects.

What do employers mean by 'critical thinking'?

When reading job advertisements and employer descriptions of skills needs for their business, you may note that the term 'critical thinking' is applied in different ways from one employer to the next. That means it is important to check exactly what is meant for each role.

Depending on the job role, employers may be looking for employees who can apply critical thinking in the following ways:

- Horizon-scanning with a critical eye, spotting problems and opportunities quickly, then taking the initiative in finding viable, creative solutions.
- Critical reflection of personal professional practice, generating improvements.
- Critical evaluation of communications and interactions with colleagues, clients, etc.
- Making good work-related decisions, based on a sound and speedy evaluation of a situation and information.
- Bringing a critical eye to business plans, new proposals, designs, changes in policy, etc.
- Analysing data, identifying what is significant.
- Being open to new ideas and perspectives.
- Applying systematic analytical skills to matters relevant to their business.

Whatever the role, employers tend to value employees who can think sensibly for themselves, recognising when they need to investigate an issue further or test things out. They want to feel confident that whoever they employ can be trusted to get on with things, using common sense in a critical way that is right for the business context, without having to be asked.

Such skills are associated with minds that are used to thinking about issues from varied angles, formulating and testing out ideas or hypotheses, hunting out information, and drawing on evidence in a suitably selective way to make sound decisions. In other words, they want the kind of skills that are usually developed through academic study or critically reflective professional practice.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

How critical thinking skills are useful in work roles

Critical thinking skills are at the core of most graduate jobs, regardless of the field. They support a range of related skills and personal qualities, such as those listed below.

- **Problem-solving skills**: the ability to generate a range of solutions, evaluate these and apply the best one to a situation.
- **Ability to make decisions under pressure:** the more well-developed your critical thinking skills, the better you are able to make quick decisions that are grounded in an analysis of what the situation demands.
- **Better communication with others:** if you are able to make sense of what is required in a work situation, you will be in a better position to communicate this clearly to colleagues and other stakeholders such as customers, patients, funders.
- **Better empathy with others:** good critical thinking skills increase your ability to see a situation from multiple perspectives and to appreciate others' viewpoints. This is particularly valuable in roles where relationships with others are central, such as service industries, sales, nursing.
- **Increased confidence:** knowing you can apply your critical thinking skills to any work situation increases your confidence in your ability to perform the role.



'When I started in my job as a communications officer, I hadn't done anything quite like that before. I was well-practised at analysing texts as a student so I decided to apply the same underlying critical skills in my work role.

A lot of the time that just means that I ask critical questions of whatever is in front of me – such as "What message are we trying to get across, here?" "Do we need to alter the way we present that for different audiences?" "The graphics look great, but do they add to, or detract from, the message we want to convey?" "What evidence have we got that people will access messages using the media we have been using?"'



Reflection: Transferring criticality

How could you apply your critical thinking skills more systematically to help you with new situations and problems that you encounter?

'As a nurse, everything I do requires me to observe, analyse, evaluate and take action. If a patient tells me that they are in pain, there is a whole list of things that I run through quickly in my head:

- What is the source of the pain? Is it from a known condition or injury or could it be a sign of something new?
- Is the patient showing physical signs of being in pain such as sweating or hand-wringing?
- When did they last have pain medication? What would the possible implications be of giving them a dose at this point, e.g. side effects, overdose?
- What is my patient's experience at the moment? Could I also ease their pain by spending a few minutes talking to them or suggesting another option such as meditation to help them manage the pain?'



© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Demonstrating critical thinking to employers

How do employers check for critical thinking skills?

If employers are interested in critical abilities, they will look at how you demonstrate these through:

- Your approach to the application process – does it reflect a critical analytical mind?
- Your awareness of how critical thinking skills are relevant to the world of work generally, and their business in particular.
- The examples you give of how you applied critical thinking in practice.
- Whether you come across as someone they would trust to think wisely about the kinds of problems and issues that will arise in the job.

Articulate strengths in relevant ways

When applying for jobs that require critical thinking:

- Think through, logically, what the job is likely to entail. What kind of day-to-day problems will you have to deal with? How would you deal with these, drawing on your critical abilities?
- Identify occasions when you have already used the kinds of critical thinking skills that will be needed in the job role – such as analysing data, writing reports, investigating issues, identifying solutions, reflecting critically on how you worked with others, etc.
- Draw parallels for yourself between ways you have applied critical thinking skills already, and the way these skills might be needed in this job. Make sure that these then come across in your application and answers at interview.
- Be as precise as you can in matching the skills you have to the work context, especially in terms of scale and complexity.

Critical thinking and empathy

For some roles, you need to apply critical thinking skills in an empathetic way. During the application process, the employer may be looking to see:

- Whether you have been able to recognise the needs of their business.
- Your understanding of their customers, clients, patients or stakeholders: how your experience or skills will be of value to them.
- Your consideration of other personnel: how your needs might impact on others.
- Difficult situations and how you would respond to the needs and feelings of others when a situation is challenging.

Criticality and forward thinking

When asking for critical thinking skills, some employers are looking for employees who:

- Are pro-active in scanning the horizon to see where opportunities and issues lie.
- Can present opportunities in a report that balances potential benefits and risks.
- Think ahead – and spot where potential problems might emerge.

Substantiate your claims

Provide evidence that demonstrates your critical abilities. A good application provides:

- at least one longer, and preferably two or three examples altogether, of when and where you have best demonstrated the attribute;
- and, if feasible, a sentence that sums up other instances to demonstrate the breadth of your abilities;
- all worded succinctly and precisely.

'In preparing an exhibition of my art work for the public, I used good critical judgement in selecting works that created the best overall effect. I wanted local people to attend so reflected that in where and how I promoted the exhibition. I provided commentary that made the work meaningful to the community. This was successful, resulting in good attendance and positive feedback.'

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Critical self-evaluation of job applications: Checklist

Apply your critical thinking skills to your own job applications and performance. Use the checklists below:

a) before submission to make sure your application is as strong as possible;
b) during the process, to make sure you remain focused on the requirements of the employer;
c) afterwards, whether successful or not, to help you to be even better next time.

A. Job application checklist		Done ✓
1.	**I demonstrate my suitability at every stage of the application process**	
a)	I am careful that skills I claim are demonstrated throughout the application process	
b)	I demonstrate good communication skills in ALL contact and writing	
c)	I have read my application carefully, rewording anything that is poorly phrased	
d)	I have checked that my online presence is suitably edited and helpful	
e)	I have updated my online profile, if relevant (such as my Linked-In account)	
2.	**I demonstrate that I am interested in what the employer needs**	
a)	My application conveys a general awareness of the business and job requirements	
b)	I avoid talking about longer-term ambitions that are not in the employer's interests	
c)	I don't sound keen about being 'away from work' (on holiday, homeworking, etc.)	
d)	I have clearly taken care with my application at all stages	
e)	I avoid criticisms of the employer, its processes, information, website, etc.	
3.	**I have used carefully all the information made available by the employer**	
a)	I have read carefully ALL the information provided by the employer or agency	
b)	I have checked the information available on the internet about the employer	
c)	I have not requested information already provided or available online	
d)	I am clear about what this job involved before making an application	
4.	**I have followed the application instructions exactly as asked**	
a)	I have completed ALL sections of the application forms, leaving no blanks	
b)	I am submitting ALL the documents required	
c)	I have removed anything that is not required	
d)	I am submitting my application in the method required	
e)	I will have met the required deadline for submitting my application	
5.	**I avoid broad statements and 'empty' answers**	
a)	I avoid simply repeating or paraphrasing the requirements in the specification	
b)	I avoid making broad generalisations about life, the world, the economy, etc.	
c)	In referring to team achievements, I make clear what my personal contribution was	

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

A. Job application checklist		Done ✓
d)	For ALL requirements, I give examples to demonstrate relevant experience/abilities	
e)	My examples of experience clarify the scale of tasks and level of seniority involved	
6.	**I demonstrate throughout that I pay close attention to detail**	
a)	I have provided all information requested (dates, salaries, qualifications, etc.)	
b)	My work history dates follow on exactly from each other, with any gaps explained	
c)	I have checked that essential information is accurate – such as the company name	
d)	I have checked my application carefully for errors, typos, spelling, etc.	
7.	**I have focused on the person specification (PS)**	
a)	I have addressed every aspect of the PS	
b)	I have taken every item in the PS seriously and in full so I don't lose points	
c)	I have treated each item in the PS separately, so it stands out from other items	
8.	**My application is tailored to this particular job and employer**	
a)	I indicate how my skills and experience are relevant to this particular job	
b)	I have adapted my CV and material to match this job	
c)	I have removed any information that refers to different applications	
9.	**I avoid 'short-cuts' that create a poor impression**	
a)	I have taken care not to contact the employer unnecessarily	
b)	I provided all information in the sections where it was requested	
c)	I have taken care not to ask the employer to look at other sections or documents	

B. Preparing for each stage of the process		Done ✓
10.	**I prepare well for assessment centres' exercises and interviews**	
a)	I have checked the kind of exercises that will be provided	
b)	I practise similar exercises several times so I am well prepared	
c)	I reread all employer and job information before meeting employers or agents	
d)	I prepare a list of possible questions and ideal answers, and practise these aloud	

C. Learning from the experience		True ✓
1.	I take up all offers of feedback on my performance in job applications	
2.	I take the feedback seriously, as helpful guidance for the next application	
3.	After interviews, I jot down the questions I was asked, to help prepare next time	
4.	I reflect for myself on where I could do better next time	

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Summary

Critical thinking skills tend to be associated with academic life and study, as higher education develops such skills in a systematic way. However, they can be developed and used in any context. That includes within the context of employment, as skills relevant to the workplace, and as skills that help you achieve a career or a job that suits you.

It is more likely that a candidate will be successful in gaining the job if there is a genuinely good match between them and the job on offer. If a candidate hasn't really thought about themselves, their own values and interests, and the kind of work that would really fit the kind of person they are, then it is likely that this will become evident during the application process. Even if it didn't, it wouldn't be helpful to enter a career path or take on work that you can't sustain: this would affect your work record. If you appear to be unable to stick out a job for a length of time, future employers will be reluctant to employ you.

This means it is important to spend time applying your critical thinking abilities in a reflective way, making an honest appraisal of what will, or won't, work for you. It means researching the labour market and considering what kind of career path would work for you. It is unlikely that your path will work out exactly as you expected, but such planning can help to focus your time more effectively when looking for work. It can also help you to avoid costly mistakes through taking the wrong qualifications or disrupting your home circumstances for a job that won't suit you, or spoiling your continuous work history.

If you only had time to focus on one aspect of the application, that should be the Person Specification. If one is not provided, then use the job description. In general, look carefully at what kind of person the employer says they are looking for. Take every aspect of the person specification seriously, as each may accrue separate points. Avoid merging two items into one response, as you may lose points. Similarly, write the information for each point in the right place – don't refer the employer back to earlier answers, a CV, or other documents. It is likely that they won't look at anything that isn't where they expect it.

Apply your critical abilities to each stage of the application process to evaluate how well you are addressing the requirements of the employer and to appraise how you would come across. Research employers and check that your answers are relevant to them. Consider what your contact with the company, in its totality, will say about you. Will it demonstrate that you are the kind of employee who puts the business first, who can be depended upon to be available, flexible, and who takes care to get things right without being directed to do so? These are practical things that are useful to almost all employers.

Use your critical self-appraisal constructively to fine-tune your application, to prepare for assessment centres and interviews, and to manage the way you present yourself, whether in person, on paper, or online. After each stage, reflect on how you could do things better next time. Keep notes to help you with your next application. Whether you were successful in gaining the job or not, there is always something that you can learn from the experience.

Further reading

Bolles, R.N. (2016) What colour is your parachute? (New York, Ten Speed Press).

Cottrell, S. (2015) Skills for Success: Personal Development and Employability, 3rd edn (London: Palgrave).

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Texts for Activities in Chapters 8, 9 and 11

The texts on pages 229–233 have been written to support the activities in Chapters 8, 9 and 11, are all available on the companion site for this book. See www.palgravehighered.com/cts3e. Names, references and data produced in the texts below are fictitious.

Text 1

It isn't really stealing to share music or video files on the internet. True artists want their work to reach as many people as possible. They are more concerned about the effect of their work on their audience than on base concerns like money. Labels and film studios are only interested in music and film with broad appeal that will bring in large profits. They overlook innovative and radical film and music that is better artistically but doesn't sell as well. Most independent artists struggle to distribute their work. People who share files with friends via the internet perform a useful service to art as they make more people aware of small artists and diverse music and film. Without this, media would be extremely bland and middle-of-the-road.

Carla: in internet chat room, Cla.media.room.host 7 September 2016; Carla is a regular file sharer of music and video.

Text 2

Neighbours are generous with the cuttings they take from plants. Up and down the country, people are exchanging cuttings from their roses, fuchsias and hostas. Many of the plants they share are registered for Plant Breeders' Rights. This entitles the person who bred or discovered the plant to a royalty. Gardeners never bother finding out which plants they must pay a royalty for. A cutting is gardeners' equivalent of file sharing for music lovers. If gardeners don't bother paying royalties on cuttings, why should other people pay royalties, such as for music downloaded from the internet?

Ivan Potter, in *Your Gardening Questions*, a popular monthly magazine published by GPX Publishers in London, vol. 6, June 2016.

Text 3

Piracy of videos, music and software is stealing, whether this is done by file sharing companies or between individuals. Some people argue that it is acceptable to make copies of works as everyone does it. I have also heard the argument that it isn't really 'stealing' as it does not take a file away from a company, it just creates a copy. They rationalise that companies have themselves to blame for charging high prices. They forget development costs and that companies are entitled to charge what the market will bear. Consumers have a choice. If they want the product enough, they should be prepared to pay for it. If not, they should go without.

Cuttle, P.D., Legal expert writing in article, 'Steal it Away'. In *National CRI Law Journal*, vol. 7, issue 4, April 2015.

Text 4

Publishers and producers of music and film are mainly concerned about large-scale copying by what amount to alternative businesses. These businesses make multiple pirate copies and sell them on at much lower prices. Media companies are not bothered about ordinary members of the public using file sharing to catch up on the latest films or TV shows.

Arnold Spratt, editorial column, in *The Middletown Argus* newspaper, 17 June 2014.

Text 5

More and more people are accessing their favourite TV shows via file sharing. The kindness of those that share files should be applauded. It is likely that by now, pretty much everyone will have downloaded a file via a file sharing site. If everyone does something, it can't be bad, and if it isn't bad, then where is the crime?

Alan Hibbs, a member of the public who usually pays for downloading media, in a letter to the editor of the *National Press Daily*, 3 November 2016.

© Stellia Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Texts for Activities in Chapters 8, 9 and 11 (continued)

Text 6

Many music and film distributers are not major business concerns. They only employ a few staff and are reliant on the overall sales of the media from many small artists. This is particularly true when distributing independent artists, where sales are always relatively low and many don't sell at all. As the market for such artists is low, even a few illegal downloads made by each potential purchaser would have a dramatic effect. Illegal file sharing is likely to contribute to the shaky financial base of the small distributors upon which the independent music and film industries depend.

Callum Kahliney, 'Is this the end of the Road?' In *Small Music Distributor*, 12 August 2013. Article in trade magazine for small distributors.

Text 7

Lawyers argue that gardeners who give away cuttings of plants that are registered for Plant Breeders' Rights (or PBRs) are cheating the people who brought the plant into the market. Breeding a new variety of plant does not come cheaply. It can take many years to develop a new strain so that it is ready for marketing. For a plant to be accepted for PBRs, it must have proved that it is stable and uniform so that those who buy it know what it will look like several years down the line. The plant has to be distinct so that it can't be confused with other plants. For every plant that succeeds, a breeder may have thousands of failures, each of which incurs a cost. Breeding can be costly, requiring investment in research, protected and controlled planting space, and specialised labour. If a breeder is lucky enough to be successful, they then have to pay a large sum to register the plant and there are further costs to renew the registration each year. After all that, the plant will last for only about 20 years, and the royalty runs out after 25 years. This means breeders need to maintain their investment in developing future strains or they will be deprived of an income. The royalty on a plant can be between 20 and 30 pence per cutting, or more. Multiply this by many thousands, and the breeders are really losing out. Whether or not they ever receive this money comes down to the average gardener's ethical sensitivity and their awareness of PBRs. It is unlikely that the police will descend to recoup the royalties: lawyers focus on the big companies. However, as the lawyers point out, that doesn't mean free cuttings are acceptable: some breeders need every penny if they are to continue to produce new varieties for us to enjoy in the future.

Anjeli Johl, 'Counting the Cost of Flowers', in the *National Press Daily* newspaper, 10 July 2016. Johl is a regular columnist in the paper's reputable law section.

© Stellla Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Texts for Activities in Chapters 8, 9 and 11 (continued)

Text 8

It doesn't make sense to argue that people shouldn't share files of music or video over the internet, even once, but that it is acceptable to copy images or music for use as ringtones or screensavers, on a daily basis if you choose. Before the internet, home taping of music from the radio was a daily occurrence. It may also be illegal to copy images and music without permission but, as with home taping, nobody bothers about this as it is almost impossible to catch people. Just because Internet Service Providers are able to detect file sharing shouldn't make it a crime. It's no worse than using an image from a film as your phone or computer screensaver.

Lee, A. (2015) 'Why Buy?' in R. Coe and B. Stepson, *Examining Media* pp. 36–57 (London: Many University Press).

Text 9

Although it is possible to devise software to catch people who file share on the internet, it is unlikely that everyone who shares files could be charged. If you can't enforce a law, then there isn't any point in passing it. If there isn't a law, then there isn't a crime.

KAZ, on AskitHere.truth, personal website, November 2015.

Text 10

Moral and ethical issues are not simply questions of right and wrong. They should be more properly regarded as dilemmas. The decision that the law, or 'justice', should ultimately decide what is right or wrong has never been made in a democratic way. The law was scrambled together over time and is often contradictory. There is too little public discussion on whether the whole concept of justice is what we really want as the basis of right and wrong. From time to time, throughout history, brave individuals have stood up to the law and, arguably, it is mainly through their defiance that the law has progressed at all. Even today, when an issue really matters to them, individuals will brave prison on the basis of their individual consciences, when the law appears to them to sanction immorality or bad ethics. Peters (1974) and Gilligan (1977) have argued that there are grounds for giving priority to other matters, such as autonomy, courage and caring about what happens to other people. Even Kohlberg (1981), who took a justice-based approach to morality, stated that being able to make judgements about justice was 'a necessary but not a sufficient condition for moral action'.

Fred Piaskin in an article, 'Moral Dilemmas in Action', in the *Joint Universities Journal of Advanced Ethics*, in 1986. Volume 8, issue 2.

Text 11

It is stealing to copy text from a book, article or the internet without acknowledging the source of the information. It is regarded as theft of the intellectual copyright of another person. This is treated very seriously by universities. However, stealing suggests you know you are taking something that is not yours to take. Many students are confused. Most know that if they use the exact words in a source, this is a quotation and they must cite the source. However, many believe, erroneously, that it is acceptable to copy whole sections as long as they change a few words here and there.

Soyinka, G. (2013) 'Plagiarism Unveiled'. In *Journal of HE Worldwide*, 27 (3), pp. 231–47.

Texts for Activities in Chapters 8, 9 and 11 (continued)

Text 12 (abbreviated version of research paper)

Ebo, T., Markham, T.H. and Malik, Y. (2014) 'The Effects of Ease of Payment on Willingness to Pay. Ethics or Ease?' *Proceedings of the Academy for Ethical Dilemmas*, vol. 3(4).

Introduction

This paper sets out to show that behaviour is affected primarily by how easy it is to act in an ethical way. It demonstrates that in the Oldlea area during 2008–13, there was a decrease in illegal media file sharing from the internet following schemes that enabled easy payment online to download music and video.

The research builds on the ground-breaking research by Mixim, Moss and Plummer (1934) which showed that some forms of theft were not based on the desire to steal but on inertia when faced with complex or onerous systems of payment. Mixim et al. found that at specific ages, people found it more difficult to queue, and had a tendency to focus on the symptoms associated with queuing rather than the requirement to pay. This resulted in them leaving shops to alleviate their discomfort, forgetting that they were carrying items for which they had not paid.

Damblin and Toshima (1974) acknowledged the theoretical framework of Mixim et al. but criticised the evidence base, which involved only 30 participants over a short time span. Damblin and Toshima (1986), using a sample of 200 senior citizens, found that there were significant differences in ethical behaviour depending on people's medical conditions. Several research studies have shown external conditions can have a greater impact on behaviour than ethical understanding (Singh, McTiern and Brauer 1991; Colby, 1994; Miah and Brauer, 1997). However, no studies have focused on people under 25 years old, nor the impact of the internet on such behaviours …

The research hypotheses are (1) that most young people who download music for free will pay a fee to download copies of the music if this is made easy, and (2) that the willingness of people to pay for music will depend on income, with high earners being more willing to pay than low earners.

Methodology

Participants were divided into three groups and two conditions. The three groups were divided into low, middle and high earners. In the first condition, facilities for quick and easy payment to download music were made available. In the second condition, the system for paying was time-consuming and complicated. The participants were 1206 people aged 15–25, matched for age, sex and ethnic background across groups and conditions. An advertisement for an alternative website where music could be downloaded for free appeared when the participant was online. This offered free downloads but carried a message that failure to pay deprived the artist of income.

Results

The results supported the first hypothesis but not the second. The results for the first hypotheses were significant at …

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Texts for Activities in Chapters 8, 9 and 11 (continued)

Discussions and conclusions

These research findings suggest, as with the older age groups, that when it is easy to pay for a service, most people aged 15–25 act in an ethical way.
When given the choice of an easy payment option
or an unethical method of free access, 78.6 per cent of purchasers selected the payment route. When payment methods were complicated, only
47 per cent of purchasers paid for their purchase, the others opting instead for the free site. Before making a purchase, almost all participants, 98 per cent, investigated the free site. This shows that they made an ethical choice when they opted to pay, rather than simply choosing the site they were allocated.

However, the second research hypothesis was not supported. This study found that 86 per cent of participants in the low-wage group paid for the music, compared with 69 per cent of those in the middle-income group and only 47 per cent in the highest-income group. This suggests that ethical responses are stronger in low-income groups and weaker amongst high earners.

References

Colby, R. (1994) 'Age, Ethics and Medical Circumstance: A Comparative Study of Behaviours in Senior Populations in West Sussex and Suffolk'. *South West Journal of Age-related Studies,* 19, 2.

Damblin, J. and Toshima, Y. (1974) 'Theft, Personality and Criminality'. *Atalanta Journal of Criminal Theory,* 134, 2.

Damblin, J. and Toshima, Y. (1986) 'Ethics and Aging'. In R. Morecambe, *Is Crime Intentional?* (Cambridge: Pillar Publications).

Miah, M. and Brauer, G.T. (1997) 'The Effect of Previous Trauma on Crime-related Behaviours'. *Atalanta Journal of Criminal Theory,* 214, 4.

Mixim, A., Moss, B. and Plummer, C. (1934) 'Hidden Consensus'. In *New Ethical Problems,* 17, 2.

Singh, K.R., McTiern, S. and Brauer, G.T. (1991) 'Context and Action: Situational Effects upon Non-typical Behaviours in Post-retirement Males'. *West African Journal of Crime Theory,* 63, 3.

Practice activities on longer texts

The following pages provide activities based on longer texts, all of which are available on the companion site for this book. See www.palgravehighered.com/cts3e.



The texts for practice activities 1 and 3 provide examples of better critical writing. These give you the opportunity to identify the features of an argument when reading a longer text. They also provide you with a basis for comparison when you go on to analyse the examples of poorer critical writing provided for Practice activities 2 and 4.

Texts 2 and 4 provide opportunities to identify examples of poor argument.

Prompts and answers are provided for each practice activity.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Practice 1: Features of an argument

- Read the passage 'Global Warming Requires a Global Solution' and identify the features of the argument, using the numbered prompts below to assist you.
- Label and number each of your answers in the *Comments* margin provided alongside the text. If you use the same numbers as those provided in the prompts table below, this will help you to check your answers.

Prompts	Done ✓ (tick when completed)
1 Identify the sentence or sentences that sum up the main argument.	
2 Identify the author's position.	
3 Identify the summative conclusion.	
4 Identify the overall logical conclusion.	
5 Identify the main reasons given to support the logical conclusion.	
6 Identify any intermediate conclusions used as reasons. In the margin, explain the purpose of the interim conclusion (i.e. why the author needed to come to an interim conclusion in order to develop the argument).	
7 Identify evidence given to support reasons.	
8 Identify descriptive text that provides background information for the reader.	
9 Identify words used to signal the development of either the main argument or arguments leading to intermediate conclusions.	
10 Identify any counter arguments put forward by the author.	
11 Identify arguments by the author to address counter arguments.	
12 Identify any use of primary sources.	
13 Identify any use of secondary sources.	

© Stellla Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Global Warming Requires a Global Solution (Text 1)



The increase in greenhouse gas emissions over the past 50 years is viewed as a major factor in global warming. Research by the leading world authorities on global warming, the Intergovernmental Panel on Climate Change (IPCC) suggests that even if all carbon dioxide emissions ceased today there could be climate change for a number of years to come, leading to water shortages for five billion people and increased flooding across Northern Europe by 2025. However, scientists have proposed a range of solutions from increasing efficient use of fossil fuels to incentives for using cleaner forms of energy, which they believe, if applied globally, would be sufficient to make a real impact on climate change.

The COP21 Paris Agreement was agreed in draft form in 2015 as a means of working towards a reduction in greenhouse gas emissions and limiting the increase in global average temperature to below two degrees Celsius. The Agreement opened for countries to sign up to in 2016 but the actions within the Agreement will not start until 2020. It is expected that 120 of the 195 countries within the United Nations will sign up to the Agreement but it is not legally enforceable and each country will decide its own national targets. Whilst several of the world's greatest polluters have indicated that they will sign the Agreement, others are more resistant. Some governments of developing nations argue that it is not in developing countries' interests to form part of a global agreement. However, I shall argue that, although it may need to be applied differentially, a global and enforced solution is ultimately essential and that developed countries need to take the lead.

Politicians, scientists and businesses in developed countries have given a number of reasons for not signing up to the Agreement. These include doubt about the real link between carbon dioxide emissions and global warming, concerns about the effect on their own economies and the rejection of a voluntary solution, which could lead to some countries not 'pulling their weight'. Meena Raman (2016) has argued that developing nations should not rush to sign the Agreement. She argues that they should wait to see how well developed nations fulfil their pre-2020 obligations under the 1997 Kyoto Protocol, and that waiting would give developing nations more political leverage in future negotiations. Whilst this argument has some validity – developed nations are the main world polluters – global warming is a problem for everyone, not just those in developed nations, and requires everyone to participate. Le Page (2015), argues that the Paris Agreement is already 'too little, too late' in terms of finding a global solution to global warming.

Comments

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Emissions from developing nations are clearly an important issue. However, for developing nations, the argument that they should sign up to climate change restrictions at the same time as developed nations does not carry weight. After watching developed nations build their wealth and power on extensive use of fossil fuels this appears to be a case of 'do what I say, not what I do'. Whilst the Agreement states that developed nations will financially support developed nations in making emission-reducing changes, many have argued that this is not enough. Raman (2015) notes that financial support has already been delayed from 2020 to 2025. Naidoo (2015), executive director of Greenpeace International, states:

> *'There's not enough in this deal for the nations and people on the frontlines of climate change. It contains an inherent, ingrained injustice. The nations which caused this problem have promised too little help to the people who are already losing their lives and livelihoods.'*

The Grassroots Global Justice Alliance (2015) accuses developed nations of 'hypocrisy', arguing that they are pushing other countries to take actions that they have not been willing to take themselves. They call the Paris Agreement 'a failure, condemning humanity to a slow and painful death'.

Former US Vice President, Al Gore (2015) suggests the Paris Agreement 'may herald the end of the fossil fuel era'. Many expect the Agreement to lead to heavy taxation of fossil fuel companies and a gradual disinvestment in their stocks and a move towards renewable energy technologies. This is likely to be easier for developed nations than developing countries. Elliot (2015) notes that developed nations have tended to outsource manufacturing and some services to developing nations, where costs are cheaper. This increases the demand for energy in developing nations – most of which still comes from fossil fuels. For the Paris Agreement to be successful in limiting global temperature increases to less than two degrees, it is estimated that between two-thirds and four-fifths of remaining fossil fuels would have to be left in the ground and unburned. Whilst several major polluters, including China and India, have committed to developing renewable sources of energy, they continue to be significant burners of fossil fuels. It is difficult to see countries giving up on fossil fuels when these are such a significant factor in their growing economic success.

In the short term, the lack of requirement for nations to sign up to the Paris Agreement and the lack of enforceability of targets will leave countries free to engage in practices that contribute to global warming. However, a failure to address climate change now voluntarily could, ultimately, lead to changes above and beyond those set out in the Agreement having to be imposed on countries. Long-term global warming is anticipated to cause significant climate changes in all countries – including those reluctant to sign up to the Agreement. These changes will impact on a range of major industries,

Comments

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

for example, causing flooding in centres of tourism and droughts in key agricultural lands. The extreme economic consequences of such changes across a global market indicate a need for all nations to sign the Agreement at the earliest possible opportunity. If voluntary action does not work, nations should be forced to sign.

Therefore, although there are economic consequences of taking action to reduce emissions and temperature increases, they are ultimately outweighed by the consequences of unwelcome climate change and long-term economic disaster if we fail to implement global action. Not all countries have played an equal role in the causation of global warming and it is fair that those that have contributed most towards global warming should also contribute most towards finding its solutions. However, given the actual and potential consequences of global warming, it does require a fully global solution and there is a role and rationale for all countries in reducing greenhouse gas emissions.

References

Elliot, L. (2015) 'Can the World Economy Survive without Fossil Fuels?' *The Guardian* 8/4/15, http://www.theguardian.com/news/2015/apr/08/can-world-economy-survive-without-fossil-fuels, downloaded 14/4/16.

Grassroots Global Justice Alliance (2015) *Call to Action: The COP21 Failed Humanity*, http://ggjalliance.org/ParisFailure2015, downloaded 16/4/16.

Jamet, M., *What do Green NGOs make of the COP21 Climate Deal?*, http://www.euronews.com/2015/12/14/what-do-green-ngos-make-of-cop21-climate-deal, downloaded 16/4/16. Jamet quotes both Raman and Naidoo.

Le Page, M. (2016) *Developing Nations Urged to Boycott Paris Agreement Signing*, Climate Home, http://www.climatechangenews.com/2016/03/29/developing-nations-urged-to-boycott-paris-agreement-signing, downloaded 15/4/16.

Raman, M. (2016) *The Signing Ceremony of the Paris Agreement in New York on 22nd April – Why There is no Need to 'Rush' into Signing*, Third World Network, https://www.scribd.com/doc/306273316/Note-on-the-Signing-Ceremony-in-New-York, downloaded 17/4/16.

United Nations Treaty Collection (2015) *The Paris Agreement*, https://treaties.un.org/pages/ViewDetails.aspx?src=TREATY&mtdsg_no=XXVII-7-d&chapter=27&lang=en, downloaded 11/4/16.

Vidal, J. and Vaughan, A. (2015) 'Paris Climate Agreement "May Signal End of Fossil Fuel Era"'. *The Guardian* 13/12/15, http://www.theguardian.com/environment/2015/dec/13/paris-climate-agreement-signal-end-of-fossil-fuel-era, downloaded 15/4/16. Quotes Al Gore.

Comments

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answer to Practice 1: Features of an argument

Global Warming Requires a Global Solution (Text 1)

The increase in greenhouse gas emissions over the past 50 years is viewed as a major factor in global warming. Research by the leading world authorities on global warming, the Intergovernmental Panel on Climate Change (IPCC)[13] suggests that even if all carbon dioxide emissions ceased today there could be climate change for a number of years to come, leading to water shortages for five billion people and increased flooding across Northern Europe by 2025. However, scientists have proposed a range of solutions from increasing efficient use of fossil fuels to incentives for using cleaner forms of energy, which they believe, if applied globally, would be sufficient to make a real impact on climate change.[8]

[8]The COP21 Paris Agreement[12] was agreed in draft form in 2015 as a means of working towards a reduction in greenhouse gas emissions and limiting the increase in global average temperature to below two degrees Celsius. The Agreement opened for countries to sign up to in 2016 but the actions within the Agreement will not start until 2020. It is expected that 120 of the 195 countries within the United Nations will sign up to the Agreement but it is not legally enforceable and each country will decide its own national targets. Whilst several of the world's greatest polluters have indicated that they will sign the Agreement, others are more resistant. Some governments of developing nations argue that it is not in developing countries' interests to form part of a global agreement. However, I shall argue that, although it may need to be applied differentially, a global and enforced solution is ultimately essential and that developed countries need to take the lead.[2]

Politicians, scientists and businesses in developed countries have given a number of reasons for not signing up to the Agreement. These include doubt about the real link between carbon dioxide emissions and global warming, concerns about the effect on their own economies and the rejection of a voluntary solution, which could lead to some countries not 'pulling their weight'.[10] Meena Raman (2016)[13] has argued that developing nations should not rush to sign the Agreement. She argues that they should wait to see how well developed nations fulfil their pre-2020 obligations under the 1997 Kyoto Protocol, and that waiting would give developing nations more political leverage in future negotiations. Whilst this argument has some validity – developed nations are the main world polluters – global warming is a problem for everyone, not just those in developed nations, and requires everyone to participate. Le Page (2015), argues that the Paris Agreement is already 'too little, too late' in terms of finding a global solution to global warming and notes the loss to flooding of some small pacific island nations.[6,5,7]

Comments

Numbers in the text and below refer to the grid on page 236.

13 The research by the IPCC provides a secondary source for this piece.

8 This descriptive opening paragraph gives essential background information on global climate change.

8 This paragraph gives essential background information about the Paris Agreement.

12 The United Nations Treaty Paris Agreement is a primary source for this piece.

2 This sentence summarises the author's position.

10 The author sets out several possible counter arguments against the main argument here.

13 Secondary source.

6 Intermediate conclusion: global warming is a concern for all and requires developing nations to play a role in reducing emissions.

5 The reason given to support this: we may already have left it too late to halt climate change and cannot risk any further delays or non-participation.

7 Evidence given for the consequences of leaving emissions unchecked.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answer to Practice 1: Features of an argument

Emissions from developing nations are clearly an important issue. However,[9] for developing nations, the argument that they should sign up to climate change restrictions at the same time as developed nations does not carry weight.[11] After watching developed nations build their wealth and power on extensive use of fossil fuels this appears to be a case of 'do what I say, not what I do'. Whilst the Agreement states that developed nations will financially support developed nations in making emission-reducing changes, many have argued that this is not enough. Raman (2015) notes that financial support has already been delayed from 2020 to 2025. Naidoo (2015),[12] executive director of Greenpeace International, states:

> *'There's not enough in this deal for the nations and people on the frontlines of climate change. It contains an inherent, ingrained injustice. The nations which caused this problem have promised too little help to the people who are already losing their lives and livelihoods.'*

The Grassroots Global Justice Alliance (2015) accuses developed nations of 'hypocrisy', arguing that they are pushing other countries to take actions that they have not been willing to take themselves. They call the Paris Agreement 'a failure, condemning humanity to a slow and painful death'.[10]

Former US Vice President, Al Gore (2015) suggests the Paris Agreement 'may herald the end of the fossil fuel era'. Many expect the Agreement to lead to heavy taxation of fossil fuel companies and a gradual disinvestment in their stocks and a move towards renewable energy technologies. This is likely to be easier for developed nations than developing countries. Elliot (2015)[13] notes that developed nations have tended to outsource manufacturing and some services to developing nations, where costs are cheaper. This increases the demand for energy in developing nations – most of which still comes from fossil fuels. For the Paris Agreement to be successful in limiting global temperature increases to less than two degrees, it is estimated that between two-thirds and four-fifths of remaining fossil fuels would have to be left in the ground and unburned. Whilst several major polluters, including China and India, have committed to developing renewable sources of energy, they continue to be significant burners of fossil fuels. It is difficult to see countries giving up on fossil fuels when these are such a significant factor in their growing economic success.[3,5,6]

In the short term, the lack of requirement for nations to sign up to the Paris Agreement and the lack of enforceability of targets will leave countries free to engage in practices that contribute to global warming.

Comments

9 'However' is used to signal development of the argument as the author moves to address the counter argument.

11 The author addresses counter argument here.

12 Naidoo is a primary source quoted in a secondary source.

10 The author hints at a counter argument here. It sounds as if it is being suggested that developed countries should take a much stronger role in funding climate change initiatives than developing countries, who have been the 'victims' of their behaviours.

13 Secondary source.

3 Summative conclusion of the argument so far: developed countries should make a greater contribution towards reducing emissions.

6 Intermediate conclusion: there are moral reasons for developed nations to support developing countries' involvement in a global solution.

5 The reasons given to support this intermediate conclusion are:

- developed countries have more resources to invest in new structures and technologies;
- developed countries behaviours have contributed to developing countries having a higher demand for energy.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answer to Practice 1: Features of an argument

However,[9] a failure to address climate change now voluntarily could, ultimately, lead to changes above and beyond those set out in the Agreement having to be imposed on countries. Long-term global warming is anticipated to cause significant climate changes in all countries – including those reluctant to sign up to the Agreement. These changes will impact on a range of major industries, for example, causing flooding in centres of tourism and droughts in key agricultural lands.[7] The extreme economic consequences of such changes across a global market indicate a need for all nations to sign the Agreement at the earliest possible opportunity.[6,5] If voluntary action does not work, nations should be forced to sign.

Therefore, although there are economic consequences of taking action to reduce emissions and temperature increases, they are ultimately outweighed by the consequences of unwelcome climate change and long-term economic disaster if we fail to implement global action.[1] Not all countries have played an equal role in the causation of global warming and it is fair that those that have contributed most towards global warming should also contribute most towards finding its solutions. However, given the actual and potential consequences of global warming, it does require a fully global solution and there is a role and rationale for all countries in reducing greenhouse gas emissions.[4]

References

Elliot, L. (2015) 'Can the World Economy Survive Without Fossil Fuels?' *The Guardian* 8/4/15, http://www.theguardian.com/news/2015/apr/08/can-world-economy-survive-without-fossil-fuels, downloaded 14/4/16.

Grassroots Global Justice Alliance (2015) *Call to Action: The COP21 failed humanity*, http://ggjalliance.org/ParisFailure2015, downloaded 16/4/16.

Jamet, M. *What do Green NGOs make of the COP21 Climate Deal?*, http://www.euronews.com/2015/12/14/what-do-green-ngos-make-of-cop21-climate-deal, downloaded 16/4/16. Jamet quotes both Raman and Naidoo.

Le Page, M. (2016) *Developing Nations Urged to Boycott Paris Agreement Signing*, Climate Home, http://www.climatechangenews.com/2016/03/29/developing-nations-urged-to-boycott-paris-agreement-signing, downloaded 15/4/16.

Raman, M. (2016) *The Signing Ceremony of the Paris Agreement in New York on 22nd April – Why There is no Need to 'Rush' into Signing*, Third World Network, https://www.scribd.com/doc/306273316/

Comments

9 'However' is used to signal a development of the main argument – that the consequences of global warming make it essential for us all to act.

7 Evidence of the consequences of global warming used to support the author's argument.

6 Intermediate conclusion: the consequences of not signing up to the Paris Agreement quickly invalidate calls to developing nations to delay signing.

5 The reasons given to support the intermediate conclusion are:
- climate change will cause flooding and drought in countries not willing to sign the Agreement;
- industries will be affected if no action is taken, undermining any potential economic advantages.

1 These sentences summarise the author's main argument that we all need to address greenhouse gas emissions but that some countries should play a greater role than others.

4 Overall logical conclusion: the consequences of not taking action outweigh the economic consequences of reducing emissions and therefore a global solution is required. This links the conclusion back to the essay title, which strengthens the presentation of the argument.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answer to Practice 1: Features of an argument

Note-on-the-Signing-Ceremony-in-New-York, downloaded 17/4/16.

United Nations Treaty Collection (2015) *The Paris Agreement*, https://treaties.un.org/pages/ViewDetails.aspx?src=TREATY&mtdsg_no=XXVII-7-d&chapter=27&lang=en, downloaded 11/4/16.

Vidal, J. and Vaughan, A. (2015) 'Paris Climate Agreement "May Signal End of Fossil Fuel Era"'. *The Guardian* 13/12/15, http://www.theguardian.com/environment/2015/dec/13/paris-climate-agreement-signal-end-of-fossil-fuel-era, downloaded 15/4/16. Quotes Al Gore.

Comments

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Practice 2: Finding flaws in the argument

- Read the second passage on global warming (Text 2), and identify flaws in the argument, using the numbered prompts below as a checklist to assist you.
- *Note*: the practice passage does not contain all the flaws on the list and some flaws occur more than once. You can use the checklist to note whether you believe the passage does or does not contain an example, to make it easier to check your answers.
- Label and number each of your answers in the *Comments* margin provided alongside the text. If you use the same numbers as those provided in the prompts table below, this will help you to check your answers.

Prompts	Example found	There is no example	See page
1 False premises			79
2 Two wrongs don't make a right			104
3 Stereotyping			84
4 Lack of consistency in the argument			59, 61
5 Unnecessary background information			52, 150
6 Lack of precision			59
7 Assumption that is not supported by the evidence			74–8
8 Incorrectly assuming a causal link			90
9 False correlation			91
10 Meeting necessary conditions			93
11 Meeting sufficient conditions			94
12 False analogy			96
13 Deflection			98
14 Complicity			98
15 Exclusion			98
16 Unwarranted leaps (e.g. castle of cards; sleight of hand)			100
17 Emotive language			101
18 Attacking the person			101
19 Misrepresentation			103
20 Trivialisation			103
21 Tautology			104
22 Poor referencing			144–5 167–8

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Practice 2: Finding flaws in the argument

Global Warming Requires a Global Solution (Text 2)



The Paris Agreement was agreed in 2015 as a means of limiting long-term climate change or 'global warming' by asking countries to sign up to reductions in greenhouse gas emissions. It seeks to limit overall increase in global temperature to below two degrees Celsius over the next few years. Although many countries, and most developed countries, have signed up to the agreement, some developing countries have not yet signed.

The Intergovernmental Panel on Climate Change (IPCC) suggests that we have probably left it too late to make the changes suggested by the Paris Agreement. Even if all carbon dioxide emissions ceased today, there would be ongoing climatic changes and global warming leading to effects such as rising sea levels and subsequent contamination of drinking water. At best, the effects will be disruptive and at worst catastrophic. We must act now and make it a crime to deny climate change.

Given the consequences of climate change, it is madness for any nation not to sign up. Those countries not signing up have given a number of reasons for this, from calling research on climate change into question and even going so far as to suggest that carbon dioxide is not a pollutant! This suggests that some countries are in denial about the cause and impact of greenhouse gas emissions. After years of struggle to get developed countries to reduce emissions we now face a situation where many developed countries are refusing to sign. Even though the Agreement builds in financial support from developed countries, some argue that developing countries need more than this and should fail to sign in order to increase their political leverage on the global stage.

This argument must really stick in the throats of developed countries. Having worked hard to reduce emissions and ratify the Paris Agreement, the very countries they have been seeking to protect and support suddenly decide that they could and should have got a better deal. Having complained about the impact of developed countries' burning fossil fuel through industry, developing countries seem to think it is reasonable for them to now increase their own emissions through industrialisation. Raman argues that developing nations should not bother to sign the Agreement, as most developed nations do not really care about climate change and will not bother to fulfil their pre-2020 obligations, made under the 1997 Kyoto Protocol. This suggests a lack of trust in developed countries and their commitment to limiting climate change. It is a bit like children who grow up complaining about their parents for smoking, but then take up a 60-cigarettes-a-day habit at the first opportunity.

Comments

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

This sense of a hollow argument is increased when one looks at the real reasons developing countries are jumpy about the Agreement – using cheap fossil fuels gives them cheap industry costs and allows them to take business away from developed countries. Even though there is clear evidence of the damage this is causing to the planet, some ignorant developing countries still believe that the most important consideration is that they are allowed to pollute their way towards full industrialisation.

Developing nations have suggested that it is not fair of developed nations to expect them to sign up to the Agreement as it would mean taking actions and introducing restrictions that developed countries have been reluctant to introduce themselves. In many cases, developing countries are only burning more fossil fuels now because developed countries have ceased many of their polluting activities but still want to buy products or services cheaply from developing nations that have a heavy use of fossil fuels. Whilst complaining about developed countries not doing enough to support renewable forms of energy, some developing countries are very keen to preserve the economic advantage that burning fossil fuels is now giving them. They have a vested interest in seeing the Agreement fail.

Ultimately, countries' failures to address greenhouse gas emissions could mean that they shoot themselves in the foot. Long-term global warming is anticipated to cause significant climate changes and these countries will have to deal with flooding in cities and tourist areas and drought in their agricultural areas. However, as neither of these may have an impact on companies or governments within these countries they can feel justified in this short-term strategy of making money and protecting their interests. This leads to them claiming that it is only fair that developing countries get their chance to do what developed countries got rich on doing for years. They are making an economic argument rather than a humanitarian one. They think that economic advantage is more important than people's lives. Given the impact their behaviour has on all human beings, this is obviously unacceptable. Everybody knows we are facing climatic meltdown. Global warming is a problem for all of us and people can't opt out just because it doesn't suit them.

Comments

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

References

United Nations Treaty Collection *The Paris Agreement* https://treaties.un.org/pages/ViewDetails.aspx?src=TREATY&mtdsg_no=XXVII-7-d&chapter=27&lang=en.

M. Le Page, (2016) - *Developing Nations Urged to Boycott Paris Agreement Signing,* Climate Home http://www.climatechangenews.com/2016/03/29/developing-nations-urged-to-boycott-paris-agreement-signing downloaded 15/4/16

Grassroots Global Justice Alliance - *Call to Action: The COP21 Failed Humanity*, http://ggjalliance.org/ParisFailure2015.

Feeling the Heat: The Politics of Climate Change in Rapidly Industrializing Countries (2012) (Basingstoke: Palgrave Macmillan). Bailey, I. and Compston, H.

Comments

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to Practice 2: Finding flaws in the argument

Prompts	Example found	There is no example	See page
1 False premises	✓		79
2 Two wrongs don't make a right		✓	104
3 Stereotyping	✓		84
4 Lack of consistency in the argument		✓	59, 61
5 Unnecessary background information		✓	52, 150
6 Lack of precision	✓		59
7 Assumption that is not supported by the evidence	✓ (3)		74–8
8 Incorrectly assuming a causal link		✓	90
9 False correlation		✓	91
10 Meeting necessary conditions	✓		93
11 Meeting sufficient conditions		✓	94
12 False analogy	✓		96
13 Deflection	✓		98
14 Complicity	✓ (2)		98
15 Exclusion		✓	98
16 Unwarranted leaps (e.g. castle of cards; sleight of hand)	✓ (2)		100
17 Emotive language	✓ (3)		101
18 Attacking the person	✓		101
19 Misrepresentation	✓		103
20 Trivialisation		✓	103
21 Tautology	✓		104
22 Poor referencing	✓ (2)		144–5 167–8

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to Practice 2: Finding flaws in the argument

Global Warming Requires a Global Solution (Text 2)

The Paris Agreement was agreed in 2015 as a means of limiting long-term climate change or 'global warming' by asking countries to sign up to reductions in greenhouse gas emissions. It seeks to limit overall increase in global temperature to below two degrees Celsius over the next few years.[6] Although many countries, and most developed countries, have signed up to the agreement, some developing countries have not yet signed.

The Intergovernmental Panel on Climate Change (IPCC) suggests that we have probably left it too late to make the changes suggested by the Paris Agreement. Even if all carbon dioxide emissions ceased today, there would be ongoing climatic changes and global warming leading to effects such as rising sea levels and subsequent contamination of drinking water. At best, the effects will be disruptive and at worst catastrophic. We must act now and make it a crime to deny climate change.[10]

Given the consequences of climate change, it is madness for any nation not to sign up.[17] Those countries not signing up have given a number of reasons for this, from calling research on climate change into question and even going so far as to suggest that carbon dioxide is not a pollutant![7] This suggests that some countries are in denial about the cause and impact of greenhouse gas emissions.[18,16] After years of struggle to get developed countries to reduce emissions we now face a situation where many developed countries are refusing to sign. Even though the Agreement builds in financial support from developed countries, some argue that developing countries need more than this and should fail to sign in order to increase their political leverage on the global stage.

This argument must really stick in the throats of developed countries.[17] Having worked hard to reduce emissions and ratify the Paris Agreement, the very countries they have been seeking to protect and support suddenly decide that they could and should have got a better deal. Having complained about the impact of developed countries' burning fossil fuel through industry, developing countries seem to think it is reasonable for them to now increase their own emissions through

Comments

Numbers in the text and below refer to the grid on page 248.

6 **Lack of precision**. The phrase 'next few years' is vague. The Paris Agreement came into effect in April 2016 and will run until 2020.

10 The necessary conditions for the argument that we must make it a crime to deny climate change have not been met. The author has not shown why, in order to reduce global warming, it is necessary to criminalise a particular point of view. The desired outcome of reduced global warming could potentially be achieved without this course of action.

17 Use of **emotive language** with the phrase 'it is madness'.

7 **Assumption**. The author assumes that there is not a valid argument for asserting that carbon dioxide is not a pollutant but gives no evidence that it is a pollutant.

18 **Attacking the person**. Referring to those who disagree as being 'in denial' undermines their argument without properly analysing their reasons.

16 **Unwarranted leap**. This argument makes an unwarranted leap in assuming that because they do not accept some research, opponents must be wrong about global warming.

17 **Emotive language** with the phrase 'stick in the throats'. (The style is also rather colloquial.)

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to Practice 2: Finding flaws in the argument

industrialisation. Raman[22] argues that developing nations should not bother to sign the Agreement, as developed nations do not really care about climate change[3] will not bother fulfil their pre-2020 obligations, made under the 1997 Kyoto Protocol.[19] This suggests a lack of trust in developed countries and their commitment to limiting climate change. It is a bit like children who grow up complaining about their parents for smoking, but then take up a 60-cigarettes-a-day habit at the first opportunity.[12]

This sense of a hollow argument is increased when one looks at the real reasons developing countries are jumpy about the Agreement – using cheap fossil fuels gives them cheap industry costs and allows them to take business away from developed countries.[7] Even though there is clear evidence of the damage this is causing to the planet, some ignorant[17] developing countries still believe that the most important consideration is that they are allowed to pollute their way towards full industrialisation![7,14]

Comments

22 **Poor referencing** of Raman source. This does not appear in the references and no date is given (compare this with the text for Practice 1).

3 It is **stereotyping** to suggest that all developed countries are not concerned about global warming.

19 **Misrepresentation**. The author misrepresents Raman's views. Raman does not suggest that developing nations should not sign up to the Agreement but suggests that developed nations should fulfil their responsibilities under the Kyoto Agreement first (see Practice 1).

12 **False analogy**. On the surface, this looks like a reasonable analogy, suggestive of hypocritical behaviour. However, it is a poor analogy as parents have a different relationship to their children from that between developed and developing countries. The issue between developed and developing resources is one of competition for limited resources – in the forms of fossil fuel, labour or products. Smoking is an issue of health and life-style and decisions about whether or not to smoke are not generally centred around the availability of cigarettes.

7 **Assumption**. The author assumes that developing countries are taking business from developed countries but gives no evidence to support this.

17 **Emotive language** is used in the phrase 'some ignorant developing countries'.

7 **Assumption**. The author assumes that developing countries' prime consideration is industrialisation but gives no evidence to support this.

14 **Complicity**. The writing style here and the use of the exclamation mark suggests the author is making the audience feel they must agree or else they might be considered 'ignorant' too.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to Practice 2: Finding flaws in the argument

Developing nations[6] have suggested that it is not fair of developed nations to expect them to sign up to the Agreement as it would mean taking actions and introducing restrictions that developed countries have been reluctant to introduce themselves.[22] In many cases, developing countries are only burning more fossil fuels now because developed countries have ceased many of their polluting activities but still want to buy products or services cheaply from developing nations that have a heavy use of fossil fuels. Whilst complaining about developed countries not doing enough to support renewable forms of energy, some developing countries are very keen to preserve the economic advantage that burning fossil fuels is now giving them. They have a vested interest in seeing the Agreement fail.[16]

Ultimately, countries' failures to address greenhouse gas emissions could mean that they shoot themselves in the foot. Long-term global warming is anticipated to cause significant climate changes and these countries will have to deal with flooding in cities and tourist areas and drought in their agricultural areas. However, as neither of these may have an impact on companies or governments within these countries they can feel justified in this short-term strategy of making money and protecting their interests.[1] This leads to them claiming that it is only fair that developing countries get their chance to do what developed countries got rich on doing for years. They are making an economic argument rather than a humanitarian one. They think that economic advantage is more important than people's lives.[22] Given the impact their behaviour has on all human beings, this is obviously unacceptable.[13] Everybody knows we are facing climatic meltdown.[14] Global warming is a problem for all of us and people can't opt out just because it doesn't suit them.

Comments

6 **Lack of Precision**. The author does not state which developing nations are referred to here.

22 **Poor referencing**. There is a reference for a relevant article in the references section but this is not clearly linked to this statement.

16 **Unwarranted leap**. The author uses sleight of hand here. There is no evidence to suggest that developing nations have a vested interest in seeing the Agreement fail.

1 **False premise**. The argument that governments and companies in developing countries would not be affected by floods or drought is based on a false premise. Both are likely to face considerable impact and disruption through climate change.

22 **Tautology**. The two sentences here rephrase the same idea in different words. This produces unnecessary repetition without taking the argument forward.

13 **Deflection**. The author uses the word 'obviously' to imply that the argument has been proved. As we have seen, this is not the case.

14 **Complicity**. The statement 'everybody knows' puts the reader in a position that makes it more difficult to disagree with the argument. The author does this through the use of language rather than through reasoning.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to Practice 2: Finding flaws in the argument

References

United Nations Treaty Collection *The Paris Agreement* https://treaties.un.org/pages/ViewDetails.aspx?src=TREATY&mtdsg_no=XXVII-7-d&chapter=27&lang=en.

M. Le Page, (2016) - *Developing Nations Urged to Boycott Paris Agreement Signing,* Climate Home http://www.climatechangenews.com/2016/03/29/developing-nations-urged-to-boycott-paris-agreement-signing downloaded 15/4/16

Grassroots Global Justice Alliance - *Call to Action: The COP21 Failed Humanity*, http://ggjalliance.org/ParisFailure2015.

Feeling the Heat: The Politics of Climate Change in Rapidly Industrializing Countries (2012) (Basingstoke: Palgrave Macmillan). Bailey, I. and Compston, H.

Comments

22 **The list of references is not complete.** References are not laid out in a correct and consistent format. No date is given for some.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Practice 3: Features of an argument

- Read the passage 'The Great Chain of Being' and identify the features of the argument, using the numbered prompts below to assist you.
- Label and number each of your answers in the *Comments* margin provided alongside the text. If you use the same numbers as those provided in the prompts table below, this will help you to check your answers.

Prompts	Done ✓ (tick when completed)
1 Identify the sentence or sentences that sum up the main argument.	
2 Identify the author's introduction to the argument.	
3 Identify the summative conclusion.	
4 Identify the overall logical conclusion.	
5 Identify the main reasons given to support the logical conclusion.	
6 Identify any intermediate conclusions used as reasons. In the margin, explain the purpose of the interim conclusion (i.e. why the author needed to make an interim conclusion in order to develop the argument).	
7 Identify evidence to support the conclusion.	
8 Identify descriptive text that provides background information for the reader.	
9 Identify words used to signal the development of either the main argument or arguments leading to intermediate conclusions.	
10 Identify any counter arguments put forward by the author.	
11 Identify arguments by the author to address counter arguments.	
12 Identify any use of primary sources.	
13 Identify any use of secondary sources.	

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

The Great Chain of Being (Text 3)



'Notions of a "Great Chain of Being" and of a natural order to the universe continued to exert ideological significance during the eighteenth century and beyond.' Discuss.

The idea of the 'Great Chain of Being' was common in medieval Europe. For those who believed in this chain, everything that existed belonged to a pre-assigned place, as if on higher or lower rungs of a ladder. Those lowest items in the universe were at the bottom of the chain whilst humans were nearer the top, below angels but above animals. In considering the influence of the chain of being in the eighteenth century, there are two aspects to consider. Firstly, whether the idea was still familiar in the eighteenth century, and secondly, whether it was used to support political or ideological positions in the important debates of the period.

First of all, it is proposed that the concept of the chain of being was very much alive in the eighteenth century. Such a view is not universally accepted. It has been argued that references to the chain of being had died out by the mid-seventeenth century (Barking, 1957; Madison, 1967). Madison claimed that such notions of the universe were replaced by more enlightened ideas based on scientific observation. It could also be argued that war and trade provided ever increasing opportunities for people to meet with new ideas and ideologies. Colley (2003), for example, refers to how impressed travellers to North African Muslim countries were by its culture and by the tolerance shown by Islam to other faiths. In earlier centuries, Christian rulers burnt people at the stake for expressing such sentiments but this was no longer the case in the eighteenth century.

However, despite such changes in people's outlook, old ideas still continued to hold sway. In the literature of the late eighteenth century, and even the early nineteenth century, it is not unusual to find references to the chain of being. Indeed, in an analysis of pamphlets produced in 1802–3, for example, Pendleton (1976) found that over one in 10 pamphlets, a significant proportion, included a reference to the 'Chain of Being'. A much higher proportion of the pamphlets alluded to related concepts, such as the 'natural order' of society. The concept of the 'chain of being', and the idea that there was an inherent order to the universe, was still prevalent in England even in the early nineteenth century.

As Pendleton's research indicates, there were still many publicists at the end of the eighteenth century who argued that the governing classes in England were a superior type of human being, higher on the great chain of being and closer to God. Many of those in power believed that their own social class was more intelligent, more beautiful, with better morals. They regarded the majority of the

Comments

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

population as less intelligent and virtuous, as uglier beings, closer to an animal state, and therefore less deserving of consideration in all respects (Lavater, 1797). People were expected to 'know their place' and to act accordingly at all times. This concept of beings on a higher level of the chain was useful in providing a justification for social superiority.

The hierarchical ordering of nature was also used to provide a powerful defence of political and economic inequalities. In the eighteenth and nineteenth centuries, very few people were allowed to vote in elections, to organise politically or even to speak out against those believed to be their 'betters'. The overwhelming majority of the population still did not have the vote and there were vast differences in wealth, health, and well-being (Thompson, 1963). The notion of a natural chain of being was used to argue that this was as nature or 'Providence' intended.

Furthermore, even after the eighteenth century, the hierarchical ordering of society was still presented as a divine plan, and all people were expected to follow the same religion. This idea of a divinely based order was used to frighten people into submission. A broadside, or poster, pasted around London in 1802 declared that 'It is the ordinance of God, that there should be infinite gradations' and that 'as one star differeth from another star in glory, so shall the plan of subordination be through the whole earthly system.' It was argued that it was natural for some people to have power and riches, and for other people to have none. For example, one pamphleteer (Pratt, 1803) claimed that if the natural order was changed, this would 'unsettle the whole system of the spheres; the planets would rush on each other ... and the earth be shrivelled, like a scroll, by a spark from the sun'. If everything in the universe formed part of a single continuous chain, then to alter a single part could disrupt the whole chain, causing all society to come to an end and even the universe to collapse. The combining of an ideology of social order with a religious philosophy made the idea of a natural order particularly significant.

Moreover, in 1802–3, such arguments were of particular importance in fostering patriotic sentiment to support war against France. The political elite encouraged each other to be active in persuading the poor where their interest lay (Ashcroft, 1977). They were worried that large sections of the population would welcome a French invasion that promised social, economic, political and religious freedom, as had been proclaimed in France after the Revolution of 1789. Some expressed fear that if they armed the English people to defend the country, they would turn their weapons against their masters (Cholmeley, 1803). Instead of taking the dangerous risk of arming the country, a propaganda campaign was launched, arguing that the 'natural order' was best and that

Comments

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

the English people should accept it rather than join the French if the country were invaded. If the natural order changed, the propaganda argued, the consequences would be famine, disease and death.

Rather than, as Madison suggests, the chain of being becoming an outmoded concept in the eighteenth century, scientists were still active in researching new possible gradations in the hierarchy. They set about measuring the bones of people of different skin colours, social classes and geographical origins, in an attempt to set down a hierarchy from best to worst, using their own skin colour as the benchmark for perfection (White, 1779). Lavater, whose writings were translated into English in 1797, referred to this as 'the transition from brutal deformity to ideal beauty' and argued that beauty was a sign of moral superiority. Lavater devised a system for measuring hierarchy, based on bone structure and appearance. His writings were widely published and highly influential in England. Over time, the use of the term 'chain of being' died out, but the belief in the natural or divine hierarchy and its use as a rationale for political and social inequalities continued to hold force.

It is important to note that the concept of a hierarchically ordered universe could be used to justify almost any kind of inequality or oppression. Indeed, Mary Wollstonecraft (1792) argued that the concept of a natural order was being used to justify all kinds of injustice such as cruelty to animals and children, the slave trade, and depriving women of political and economic rights. Literature that used the idea of natural hierarchies also made use of comparisons between all types of people who did not form part of the English ruling oligarchy and growing middle classes. The *Encyclopaedia Britannica*, for example, in its 1797 edition, compared the behaviour of Africans, the English working class, and French revolutionaries, arguing that they shared common characteristics such as 'a lack of moral principles' and an absence of 'natural affections'. Over the next century and a half, these ideas were drawn upon and extended by others to justify policies based on racial and social injustice in many parts of the world.

Ideas based on the chain of being, therefore, far from waning in the eighteenth century, were further developed and extended. The dangers posed by the French Revolution and the proposed invasion of England, added force to the long-established concept of a natural order, especially after these were widely publicised in the anti-invasion and anti-revolutionary propaganda of 1802–4. The scientific methodologies of the eighteenth century resulted in findings that appeared to justify the concept of a natural hierarchy. Although the vocabulary of the chain of being began to die out, the underlying concept was reinforced and used to reinforce negative social, gender and racial stereotypes in the next two centuries. Hence, the concept of the 'great chain of being' did continue to exert ideological significance during the eighteenth century and beyond.

Comments

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Practice 3: Features of an argument

References

Primary sources

Anon. (1803) *Such is Buonaparte* (London: J. Ginger).

Ashcroft, M.Y. (1977) *To Escape a Monster's Clutches: Notes and Documents Illustrating Preparations in North Yorkshire to Repel the Invasion.* North Yorkshire, CRO Public No. 15.

Cholmeley, C. (1803) Letter of Catherine Cholmeley to Francis Cholmeley, 16 August 1803. In M.Y. Ashcroft (1977) *To Escape a Monster's Clutches: Notes and Documents Illustrating Preparations in North Yorkshire to Repel the Invasion.* North Yorkshire, CRO Public No. 15.

Encyclopaedia Britannica (1797) 3rd edition, Edinburgh.

Lavater, J.K. (1797) *Essays on Physiognomy*, translated by Rev. C. Moore and illustrated after Lavater by Barlow (London: London Publishers).

'Pratt' (1803) *Pratt's Address to His Countrymen or the True Born Englishman's Castle* (London: J. Asperne).

White, C. (1779) *An Account of the Infinite Gradations in Man, and in Different Animals and Vegetables; and from the Former to the Latter.* Read to the Literary and Philosophical Society of Manchester at Different Meetings (Manchester: Literary and Philosophical Society).

Wollstonecraft, M. (1792) *Vindication of the Rights of Women.* (Republished in 1975 by Penguin, Harmondsworth, Middlesex.)

Secondary sources

Barking, J.K. (1957) *Changes in Conceptions of the Universe* (Cotteridge: Poltergeist Press).*

Colley, L. (2003) *Captives: Britain, Empire and the World 1600–1850* (London: Pimlico).

Madison, S. (1967) 'The End of the Chain of Being: the Impact of Descartian Philosophy on Medieval Conceptions of Being'. *Journal of Medieval and Enlightenment Studies*, 66, 7.*

Pendleton, G. (1976) 'English Conservative Propaganda During the French Revolution, 1780–1802', Ph.D. (unpub.), Emory University.

Thompson, E.P. (1963) *The Making of the English Working Class* (Harmondsworth, Middlesex: Penguin).

*These two sources are hypothetical and provided for the purpose of the practice activity; the other sources are genuine.

Comments

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to Practice 3: Features of an argument

The Great Chain of Being (Text 3)

'Notions of a "Great Chain of Being" and of a natural order to the universe continued to exert ideological significance during the eighteenth century and beyond.' Discuss.

The idea of the 'Great Chain of Being' was common in medieval Europe. For those who believed in this chain, everything that existed belonged to a pre-assigned place, as if on higher or lower rungs of a ladder. Those lowest items in the universe were at the bottom of the chain whilst humans were nearer the top, below angels but above animals.[8]

In considering the influence of the chain of being in the eighteenth century, there are two aspects to consider. Firstly, whether the idea was still familiar in the eighteenth century, and secondly, whether it was used to support political or ideological positions in the important debates of the period.[2]

First of all,[9] it is proposed that the concept of the chain of being was very much alive in the eighteenth century.[1] Such a view is not universally accepted. It has been argued that references to the chain of being had died out by the mid-seventeenth century (Barking, 1957; Madison, 1967). Madison claimed that such notions of the universe were replaced by more enlightened ideas based on scientific observation.[10, 13] It could also be argued that war and trade provided ever increasing opportunities for people to meet with new ideas and ideologies. Colley (2003), for example, refers to how impressed travellers to North African Muslim countries were by its culture and by the tolerance shown by Islam to other faiths. In earlier centuries, Christian rulers burnt people at the stake for expressing such sentiments but this was no longer the case in the eighteenth century.[10, 13]

However, despite such changes in people's outlook, old ideas still continued to hold sway.[11] In the literature of the late eighteenth century, and even the early nineteenth century, it is not unusual to find references to the chain of being. Indeed, in an analysis of pamphlets produced in 1802–3, for example, Pendleton (1976) found that over one in 10 pamphlets, a significant proportion, included a reference to the 'Chain of Being'.[7, 13] A much higher proportion of the pamphlets alluded to related concepts, such as the 'natural order' of society. The concept of the 'chain of being' and the idea that there was an inherent order to the universe, was still prevalent in England even in the early nineteenth century.[6]

Comments

Numbers in the text and below refer to the grid on p. 253.

8 This is descriptive text that tells the reader, briefly, what the 'chain of being' was. This is necessary background information.

2 These two sentences set out how the author intends to approach the argument, breaking it into two sections to help the reader recognise these stages in the argument when they are introduced later.

9 Signal word to introduce the first of the author's reasons.

1 This sentence sums up the main argument.

10 The author considers here counter arguments to the main line of reasoning. In this case, the counter arguments are raised and dealt with early in the line of reasoning as, if it were true that the idea had already died out, there wouldn't be much point continuing with the rest of the argument.

13 Examples of secondary sources (see p. 108).

11 This paragraph addresses the counter argument raised by Barking that references to the chain of being had died out. The counter argument raised by Madison is addressed throughout the essay, and in a separate paragraph below.

7 This evidence helps to support the overall conclusion that the concept did continue to have ideological significance.

13 Examples of secondary sources. (The pamphlets themselves are primary sources. See p. 108.)

6 Intermediate conclusion used as a reason: the author establishes first that the notion of a 'great chain of being' was still current. The reasons to support this are:

- it is not uncommon to find examples;
- Pendleton's research.

© Stellla Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to Practice 3: Features of an argument

As Pendleton's research indicates, there were still many publicists at the end of the eighteenth century who argued that the governing classes in England were a superior type of human being, higher on the great chain of being and closer to God. Many of those in power believed that their own social class was more intelligent, more beautiful, with better morals. They regarded the majority of the population as less intelligent and virtuous, as uglier beings, closer to an animal state, and therefore less deserving of consideration in all respects (Lavater, 1797).[12] People were expected to 'know their place' and to act accordingly at all times. This concept of beings on a higher level of the chain was useful in providing a justification for social superiority.[5,6]

The hierarchical ordering of nature was also[9] used to provide a powerful defence of political and economic inequalities.[5,6] In the eighteenth and nineteenth centuries, very few people were allowed to vote in elections, to organise politically or even to speak out against those believed to be their 'betters'. The overwhelming majority of the population still did not have the vote and there were vast differences in wealth, health, and well-being (Thompson, 1963).[7,8,13] The notion of a natural chain of being was used to argue that this was as nature or 'Providence' intended.

Furthermore,[9] even after the eighteenth century, the hierarchical ordering of society was still presented as a divine plan, and all people were expected to follow the same religion. This idea of a divinely based order was used to frighten people into submission.[6] A broadside, or poster, pasted around London in 1802 declared that 'It is the ordinance of God, that there should be infinite gradations' and that 'as one star differeth from another star in glory, so shall the plan of subordination be through the whole earthly system.'[12,7] It was argued that it was natural for some people to have power and riches, and for other people to have none. For example, one pamphleteer (Pratt, 1803) claimed that if the natural order was changed, this would 'unsettle the whole system of the spheres; the planets would rush on each other ... and the earth be shrivelled, like a scroll, by a spark from the sun'.[12] If everything in the universe formed part of a single continuous chain, then to alter a single part could disrupt the whole chain, causing all society to come to an end and even the universe to collapse. The combining of an ideology of social order with a religious philosophy made the idea of a natural order particularly significant.[7]

Comments

12 Primary source. See p. 108.

5 and 6 Intermediate conclusion used as a reason: the author has established the significance of the concept of the chain of being to maintaining the social structure. This then provides a reason that supports the overall conclusion that the concept is still of significance.

9 Signal word used to indicate to the reader that the author is adding more reasons to support the line of reasoning.

5 and 6 Intermediate conclusion used as a reason: the author has established the significance of the concept of the chain of being to defending the political and economic status quo. This then provides a reason that supports the conclusion.

8 Necessary but brief description of eighteenth-century society, to support the reasoning and to illustrate the significance of the political use of the concept of the chain of being.

7/13 Secondary source. This refers the reader to the source of evidence for the assertions made.

9 Signal word to indicate the argument is continuing in a similar direction.

6 Intermediate conclusion: the concept was used to rouse fear and submission.

12 These are examples of primary source materials, see p. 108.

7 Source materials used in this paragraph are used as evidence to support the conclusion that the concept had ideological significance.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to Practice 3: Features of an argument

Moreover,[9] in 1802–3, such arguments were of particular importance in fostering patriotic sentiment to support war against France.[6] The political elite encouraged each other to be active in persuading the poor where their interest lay (Ashcroft, 1977).[12/13] They were worried that large sections of the population would welcome a French invasion that promised social, economic, political and religious freedom, as had been proclaimed in France after the Revolution of 1789.[8] Some expressed fear that if they armed the English people to defend the country, they would turn their weapons against their masters (Cholmeley, 1803).[12] Instead of taking the dangerous risk of arming the country, a propaganda campaign was launched, arguing that the 'natural order' was best and that the English people should accept it rather than join the French if the country were invaded.[5] If the natural order changed, the propaganda argued, the consequences would be famine, disease and death.

Rather than, as Madison suggests, the chain of being becoming an outmoded concept in the eighteenth century, scientists were still active in researching new possible gradations in the hierarchy.[11] They set about measuring the bones of people of different skin colours, social classes and geographical origins, in an attempt to set down a hierarchy from best to worst, using their own skin colour as the benchmark for perfection (White, 1779).[12] Lavater, whose writings were translated into English in 1797, referred to this as 'the transition from brutal deformity to ideal beauty' and argued that beauty was a sign of moral superiority. Lavater devised a system for measuring hierarchy, based on bone structure and appearance. His writings were widely published and highly influential in England. Over time, the use of the term 'chain of being' died out, but the belief in the natural or divine hierarchy and its use as a rationale for political and social inequalities continued to hold force.[6, 5]

Comments

9 The word 'moreover' is used to signal that the same line of reasoning is being continued.

6 The intermediate conclusion is that this concept was significant at a particular historical moment.

12/13 This is a collection of primary sources published in 1977. (A modern date does not automatically indicate a secondary source.)

8 Necessary background to illustrate the significance of the concept at a particularly important political moment.

12 Primary source.

5 The reason given to support the intermediate conclusion for this paragraph is:

- propaganda making use of the chain of being was used as an anti-French device, rather than arming the country.

11 This addresses a counter argument raised in the second paragraph above.

12 Primary sources.

6 Intermediate conclusion used as a reason: the author establishes that the assumptions underlying the chain of being were further developed by scientists and given new life. The reasons to support this interim conclusion are provided by the examples of research and by the acknowledgement that although the term 'chain of being' died out, its assumptions remained in force.

5 The intermediate conclusion is also a main reason used to support the overall conclusion, that the concept retained significance.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to Practice 3: Features of an argument

It is important to note that[9] the concept of a hierarchically ordered universe could be used to justify almost any kind of inequality or oppression.[6, 5] Indeed, Mary Wollstonecraft (1792) argued that the concept of a natural order was being used to justify all kinds of injustice such as cruelty to animals and children, the slave trade, and depriving women of political and economic rights.[12] Literature that used the idea of natural hierarchies also made use of comparisons between all types of people who did not form part of the English ruling oligarchy and growing middle classes. The *Encyclopaedia Britannica*, for example, in its 1797 edition, compared the behaviour of Africans, the English working class, and French revolutionaries, arguing that they shared common characteristics such as 'a lack of moral principles' and an absence of 'natural affections'.[12] Over the next century and a half, these ideas were drawn upon and extended by others to justify policies based on racial and social injustice in many parts of the world.

Ideas based on the chain of being, therefore,[9] far from waning in the eighteenth century, were further developed and extended.[3] The dangers posed by the French Revolution and the proposed invasion of England added force to the long-established concept of a natural order, especially after these were widely publicised in the anti-invasion and anti-revolutionary propaganda of 1802–4. The scientific methodologies of the eighteenth century resulted in findings that appeared to justify the concept of a natural hierarchy. Although the vocabulary of the chain of being began to die out, the underlying concept was reinforced and used to reinforce negative social, gender and racial stereotypes in the next two centuries. Hence, the concept of the 'great chain of being' did continue to exert ideological significance during the eighteenth century and beyond.[4]

References

Primary sources

Anon. (1803) *Such is Buonaparte* (London: J. Ginger).

Ashcroft, M.Y. (1977) *To Escape a Monster's Clutches: Notes and Documents Illustrating Preparations in North Yorkshire to Repel the Invasion.* North Yorkshire, CRO Public No. 15.

Cholmeley, C. (1803) Letter of Catherine Cholmeley to Francis Cholmeley, 16 August 1803. In Ashcroft, M.Y. (1977) *To Escape a Monster's Clutches: Notes and Documents Illustrating Preparations in North Yorkshire to Repel the Invasion.* North Yorkshire, CRO Public No. 15.

Comments

9 This phrase is used to signal a further aspect of the argument, building on previous reasons.

6 The intermediate conclusion is that the concept was used to justify many kinds of oppression. The reasons to support this interim conclusion are evidence drawn from contemporaries such as Mary Wollstonecraft and the *Encyclopaedia Britannica* of 1797, and the uses to which the ideas were put.

5 The intermediate conclusion is another reason to support the overall conclusion that the concept remained significant in the eighteenth century and beyond.

12 These are examples of primary source materials, see p. 108.

9 Signal word used to indicate the conclusion.

3 This paragraph is mainly a summative conclusion – summarising key points from the preceding paragraphs.

4 The final sentence provides the logical conclusion here: it makes a deduction drawn from all the reasoning given above.

NB Linking the final sentence back to the title signals to the reader that the main question posed in the title has been addressed.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Encyclopaedia Britannica (1797) 3rd edition, Edinburgh.
Lavater, J.K. (1797) *Essays on Physiognomy*, translated by Rev. C. Moore and illustrated after Lavater by Barlow (London: London Publishers).
'Pratt' (1803) *Pratt's Address to His Countrymen or the True Born Englishman's Castle* (London: J. Asperne).
White, C. (1779) *An Account of the Infinite Gradations in Man, and in different Animals and Vegetables; and from the Former to the Latter.* Read to the Literary and Philosophical Society of Manchester at Different Meetings (Manchester: Literary and Philosophical Society).
Wollstonecraft, M. (1792) *Vindication of the Rights of Women.* (Republished in 1975 by Penguin, Harmondsworth, Middlesex.)

Secondary sources

Barking, J.K. (1957) *Changes in Conceptions of the Universe* (Cotteridge: Poltergeist Press).*
Colley, L. (2003) *Captives: Britain, Empire and the World, 1600–1850* (London: Pimlico).
Madison, S. (1967) 'The End of the Chain of Being: the Impact of Descartian Philosophy on Medieval Conceptions of Being'. *Journal of Medieval and Enlightenment Studies*, 66, 7.*
Pendleton, G. (1976) 'English Conservative Propaganda During the French Revolution, 1780–1802', Ph.D. (unpub.), Emory University.
Thompson, E.P. (1963) *The Making of the English Working Class* (Harmondsworth, Middlesex: Penguin).

*These two sources are hypothetical and provided for the purpose of the practice activity; the other sources are genuine.

Comments

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Practice 4: Finding flaws in the argument

- Read this second passage on the Great Chain of Being, and identify flaws in the argument, using the numbered prompts below as a checklist to assist you.
- *Note*: the practice passage does not contain all the flaws on the list but there is more than one example of some. You can use the checklist to note whether you believe the passage does or does not contain an example, to make it easier to check your answers.
- Label and number each of your answers in the *Comments* margin provided alongside the text. If you use the same numbers as those provided in the prompts table below, this will help you to check your answers.

Prompts	Example found	There is no example	See page
1 False premises			79
2 Two wrongs don't make a right			104
3 Stereotyping			84
4 Lack of consistency in the argument			59, 61
5 Unnecessary background information			52, 150
6 Lack of precision			59
7 Assumption that is not supported by the evidence			74–8
8 Incorrectly assuming a causal link			90
9 False correlation			91
10 Meeting necessary conditions			93
11 Meeting sufficient conditions			94
12 False analogy			96
13 Deflection			98
14 Complicity			98
15 Exclusion			98
16 Unwarranted leaps (e.g. castle of cards; sleight of hand)			100
17 Emotive language			101
18 Attacking the person			101
19 Misrepresentation			103
20 Trivialisation			103
21 Tautology			104
22 Poor referencing			144–5 167–8

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

The Great Chain of Being (Text 4)



'Notions of a "Great Chain of Being" and of a natural order to the universe continued to exert ideological significance during the eighteenth century and beyond.' Discuss.

The concept of the 'Great Chain of Being' dominated thinking and writing for many centuries before the eighteenth century. Indeed, Shakespeare and other great writers of the seventeenth century drew on the idea for inspiration. By the eighteenth century, things had started to change radically. This was a period of expansion intellectually and geographically for European states, including Britain. Old ideas were dying out as soldiers travelled the world during the wars against the American colonies and the expanding empire (Colley, 2003), and merchants traded more extensively with the East. Barking (1957) and Madison (1967) argue that enlightenment ideas and scientific observation replaced more traditional ideas. A revolution in taste took place as the homes of the rich filled with *chinoiserie*, art from China. Young people came of age through making a 'Grand Tour' of Europe. The concept of the chain of being was being supplanted by other ideas more familiar to our modern world.

In this period of exploration and change, the ongoing wars between England and revolutionary France led to an unusually large production of political propaganda. Pendleton's analysis of this showed that many pamphlets used the concept of the 'chain of being' to encourage the population to support the war. There were many ways that publicists referred to the idea of a natural order to encourage people to refuse the revolutionary ideologies espoused by the French and to encourage them to defend England in the event of an invasion. Those producing pamphlets and other pro-war literature referred to the notion of a natural order to decry French theories of liberty and equality and to argue that English people should take on a patriotic defence of the realm.

The propaganda was very insulting about the French and their new ideas and could easily have caused a diplomatic incident and an earlier outbreak of war. Fortunately, England and France are hundreds of miles apart, and such distances were more significant in the eighteenth century. This means that the French leader, Buonaparte, wouldn't have seen the propaganda and so didn't launch a full-scale invasion.

Comments

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Practice 4: Finding flaws in the argument

The idea of a natural order was used to bolster the authority of those with social and economic power. Mary Wollstonecraft (1792) argued that the concept of a natural order was used to justify all kinds of injustice.

She argued that people who were cruel to animals and children, were also likely to agree with the slave trade, and the oppression of women, which she opposed. However, she clearly thought it was acceptable to lump humans who lacked money and power into the same bracket as animals. As animals were lower down the chain of being at that time, her comparison of animals with humans who lacked power shows she thought of poor people and slaves as being lower types of being. Her prejudices are typical of ruling class women from that period.

Clearly, rich people in the eighteenth century found the idea of a natural order beneficial. This is particularly outrageous when one considers how vulnerable the poor were at the time, how sad their lives and how dependent on a kind word from their social betters. People were taught to regard those richer than themselves as their 'betters' and to refer to them as their masters. People were meant to accept that they must regard others as superior by virtue of their birth, and to defer to them in all things.

The idea of a natural order was strong even in the beginning of the twentieth century. After the Great War of 1914–18, working men and women gained the vote and social mobility increased. Far fewer people worked as domestic servants after the war. Having a vote on equal terms made people realise that democracy was a good thing and seems to have made them less keen to do jobs as servants. If everyone had the vote, then they were equal before the law, and if they were all equal, then there evidently wasn't a natural order, so the idea of a natural order was bound to die out and the vote would bring about the end of social hierarchies.

Such change would be welcomed. Many judges, priests, politicians and educators, argued that the chain of being was part of God's plan and this effectively frightened people into compliance with the way the country was run. Clerics such as Watson, the Bishop of Llandaff, wrote that it was God who let people get rich and powerful, signs of his favour and proof of their superiority. Other writers said similar things. For example, a poster in 1802 wrote about it being 'the ordinance of God' that the world was graded into different levels of being. Another, a pamphleteer (Pratt, 1803), argued that changing the order established by God would 'unsettle the whole system of the spheres; the planets would rush on each other … and the earth be shrivelled, like a scroll, by a spark from the sun'. However, Pratt was obviously not very bright and had a very poor grasp of science so was not likely to be taken seriously by his contemporaries.

Comments

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

One person who contributed most to perpetuating ideas of a natural order was the Swiss scientist Kaspar Lavater. His work was translated into many languages and used as a manual by the educated classes when they were employing new servants or making judgements about new acquaintances. Lavater invented a new science known as physiognomy which set out to prove that a person's character could be read from their facial features and the shape of the skull. Lavater (1797) argued that certain features were typical of a higher class of people, who were more moral and typical of the European ruling classes. He argued that other features, such as those shared by poorer people, and people with darker skins, were signs of an inferior nature, closer to the animals. Obviously, this was nonsense and no right-minded person would believe that physical features such as your skull would reflect your morals or worth. This would be like assuming that the way people walk tells you how healthy they are. However, many people at that time believed strongly in this method of working out who was superior and who inferior.

In the eighteenth century, people were more likely to believe in progress and change in the surrounding world, rather than a static concept such as the natural order. There were people who used the concept of the chain of being in an instrumental way, to frighten or coerce people into accepting that there was nothing they could do to change their lot. Certain applications of the idea of 'natural order' were adopted by richer people, but this practice was likely to have been a fad or fashion, like doing a quiz in a magazine today. Others used the idea to bolster their own sense of superiority. However, it is not likely that most people took such ideas seriously in the way they led their lives and made choices. In this respect, notions of the great chain of being and the natural order were not significant by the end of the eighteenth century.

References [compare with Practice 3, p. 257]

Primary sources

Anon (1802) *Such is Buonaparte*, London.

Kaspar Lavater *Essays on Physiognomy*, Translated by Rev. C. Moore and illustrated after Lavater by Barlow, London, 1797.

Pratt, *Pratt's Address to His Countrymen or the True Born Englishman's Castle.* London.

Bishop of Llandaff

Mary Wollstonecraft (1792) *Vindication of the Rights of Women.* Middlesex.

White, C. *An Account of the Infinite Gradations in Man* (Read to the Literary and Philosophical Society of Manchester at Different Meetings) (1779).

Comments

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Practice 4: Finding flaws in the argument

Secondary sources

Madison. (1967) 'The end of the Chain of Being: the impact of Descartian. *Journal of Medieval and Enlightenment Studies*, 66; 7.*

Barking, J.K. (1957) *Changes in Conceptions of the Universe.* Cotteridge: Poltergeist Press*

Linda Colley (2003) Captives.

Holmes, Geoffrey. (1977) 'Gregory King and the social structure of pre-industrial England' *Transactions of the Royal History Society*, 27

Pendleton

E.P. Thompson *The Making of the English Working Class* (1963) Middlesex: Penguin

*These two sources are hypothetical and provided for the purpose of the practice activity; the other sources are genuine.

Comments

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to Practice 4: Finding flaws in the argument

Prompts	Example found	There is no example	See page
1 False premises	✓		79
2 Two wrongs don't make a right		✓	104
3 Stereotyping	✓		84
4 Lack of consistency in the argument	✓ (3)		59, 61
5 Unnecessary background information	✓ (2)		52, 150
6 Lack of precision	✓		59
7 Assumption that is not supported by the evidence	✓ (3)		74–8
8 Incorrectly assuming a causal link	✓		90
9 False correlation	✓		91
10 Meeting necessary conditions	✓		93
11 Meeting sufficient conditions			94
12 False analogy	✓		96
13 Deflection	✓		98
14 Complicity	✓		98
15 Exclusion		✓	98
16 Unwarranted leaps (e.g. castle of cards; sleight of hand)	✓ (2)		100
17 Emotive language	✓		101
18 Attacking the person	✓		101
19 Misrepresentation	✓		103
20 Trivialisation		✓	103
21 Tautology	✓		104
22 Poor referencing	✓		144–5 167–8

© Stellla Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to Practice 4: Finding flaws in the argument

The Great Chain of Being (Text 4)

'Notions of a "Great Chain of Being" and of a natural order to the universe continued to exert ideological significance during the eighteenth century and beyond.' Discuss.

The concept of the 'Great Chain of Being' dominated thinking and writing for many centuries before the eighteenth century. Indeed, Shakespeare and other great writers of the seventeenth century drew on the idea for inspiration. By the eighteenth century, things had started to change radically. This was a period of expansion intellectually and geographically for European states, including Britain. Old ideas were dying out as soldiers travelled the world during the wars against the American colonies and the expanding empire (Colley, 2003), and merchants traded more extensively with the East. Barking (1957) and Madison (1967) argue that enlightenment ideas and scientific observation replaced more traditional ideas. A revolution in taste took place as the homes of the rich filled with *chinoiserie*, art from China. Young people came of age through making a 'Grand Tour' of Europe.[5] The concept of the chain of being was being supplanted by other ideas more familiar to our modern world.[4]

In this period of exploration and change, the ongoing wars between England and revolutionary France led to an unusually large production of political propaganda. Pendleton's[22] analysis of this showed that many[6] pamphlets used the concept of the 'chain of being' to encourage the population to support the war.[4] There were many ways that publicists referred to the idea of a natural order to encourage people to refuse the revolutionary ideologies espoused by the French and to encourage them to defend England in the event of an invasion. Those producing pamphlets and other pro-war literature referred to the notion of a natural order to decry French theories of liberty and equality and to argue that English people should take on a patriotic defence of the realm.[21]

[5]The propaganda was very insulting about the French and their new ideas and could easily have caused a diplomatic incident and an earlier outbreak of war. Fortunately, England and France are hundreds of miles apart, and such distances were more significant in the eighteenth century.

Comments

5 Unnecessary background information, especially as this is not used to look specifically at the idea of the chain of being. On the other hand, important background information, such as explaining what is meant by the 'chain of being', is not provided.

4 Inconsistency. This paragraph suggests the concept of the chain of being was waning. The next paragraph suggests it was still widely used (see [4] below). The author doesn't show how these two apparently contradictory ideas could both be true. For example, the idea could have been used in the propaganda for political purposes even if many people no longer believed in it. Apparent contradictions such as this need to be explained and resolved.

22 Poor reference. No date is provided here and Pendleton is not fully detailed in the references, so it would be hard for the reader to check this source of information for themselves.

6 Lack of precision. 'Many' is a vague term. The reader needs to know how many? What proportion?

21 Tautology. This paragraph repeats the same basic idea three times but in different words: i.e. that publicists used the idea of natural order to encourage a pro-war patriotic stance rather than support France and its ideas of liberty and equality. The final sentence, for example, does not take the argument forward or provide any new information for the reader.

5 This paragraph provides unnecessary background about the impact of the propaganda on the conduct of the war by the French, which is not what the question asks.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to Practice 4: Finding flaws in the argument

This means that the French leader, Buonaparte, wouldn't have seen the propaganda and so didn't launch a full-scale invasion.[1,7,16]

The idea of a natural order was used to bolster the authority of those with social and economic power. Mary Wollstonecraft (1792) argued that the concept of a natural order was used to justify all kinds of injustice. She argued that people who were cruel to animals and children, were also likely to agree with the slave trade, and the oppression of women, which she opposed. However, she clearly thought it was acceptable to lump humans who lacked money and power into the same bracket as animals. As animals were lower down the chain of being at that time, her comparison of animals with humans who lacked power shows she thought of poor people and slaves as being lower types of being.[19] Her prejudices are typical of ruling class women from that period.[3]

Clearly, rich people in the eighteenth century found the idea of a natural order beneficial.[13] This is particularly outrageous when one considers how vulnerable the poor were at the time, how sad their lives and how dependent on a kind word from their social betters.[17] People were taught to regard those richer than themselves as their 'betters' and to refer to them as their masters. People were meant to accept that they must regard others as superior by virtue of their birth, and to defer to them in all things.

Comments

1 False premises. The argument proposed for the lack of an invasion is based on false premises: it is factually inaccurate that England and France are hundreds of miles apart so this would not be a reason for the propaganda not being seen in France. The first paragraph of this piece suggests there was a lot of travel and exchange of ideas, which, if true, would make it more likely that the propaganda would have been seen in France.

7 Unsupported assumption about why Buonaparte didn't launch a full-scale invasion.

16 Unwarranted leaps. The author jumps from an unsubstantiated point (that the propaganda could have resulted in invasion – we don't know this) to another (that Buonaparte couldn't have seen it), to an unsubstantiated conclusion about why a full-scale invasion didn't happen.

19 Misrepresentation. Mary Wollstonecraft draws a comparison between different kinds of oppression, because she saw a common pattern of cruelty, which she opposed. The author misrepresents her intentions by claiming she regarded the poor and slaves as more animal like, simply because the idea of the chain of being was used by others at that time.

3 Stereotyping. Although many women of her class may have held such prejudiced opinions, the author stereotypes Mary Wollstonecraft by assuming she held the same ideas, without giving any evidence of this.

13 Deflection. The word 'clearly' suggests that the author has established how the rich people used the idea of natural order. This can deflect the reader from noticing that sufficient evidence has not yet been provided to prove the argument.

17 Emotive language. Use of words such as 'outrageous' and phrases such as 'sad lives' and 'dependent on a kind word' appeal to the emotions rather than relying on facts and reasons to advance the argument.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to Practice 4: Finding flaws in the argument

The idea of a natural order was strong even in the beginning of the twentieth century. After the Great War of 1914–18, working men and women gained the vote and social mobility increased. Far fewer people worked as domestic servants after the war. Having a vote on equal terms made people realise that democracy was a good thing and seems to have made them less keen to do jobs as servants.[8, 9] If everyone had the vote, then they were equal before the law, and if they were all equal, then there evidently wasn't a natural order, so the idea of a natural order was bound to die out and the vote would bring about the end of social hierarchies.[16, 7]

Such change would be welcomed. Many judges, priests, politicians and educators, argued that the chain of being was part of God's plan and this effectively frightened people into compliance with the way the country was run. Clerics such as Watson, the Bishop of Llandaff,[22] wrote that it was God who let people get rich and powerful, signs of his favour and proof of their superiority. Other writers said similar things. For example, a poster in 1802 wrote about it being 'the ordinance of God' that the world was graded into different levels of being. Another, a pamphleteer (Pratt, 1803), argued that changing the order established by God would 'unsettle the whole system of the spheres; the planets would rush on each other ... and the earth be shrivelled, like a scroll, by a spark from the sun'. However, Pratt was obviously not very bright and had a very poor grasp of science so was not likely to be taken seriously by his contemporaries.[7, 18, 4]

Comments

8 Incorrectly assuming a causal link. The author assumes a causal link between the extension of voting rights and the reduction in the number of domestic servants. However, there is no obvious reason why having a vote should create a different set of work opportunities for people. The reduction in the number of servants is more likely to be the result of economic changes, such as new kinds of job with better wages becoming available, or families no longer being able to afford to pay competitive wages.

9 This is also an example of a false correlation: mistaking the indirect correlation of increased suffrage (more people having the vote) and decreasing numbers of servants as directly connected.

16 and 7 Unwarranted leaps (castle of cards) and assumptions. The last sentence is another example of the author jumping from one unsubstantiated claim to another, such as that equality before the law automatically brings about equality of other kinds, such as social equality. However, social hierarchies are usually related to other things such as attitudes to ancestry, occupation, income, geography and ethnicity rather than depending on whether someone has the vote. Therefore, the author is wrong to draw the conclusion that the vote brought about the end of social hierarchies.

22 Poor referencing. No date and no reference given below so the reader can't check this for accuracy.

7 Unsupported assumption. The author makes an assumption here that Pratt's contemporaries would not take him seriously but does not provide any evidence to support this. Pratt's views might have been shared by others at that time. (See next page.)

18 This amounts to a personal attack on Pratt rather than a reasoned analysis of his views.

4 Inconsistency. The evidence is presented in a confusing, inconsistent way. The paragraph opens by arguing that references to God were effective. The author seems to cite Pratt as evidence of this effectiveness, but then states that nobody was likely to believe Pratt.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to Practice 4: Finding flaws in the argument

One person who contributed most to perpetuating ideas of a natural order was the Swiss scientist Kaspar Lavater. His work was translated into many languages and used as a manual by the educated classes when they were employing new servants or making judgements about new acquaintances. Lavater invented a new science known as physiognomy which set out to prove that a person's character could be read from their facial features and the shape of the skull. Lavater (1797) argued that certain features were typical of a higher class of people, who were more moral and typical of the European ruling classes. He argued that other features, such as those shared by poorer people, and people with darker skins, were signs of an inferior nature, closer to the animals. Obviously, this was nonsense and no right-minded person would believe that your physical features such as your skull would reflect your morals or worth.[14] This would be like assuming that the way people walk tells you how healthy they are.[12] However, many people at that time believed strongly in this method of working out who was superior and who inferior.

In the eighteenth century, people were more likely to believe in progress and change in the surrounding world, rather than a static concept such as the natural order.[7] There were people who used the concept of the chain of being in an instrumental way, to frighten or coerce people into accepting that there was nothing they could do to change their lot. Certain applications of the idea of 'natural order' were adopted by richer people, but this practice was likely to have been a fad or fashion, like doing a quiz in a magazine today.[6,10] Others used the idea to bolster their own sense of superiority. However, it is not likely that most people took such ideas seriously in the way they led their lives and made choices.[6] In this respect, notions of the great chain of being and the natural order were not significant by the end of the eighteenth century.[11]

Comments

14 The author relies on complicity here, writing as if the audience would automatically agree. If the author thinks this is so obvious, then there is no need to state that it is obvious. If the author thinks the audience might not find this approach to be nonsense, then reasons for not accepting it are needed.

12 False analogy. The analogy is not useful as the writer argues that Lavater's system is nonsense, whereas the way a person walks can tell you a great deal about some illnesses.

7 This paragraph contains unsupported assumptions about what people believed. Not enough evidence has been included to support these assumptions. The assumptions are then used as reasons to support the conclusion proposed in the final sentence.

6 Lack of precision. Not enough detail provided.

10 Meeting necessary conditions. The necessary conditions for establishing this interim conclusion are not met. To substantiate that the idea of a natural order was used only as a fashionable pursuit, the author would have to do the following:

- provide evidence that the idea was widely used amongst a certain group, and therefore constituted a 'fashion';
- provide evidence that the idea was used in a particular way only for a certain time (as fashions are time-bound);
- provide evidence that those who did use the concept of natural order in one aspect of their life, then acted in a contrary way in other aspects: that is, that the notion was not core to their belief system to such an extent that it ruled their behaviour.

11 The conclusion does not meet sufficient conditions as the evidence provided does not adequately support it. (See previous page.)

© Stellla Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to Practice 4: Finding flaws in the argument

References [compare with p. 257][22]

Primary sources

Anon (1802) *Such is Buonaparte*, London.[c]

Kaspar Lavater[a] *Essays on Physiognomy*, Translated by Rev. C. Moore and illustrated after Lavater by Barlow, London, 1797.

Pratt, *Pratt's Address to His Countrymen or the True Born Englishman's Castle.* London.

Bishop of Llandaff[d,e,f]

Mary Wollstonecraft (1792) *Vindication of the Rights of Women.* Middlesex.[a,f]

White, C. *An Account of the Infinite Gradations in Man* (Read to the Literary and Philosophical Society of Manchester at Different Meetings) 1779.[a,c,e]

Secondary sources

Madison.[d,e,f] (1967) 'The end of the Chain of Being: the impact of Descartian. *Journal of Medieval and Enlightenment Studies*, 66; 7.*

Barking, J.K. (1957) *Changes in Conceptions of the Universe.* Cotteridge: Poltergeist Press*

Linda Colley (2003) Captives.[a,d,e]

Holmes, Geoffrey. (1977) 'Gregory King and the social structure of pre-industrial England' *Transactions of the Royal History Society*, 27 [a,c]

Pendleton[d,e]

E.P. Thompson *The Making of the English Working Class* (1963) Middlesex: Penguin[a,c]

* These two sources are hypothetical and provided for the purpose of the practice activity; the other sources are genuine.

Comments

22 Compare the details of these references with those for Practice 3. Note:

(a) The order of items within each reference is not consistent from one reference to another, such as the order of the date, and whether initials or names are used.

(b) Not all the references used in the text are detailed in this list of references.

(c) Some items in the reference list do not appear in the text so should not appear here. It is possible that the author has used this source but not referenced it properly in the text.

(d) The information about the author is incomplete for some references so the reader cannot look these up.

(e) Some titles are not written in full (see list of references from Practice 3).

(f) Items are not in alphabetical order.

© Stellla Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Appendix

Selected search engines and databases for online literature searches

Cinhal – www.ebscohost.com
Nursing and health care database.

Embase – www.embase.com
Subscription-based biomedical and pharmaceutical database.

Google Scholar – http://scholar.google.co.uk
A good starting place to look up academic articles.

Ingenta Connect – www.igentaconnect.com
Finds abstracts for articles for over 30,000 journals and other publications. From university sites, you can also read the full text of these articles, if the university has subscribed to them.

Magazines – www.magportal.com
Lets you read old magazine articles up to January 2016 online.

PsycInfo – www.apa.org/psycinfo
Subscription-based database of psychological articles dating back to the 1800s.

PubMed – www.ncbi.nlm.nih.gov//pubmed/
Large biomedical and life sciences database.

Questia – www.questia.com
Subscription-based collection of journal articles.

ScienceDirect – www.sciencedirect.com
Search engine dedicated to science and technology. Searches books, journals and open access content.

World Wide Arts Resources – http://wwar.com
Resources on art news, art history and contemporary artists.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 2

Assess your thinking skills (p. 18)

Use the scoring sheet on page 22 to record your answers.

Comparison (p. 18)

A The odd box is 5. In all the other boxes, the line of circles in the second row is larger than in the other rows. Score 1 for a correct answer.

B The odd box is 1. In all the other boxes, the set of scissors in the bottom right-hand corner has white blades whereas the set in box 1 has black blades. Score 1 for a correct answer.

C The odd box is 4. The left-hand image in the bottom row is presented back to front in this box. Score 1 for a correct answer.

D The odd box is 6. The middle item in this box is different. Score 1 for a correct answer.

Sequence (pp. 18–19)

A Selection 1. Score 1 for a correct answer. The first three boxes repeat a pattern, with the tail to the right. The next set repeat the pattern with the tail to the left. Score 1 for identifying the reason correctly.

B Selection 4. Score 1 for a correct answer. Every second box has the same pattern. Score 1 for identifying the reason correctly.

C Selection 5. Score 1 for a correct answer. In each successive box, the top row from the previous box becomes the bottom row. Score 1 for identifying the reason correctly.

D Selection 2. Score 1 for a correct answer. Stars increase by one for the first three boxes then start again from one. Stars also move from top to bottom in alternate boxes. Boxes 4 and 5 start to repeat the same sets of lines (≡ ≤ and ≥) as earlier boxes, but these alternate between being at the top or the bottom of the box. Score 2 for identifying the reason correctly.

Categorising (p. 19)

A (i) Parts of a computer: mouse, drive, printer, monitor, screen.

(ii) Verbs (or words ending in -ing): typing, talking, scrolling, eating.

Score 1 for categorising items correctly.

Score 1 for identifying why group (i) items belong together.

Score 1 for identifying why group (ii) items belong together.

B (i) Items associated with Egypt: pyramid, oasis, palm-tree, desert, Nile.

(ii) Words that indicate that something is large: immense, vast, massive, enormous, gigantic.

Score 1 for categorising items correctly.

Score 1 for identifying why group (i) items belong together.

Score 2 for identifying why group (ii) items belong together.

C (i) Precious stones: agate, topaz, ruby, opal.

(ii) Precious metals: gold, silver, platinum.

Score 1 for categorising items correctly.

Score 1 for identifying why group (i) items belong together.

Score 2 for identifying why group (ii) items belong together.

D (i) Words beginning with a capital letter: Empty, Gate, Shoal, Divan, Kenya, Pound.

(ii) Words that begin in the lower case (i.e. without a capital letter): chops, hertz, micro, burst.

Score 1 for categorising items correctly.

Score 1 for identifying why group (i) items belong together.

Score 1 for identifying why group (ii) items belong together.

© Stellla Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 2 (continued)

Following directions (p. 19)

The aim of this activity was to enable you to check how closely you followed directions, especially when not forewarned. Academic work and exams usually require you to answer the question exactly as it is worded.

A Answer 4 is correct: a cow has four legs. If you answered C, check the exact wording of the question again: it only asks how many legs a cow has. Score 2 for a correct answer.

B Answer 1 is correct: water is comprised of oxygen and hydrogen. This is all that the question asks. Although the other responses may be true, they do not answer the exact question posed. Score 2 for a correct answer.

Close reading (pp. 20–1)

The Arctic

Score 1 for each correct answer.

1 **B** The passage does state that summers are short but the main argument is that conditions in the Arctic are harsh.

2 **C** This may be true or false: the passage does not say.

3 **B** Untrue. The passage says the sun doesn't appear for three months – but it can during the rest of the year.

4 **A** True – this follows logically. The passage states that there is no daylight for three months.

5 **C** The passage does not give enough information about non-natural forms of energy.

George Washington Carver

6 **B** Untrue. This is not stated in the passage and does not follow logically from what is stated. Score 1 point.

7 **C** The passage does not give exact dates so the monument could have been built while Carver was still alive. Score 2 points.

8 **B** Untrue. The passage states that there are mythic aspects to Carver's life, and implies that this is because he was a great man. It does not state that Carver's life, in its entirety, was a myth. Score 1 point.

9 **A** True. The passage states that religious groups believed this because Carver declared that God was the inspiration for his discoveries. Score 1 point.

10 **C** We would need more information. The passage refers only to Carver's own college, not to all the other colleges in the USA. Score 2 points.

11 **B** Untrue. The passage states the opposite, that this was the case as a result of Carver's inventions. Score 1 point.

12 **C** The passage states that Carver helped to develop over 100 industrial applications from agricultural products *such as* soybeans: we don't know how many of these were from soybeans themselves and how many from other agricultural products. Score 2 points.

Recognising similarities (p. 21)

For each passage, score 1 for identifying the correct option, and score 1 for finding the reason.

Option 1 is the nearest in meaning to passage 2.1 (*The Arctic*). Option 2 states that inhabitants like living there, which we do not know from the passage.

Option 5 is the nearest in meaning to the *George Washington Carver* passage, and is a summary of its key points. Options 3 and 4 focus on only some aspects of the passage, and make assumptions that are not in the passage. Option 3 expresses religious views that are not stated in the passage. Option 4 is inaccurate in stating why the monument was set up: the exact reasons are not given in the passage.

© Stellla Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 2 (continued)

Focusing attention

Find the 't' (p. 23)

There are 14 't's. These are highlighted.

Terrifying torrents and long dark tunnels are used to create the excitement of the thrilling train ride at the park.

If you counted less than 14 the first time round, this is not unusual. It is likely that your brain used its 'auto-pilot' to read the short words such as 'to' and 'the' automatically as a whole unit, rather than as a sequence of letters, even when you were trying to focus on the individual letters. If this happened to you, it may indicate that your brain works efficiently for most reading purposes.

Identifying difference (p. 24)

1 = f	5 = f	9 = f
2 = b	6 = e	10 = b
3 = a	7 = a	11 = b
4 = f	8 = e	12 = d

Recognising sequence (pp. 25–6)

1 = 1b An additional symbol is added in each successive box.

2 = 2e Boxes a and b are repeated in each successive pair of boxes.

3 = 3c Boxes a and b are repeated in each successive pair of boxes.

4 = 4d Each box adds one item, first to the left-hand column, then to the right-hand column.

5 = 5e Each symbol travels anti-clockwise around the box, moving one place in each successive box.

6 = 6b Each new box is repeated once.

7 = 7f The bottom row doubles in the next box along and also moves up one place.

8 = 8a

✸✠✸ moves up one place each row.

O moves along the row from left to right and descends a row each box.

➲✸⌑ at top of box for two boxes then at bottom for two, then top for two.

◣ repeats for two boxes then shifts one place.

9 = 9d The symbol ◎ moves from left to right along each row in turn. All the other symbols do the same. Once a symbol arrives at the bottom right-hand corner, it starts again in the top left-hand box.

10 = 10f The top row alternates in each successive box. The last three boxes of the middle row form a mirror image to the first three boxes of that row. The third row adds one dot in each successive box, for a sequence of three, then starts again.

11 = 11e

■▪■▪ moves down a row in each successive box for four boxes (and then starts again from the top).

The diamond pattern ⬩◆⬩ moves down a row in each successive box for three boxes (and then starts again from the top). It also alternates to a larger diamond coming first, ◆⬩◆t, in alternate boxes.

The □O◎ pattern moves down a row in each successive box for four boxes (and then starts again from the top). Within the pattern, the square moves one place to the right in each successive box.

☽ moves down a row in each successive box for four boxes (and then starts again from the top). It also doubles on alternate boxes.

12 = 12b

The row of five squares, ▪▪▪▪▪, moves clockwise around the edge of the box from one box to the next.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 2 (continued)

♦♦♦ alternates between the top and second lines, and changes to ♦♦ on the top line.
•• alternates between the third and second lines, and changes to ••• on the second line.
✸ ✪ alternates between the fourth and third lines, and changes to ✪ ✸ in alternate boxes.

Categorising

Categories (p. 27)

(a) Bodies of water
(b) Nationalities
(c) Animal habitats
(d) Science subjects
(e) Seven-letter words
(f) Verbs with the prefix (i.e. starting with) 'de'
(g) Words containing 'eve'
(h) Cognitive (thinking) skills
(i) Inflammatory conditions of bodily organs
(j) Palindromes: words that read the same backwards and forwards
(k) Terms that refer to the development of a soil profile
(l) Multiples of 7
(m) Forms of government
(n) Collective nouns for types of animal

Categorising text (p. 28)

Passage 2.3 Matter
Passages (a) and (b) are the most similar as they merely describe the two sets of classification. Passage (c) makes value judgements about the systems.

Passage 2.4 Anointing oil
Passages (b) and (c) are the most similar. Passage (a) makes a claim that the mixture of oils was invented in Egforth's reign whereas the other two passages express this only as a possibility.

Passage 2.5 The right hemisphere
Passages (b) and (c) are the most similar. Passage (a) attributes difficulty in differentiating reality to both hemispheres whereas (b) and (c) attribute this only to the right hemisphere. Passage (a) also claims that right-hemisphere damage leads to loss of imagination in general, which the other two do not.

Close reading (pp. 29–31)

Passage 2.6 Traditional legends

1 A This follows logically from the final line of the passage.
2 A This follows from what is said in the second sentence.
3 C There is not enough evidence in the passage to know this, as the passage refers only to myths of the Americas.
4 B No, the passage does not say anything about a sense of direction.
5 B This does not follow from the passage.
6 A This is consistent with what is written in the passage.

Passage 2.7 Transformation

7 B The passage does not say anything about people who travel a lot.
8 B The passage does not state this and it does not follow logically from information in the passage.
9 C The passage does not state this and we would need more data to know if most people did find illness to be transformative.
10 A This is consistent: it paraphrases what the passage states.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 2 (continued)

Passage 2.8 Clinical trials

11 A The statement is consistent with the passage, which argues that depression is wrongly attributed to levels of serotonin.

12 A This is consistent with the passage, as it states that raw data are rarely published whereas the results of beneficial tests are more likely to be published.

13 C The passage does not provide information about ageing effects.

14 B This is not consistent with the passage. The passage suggests that academic articles are normally just as inaccurate as they are based on the same data.

15 B This is not consistent with the information given in the passage, which argues that there is insufficient evidence about the links between serotonin levels and depression.

Passage 2.9 Helping others

16 C No data are provided on this in the passage.

17 B This contradicts the message of the passage, which is that in most circumstances there are no obvious racial differences in patterns of assistance.

18 A This is consistent with the third sentence, which states that a third more people are likely to help someone who is lame.

19 A This is consistent with the sentence in the middle of the passage, which says people are more likely to help if they think they will be effective.

20 A This is consistent with the first sentence, about the speed with which people help those with canes compared with others, and the sentence in the middle of the passage, about when people are more likely to offer help.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 3

Capturing the author's position (p. 35)

Passage 3.1

The author's position is: *It is possible to find a legal job that suits your preferences for court work.* The passage provides advice for aspiring barristers on selecting their field according to how much time they want to spend in court.

Passage 3.2

The author's position is: *Koch contributed one of the most important methodological advances in the the past of medicine.* The passage supports this with information about experiments Koch developed that were able to prove the germ theory of disease.

Passage 3.3

The author's position is: *The Sahara is an area worth investigating by those who are interested in the past.* The passage supports this by referring to traces of past cultures hidden beneath the sands.

Passage 3.4

The author's position is: *Even young children can perform tasks considered too advanced for them, as long as these are set up in ways that make sense to them.* The passage supports this by arguing that children perform better if the task is presented in language they can understand.

Identifying simple arguments (p. 40)

Passage 3.7

This is an argument. The conclusion is that this is a good picture. The supporting reasons are: the use of colour and the well-drawn and interesting figures.

Passage 3.8

This is not an argument. Even if you rearrange the sentences, none of them could act as a conclusion based on reasons provided by the others.

Passage 3.9

This is not an argument even though all the statements are true. If you rearrange the sentences, none of them could act as a conclusion based on reasons provided by the others. Just because the subject sounds scientific, this doesn't make it any more of an argument.

Passage 3.10

This is an argument. The conclusion comes at the end: *The piper's action was one of revenge.* The supporting reasons for the argument are: he was angry at the townspeople as they hadn't paid for his services; he led their children away deliberately and caused their deaths.

Passage 3.11

Although short, this is still an argument. The conclusion is that there must have been a signal failure. The reason for thinking this is that the train is late.

Passage 3.12

This is just a collection of related information and is not an argument. None of the statements serves as a conclusion to the rest.

Passage 3.13

This is an argument, even if you may not agree with the conclusion. The conclusion comes at the end: *It must have been a ghost.* The supporting reasons are the rattling windows, banging doors, charged air and feelings of fear.

Passage 3.14

This is an argument. The conclusion is at the beginning: *Many adults learn to read later in life.* The supporting reasons for the argument are that: John and Miranda caught up with their peers in reading; almost a million adults have improved their literacy skills.

© Stellla Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 3 (continued)

Passage 3.15

This is not an argument. The author is not attempting to persuade the audience to accept a position or a particular conclusion. This passage describes a process.

Reasons and conclusions (p. 41)

Passage 3.16

Main argument: The swollen river dislodged the skeleton.

Reasons that support the argument:

- Reason 1: A skeleton was found near the river.
- Reason 2: The police have ruled out the possibility that a suspected local family was involved.
- Reason 3: The bones are believed to be several hundred years old.
- Reason 4: Historians confirm that the river passes close to ancient burial grounds.
- Reason 5: Other bodies are known to have been carried away by the river in the past.
- Reason 6: Recent storms have caused the river to rise by over half a metre.

Conclusion: It is probable that the skeleton was dislodged from its resting place by the swollen river rather than murdered by the local family.

Passage 3.17

Main argument: Sea grasses are important.

Reasons that support the argument:

- Reason 1: In shallow waters, sea grasses are the dominant form of vegetation.
- Reason 2: They support a host of marine life.
- Reason 3: The grasses act as nurseries for fish, including commercial varieties.
- Reason 4: Without sea grasses, the bio-diversity of coastal regions would be severely impoverished.

Conclusion: Sea grasses make important contributions to coastal ecosystems.

Passage 3.18

Main argument: We are unhappy because we focus too much on how the world is and on the gratification of short-term objectives, rather than thinking about how we could live more in harmony with other people and the environment over the longer term.

Reasons that support the argument:

- Reason 1: We forget to be kind and supportive.
- Reason 2: We satisfy our wants first without thinking of people who have greater needs.
- Reason 3: We ignore what we know about the needs of the environment.
- Reason 4: We focus on instant gratification and short-term gains rather than considering the longer-term consequences.

Conclusion: The challenge facing humans is to find a way of acting more co-operatively with each other and more in harmony with our universe.

Passage 3.19

Main argument: Pregnant women and those with suppressed immune systems need to be aware of the potential danger of toxoplasmosis passed on by cats.

Reasons that support the argument:

- Reason 1: Cats are hosts to *toxoplasma gondii*, the protozoa that can cause disease in humans.
- Reason 2: If pregnant women become infected with the protozoa, their foetus could lose its eye-sight or have motor deficits.
- Reason 3: In people with poor immune systems or AIDS, toxoplasmosis can cause seizures and death.
- Reason 4: Infected cats show no symptoms.

Conclusion: It is important that pregnant women and those with poor immune systems become aware of the potential risks posed by cats.

© Stellla Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 4

Argument or disagreement (p. 47)

Passage 4.1

A Argument. The overall argument is: *Bilingualism and multilingualism confer many benefits.* The reasons given are: (1) that speakers of more than one language have a better understanding of how languages are structured; (2) a second language can help people to understand a first language.

Passage 4.2

B The final line expresses disagreement with the idea that complementary therapies are the equivalent of medical treatments. No reasons for this are given so this is not an argument.

Passage 4.3

B The final line expresses disagreement with the idea that employers cannot do more to help save lives in the workplace. No reasons for this are given so this is not an argument.

Passage 4.4

A This is an argument. The conclusion is in the first line: *People are less politically aware now than they have been at any time in the past.* The reasons given are: (1) people used to fight for causes from which they didn't gain personally; (2) people took more risks for political issues; (3) rallies had a more international perspective; (4) fewer people vote now in elections.

Passage 4.5

B The final line expresses disagreement with arguments against global warming. No reasons for this are given so this is not an argument.

Passage 4.6

A Argument. The conclusion is in the second line: *Of course it harms them, both physically and emotionally,* referring back to the issue in the first line about smacking. The reasons given to persuade us are that (1) it is assault; (2) assaults on adults are not accepted; (3) smacking perpetuates a cycle of violence.

What type of message? (p. 50–1)

Passage 4.7

Description of key aspects of space launches.

Passage 4.8

Argument that babies may benefit from sleeping with their mothers.

Passage 4.9

Summary, by Farrar (2004a) of an article by Platek et al. See Bibliography.

Passage 4.10

Description of the location of a village.

Passage 4.11

Explanation. The text explains why the dog needed to use smell rather than shape or colour to identify his toy mouse.

Passage 4.12

Summary of the plot of a Shakespeare play.

Passage 4.13

Explanation of why the student was late.

Passage 4.14

Explanation of why the cave drawings were identified so recently.

Passage 4.15

Argument that there were greater cultural links between continental Europe and Britain during the Ice Age than was formerly believed.

Passage 4.16

Description of specialists' responses to the cave drawings.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 5

How clear is the author's position? (p. 58)

Passage 5.1

The author's position is not clear. It could be clarified, for example, by using either the opening sentences to introduce the argument and/or the final sentence to sum it up. The author uses too many questions without providing answers to them. There are many facts, but these do not help clarify the position. The author needs to provide more guidance to the reader about the direction of the argument.

Passage 5.2

The author's position is not clear. The author is aware of different viewpoints, which is good. However, the writing wanders back and forth between different standpoints without being clear about which point of view the author wants the audience to accept. The author doesn't fully agree or disagree with either point of view and does not suggest an alternative third point of view. The author needs to sort the issues so that similar points are considered together, and to order them so that they lead towards a conclusion. The passage reads as though the author doesn't know what to believe. In such cases, an author needs to take up a position for the duration of presenting the argument, even if only to indicate how one point of view has certain advantages over the other.

Passage 5.3

The author's position is not clear. The purpose of the report was to clarify whether a sports centre should be built. The passage looks at points for and against building the sports centre, which is appropriate, but the points are jumbled. It would have been clearer if those for building the centre were given first, and then those against. The relative weighting might have come across better. The author needs to give some indication of whether the sports centre should be built or not, in order for their position to come across.

Internal consistency (p. 60)

Passage 5.4

B: inconsistent. The author argues that performance enhancing drugs should be banned on the grounds that they give an unfair advantage, not on whether someone intended to cheat or not. By the end of the passage, the 'unfair advantage' argument is replaced by arguments about medical need and intention. To be consistent, the author should maintain the position that taking performance enhancing drugs is always wrong, or else argue a more moderate position as in Passage 5.5.

Passage 5.5

A: consistent. In this case, the author argues consistently that drugs should be generally discouraged on health grounds but permitted on an individual basis for health reasons.

Passage 5.6

B: inconsistent. The author argues that reality TV is not giving the public what it wants, but then points out that 'almost the whole nation' is watching it, which suggests it is popular. The author could have made the argument more consistent by, for example:

- offering an explanation for why people watched programmes they did not want;
- giving evidence that there were no other choices;
- presenting evidence of surveys that show people would prefer to watch a good alternative type of programme.

Passage 5.7

B: inconsistent. The author argues that the countryside is disappearing but cites a figure of only around 8 per cent of the countryside as built up so far. To be consistent, the author would need to present further arguments to show why the other 92 per cent is really at risk of disappearing.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 5 (continued)

Passage 5.8

B: inconsistent. The author argues that before Columbus, 'everyone believed the world was flat'. However, several examples are given of people who didn't believe the world was flat. It is not unusual for people to include this sort of inconsistency in their arguments. People often repeat a commonly held belief, such as the medieval church's belief that the world is flat, without noticing that they are citing contradictory evidence. To be consistent, the author could argue that Columbus was courageous on other grounds than that of other people's belief in a flat earth. For example, it could be argued that he was courageous to persist with the voyage when the distances involved, and consequences of these, were not known.

Logical consistency (p. 62)

Passage 5.9

B: not logically consistent. It does not follow logically that because some animals can survive without light, all animals can do so.

Passage 5.10

B: not logically consistent. The reasons given do not support the conclusion that the number of accidents will rise over the next year.

Passage 5.11

B: not logically consistent. A more logical conclusion from the reasons given is that more status should be given to subjects such as sports, media and popular culture. If a subject's low status follows the social class of the students, then if the students change subject, the status of the subject they take might fall, perpetuating the same problems.

Passage 5.12

A: logically consistent. The igneous rock could only cut across the layers of sediment if they were already there. They must be older, and the igneous rocks more recent.

Passage 5.13

B: not logically consistent. It may be true that it is impossible to find a place of absolute silence but that does not mean noise pollution is increasing. Noise levels may be the same as in previous times but with different causes: we cannot tell from the arguments presented.

Passage 5.14

B: not logically consistent. The conclusion is that computers will one day be able to outperform humans at everything. However, the author has argued that computers lack the qualities needed for empathy. This contradicts the idea of computers being better at 'everything'.

Independent reasons and joint reasons (p. 64)

Passage 5.15

Joint reasons. The reasons are mutually supporting of the rights and responsibilities of young people.

Passage 5.16

Independent reasons. The reasons given concern the environment (litter), value (few discoveries for the number of expeditions taken), economics (effect on local economy), and safety.

Passage 5.17

Independent reasons. Lying is defended on the basis of different arguments: (a) the truth hurts; (b) it provides a useful coping mechanism; (c) it isn't always possible to tell a lie from the truth; (d) the social benefits of lying.

Passage 5.18

Independent reasons. The argument is that the book is a faithful representation of a rock band. The reasons given are based on (1) knowledge: the author's close knowledge of the band; (2) experience: her experience of being in a band herself; (3) objectivity: reasons why the author was able to be objective.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Passage 5.19

Independent reasons. The reasons given are related to (1) effective use of resources; (2) public image; (3) support for staff.

Passage 5.20

Joint reasons: all support the argument that Magritte gave very few clues to help others to interpret his work.

Intermediate conclusions (p. 67)

Passage 5.21

Overall argument. This is at the end of the passage: *The government should take strong action to raise awareness of the risks of smoking and to ban it in public places.*

The intermediate conclusions are highlighted in bold.

Although most smokers say they enjoy smoking, many smokers wish they didn't smoke. 'It feels as if I am setting light to my money,' wrote one correspondent. Cigarettes can account for up to a half of an individual's total spending. As people are borrowing more money in general, and paying interest on it, the overall cost of cigarettes is sometimes hidden. However, as many smokers are all too aware, **smoking does not make good financial sense.**

The effects on long-term health are equally devastating. Just as smokers are often building up debts in the bank, they are also accruing unseen deficits in terms of their health. It is easy to forget the health implications of smoking. Warnings about illness and death can seem a long way away. Unfortunately, once cancer of the bowel, the lung, the throat, or the stomach sets in, it is often too late to take any action. Moreover, these diseases can strike unexpectedly whilst people are still young.

Smokers spread strong, unpleasant odours all around them, affecting other people without their consent. Smoking impairs the sense of smell so smokers do not realise how much they are inflicting awful odours on others. Some believe that smoking outdoors washes all those nasty odours away, but this is clearly not the case.

Furthermore, studies of the houses of people who always smoke outdoors, have found that the chemicals found in cigarettes are over seven times as prevalent as in the houses of non-smokers. Noxious chemicals linger, affecting the health of other people, sometimes fatally. **Whether outdoors or in, smoking doesn't simply kill the smoker, it kills other people and this should not be permitted.**

Intermediate conclusions used as a reason (p. 67)

The two intermediate conclusions for each passage are highlighted in bold.

Passage 5.22

It is a legal offence to assault other people. Hitting and slapping people are forms of assault and cause psychological, if not physical, damage. **They should always be considered as examples of legal assault.** Although this rule is applied to adults, it is often not recognised in the case of children. Slapping is defended as a useful and necessary form of discipline. It is also argued that children are not independent beings. This is not a valid argument. **Children** may be dependent on adults but they **are still people**. Therefore, slapping a child should also count as legal assault.

In this case, in order to argue that slapping a child should count as legal assault, the author has first to establish that:

(1) slapping should always count as legal assault;

(2) children should count as people.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Passage 5.23

Many people speak out in discussion too quickly because they are anxious about leaving a silence. When questioned, people often acknowledge that they spoke early in order to ensure there was no gap in the discussion. **They are not used to silences in conversation and don't know how to manage them skilfully.** They can find silences in discussion to be unnerving and embarrassing. **However, silence can be productive**. First of all, it allows time for reflection so that speakers can construct a more considered and accurate response, making a more useful contribution to the debate. Secondly, it gives more people the opportunity to speak first. For more productive discussions, we need to be skilled in managing silences.

The author has to establish two interim conclusions that can be used as reasons or arguments in their own right:

- The reason people speak too early is because they don't know how to manage silence.
- Silence can be productive in improving discussion.

(1) The reason people speak too early is because they don't know how to manage silence. If this can be established, then it supports the conclusion that skilful management of silence will improve discussion. The author establishes the interim conclusion by (a) citing people's own acknowledgements that this is accurate; and (b) giving the reason that people are not used to silences in conversation so cannot manage them skilfully.

(2) Silence can improve discussion. The author does this by offering two independent reasons: (a) silences allow thinking time so that responses are better constructed; (b) more people get a chance to speak first.

Summative and logical conclusions (p. 69)

Passage 5.24

Logical conclusion. The author weighs two different sets of arguments and draws a conclusion, or deduces, that the environment is more influential than genes in forming criminal behaviour, so the passage forms an argument.

Passage 5.25

Summative conclusion. The author summarises two positions but does not draw a conclusion about whether reality shows are good for television or not. As there is not a logical conclusion based on the reasons, this is not an argument.

Passage 5.26

Logical conclusion. The author makes a judgement about the level of costs that would be borne by banks if debts in developing countries were cancelled. This conclusion is deduced from the reasons, so this passage constitutes an argument.

Passage 5.27

Summative conclusion. The author merely summarises two points of view without making a judgement about whether organic food tastes better. There isn't a logical conclusion based on reasons so this is not an argument.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Logical order (p. 71)

Passage 5.28 *Circadian rhythms*

The passage is badly organised for the following reasons:

- The author hops back and forward between points rather than grouping similar points together into separate sections.
- There is no obvious introduction.
- The conclusion and the author's position are not obvious.
- The passage lacks words to link each new point to highlight the direction of the argument.

Compare the original version with the version below. This contains almost identical material but is ordered differently and phrases are added to indicate the logical links. These are indicated in bold.

5: Our bodies remain more responsive to biological rhythms than to the demands of clock time or the distractions of the outside world. 10: These biological rhythms are known as circadian rhythms and we know they are particularly strong in birds. 11: In humans they are particularly controlled by the suprachiasmatic nucleus (SCN) in the anterior hypothalamus at the base of our brains. 12: **We know this because**, if this part of the brain is damaged, a person loses all sense of a natural 24-hour clock, where sleep coincides with night-time. 13: In other people, circadian rhythms are much stronger than was expected. 1: **For example**, in experiments, human volunteers spent several weeks underground in constant light. 2: At first, their natural clock and sleep patterns were disrupted. 3: **However,** after a few weeks, they reverted back to the natural circadian rhythm with a 24-hour clock more or less in line with the outside world. 4: **Nonetheless,** our natural clocks are helped to adjust by exposure to sunlight and do respond to patterns of light and dark. 14: Astronauts, who lose this connection to the sun's rhythms for a long time, find it hard to adjust. 15: Many require medication to help them sleep. 16: Night-workers, even after 20 years on shift patterns, do not adjust circadian rhythms to suit the demands of night working.17: Certain illnesses such as peptic ulcers and heart disease, as well as increased risk of car crashes, are much more common to night-shift workers.

6: Since the mapping of human genes as part of the genome project, we have a greater understanding of circadian rhythms and their role in genetic conditions. 7: Some families have genetic conditions which make them less sensitive to circadian rhythms. 8: This may help explain patterns of sleep disturbances found in those families. 19: It may be that conditions associated with mental ill-health, such as schizophrenia and bi-polar disorders, are also linked to malfunctions in circadian rhythms.

9: Our work patterns, leisure patterns, architecture, lighting, food, drugs and medication compete with our natural clocks. 18: As the long-term effects of disrupting circadian rhythms are yet to be discovered, we should take care to ensure the health of shift-workers and those with genetic conditions that make them less sensitive to the biological 24-hour clock.

This is not the only possible alternative. Another option would be to order the sentences as:

5, 10, 11, 12
6, 7, 8, 13
9, 1, 2, 3, 4, 14, 15
16, 17, 19, 18

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

This would then read:

Passage 5.28 Circadian rhythms

5: Our bodies remain more responsive to biological rhythms than to the demands of clock time or the distractions of the outside world. 10: These biological rhythms are known as circadian rhythms and we know they are particularly strong in birds. 11: In humans they are particularly controlled by the suprachiasmatic nucleus (SCN) in the anterior hypothalamus at the base of our brains. 12: If this part of the brain is damaged, a person loses all sense of a natural 24-hour clock, where sleep coincides with night-time.

6: Since the mapping of human genes as part of the genome project, we have a greater understanding of circadian rhythms and their role in genetic conditions. 7: Some families have genetic conditions which make them less sensitive to circadian rhythms. 8: This may help explain patterns of sleep disturbances found in those families. 13: In other people, circadian rhythms are much stronger than was expected.

9: Our work patterns, leisure patterns, architecture, lighting, food, drugs and medication compete with our natural clocks. 1: In experiments, human volunteers spent several weeks underground in constant light. 2: At first, their natural clock and sleep patterns were disrupted. 3: After a few weeks, they reverted back to the natural circadian rhythm with a 24-hour clock more or less in line with the outside world. 4: Nonetheless, our natural clocks are helped to adjust by exposure to sunlight and do respond to patterns of light and dark. 14: Astronauts, who lose this connection to the sun's rhythms for a long time, find it hard to adjust. 15: Many require medication to help them sleep.

16: Night-workers, even after 20 years on shift patterns, do not adjust circadian rhythms to suit the demands of night working. 17: Certain illnesses such as peptic ulcers and heart disease, as well as increased risk of car crashes, are much more common to night-shift workers. 19: It may be that conditions associated with mental ill-health, such as schizophrenia and bi-polar disorders, are also linked to malfunctions in circadian rhythms. 18: As the long-term effects of disrupting circadian rhythms are yet to be discovered, we should take care to ensure the health of shift-workers and those with genetic conditions that make them less sensitive to the biological 24-hour clock.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 6

Identify the underlying assumptions (p. 75)

Passage 6.1

Underlying assumption: *Campaigning against nuclear weapons is an accurate measure of how politically-minded a group is.* However, it could be that other political issues are just as important to different generations.

Passage 6.2

Underlying assumption: *Whenever house prices rise quickly, there will always be a slump in which people lose money.* It may be, for example, that patterns of investment or interest rates vary during different periods of rapid house-price rises, so that a slump or loss of money might not automatically follow them.

Passage 6.3

Underlying assumption: *Advertising aimed at children is to blame for peer pressure.* This may be true or untrue. The link between advertising and peer pressure isn't established in the passage itself.

Passage 6.4

Underlying assumption: *A high number of searches on the internet means that 'everyone' must know about the subject.* It may be true, but it is probable that many people haven't heard about Emeagwali. A web page which receives many 'hits' is, nonetheless, visited by a relatively small proportion of people. Also, the same people may have visited the website many times.

Passage 6.5

Underlying assumption: *All jobs could be moved to lower-wage economies.* This assumption is needed for the conclusion that there would be 'no jobs' left. Some reflection would indicate that this is unlikely to be the case. Many jobs, such as medicine, catering, retailing, teaching and caring services, need to be delivered locally so it is unlikely that 'no jobs' would be left in high-wage economies. The passage also assumes that only 'companies' offer jobs, but other organisations and individuals could also be employers.

Passage 6.6

Underlying assumption: *Some consumers do not understand the information they read about E numbers.* If this was not the assumption, then the conclusion that 'putting information on the label is not necessarily helpful: people need to know what it means' could not be drawn. E numbers mean 'approved for use in every country in Europe' and include chemicals such as vitamins as well as those considered unhealthy. E300 is vitamin C. There is also an assumption that consumers want to eat more healthily, which may not be the case.

Implicit assumptions used as reasons (p. 78)

Passage 6.7

Conclusion: *As there has been so little advance on humanoid robots assisting with housework and construction, it will probably never be achieved.*

The implicit assumptions used as reasons are:

(1) *Just because a robot was designed a long time ago, there have been continuous efforts since then to design a robot to deal with certain kinds of work.* No evidence is given to show that this is what Leonardo or inventors since him set out to do.

(2) *If something hasn't been done before a certain time, it never can be.* In the case of designing the robot described, the author doesn't prove this.

These assumptions may be true but are not supported in the passage by evidence.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 6 (continued)

Passage 6.8

The conclusion is: *Ending postal voting will ensure a return to fair elections.*

It may be true or untrue that postal voting is less fair than other forms of voting. However, the implicit assumption is:

(1) *Elections were fair before postal voting was introduced.* This is not proved in the passage. For example, some people might not consider that elections are fair if those who work away from home on the day of an election through no choice of their own, or those in hospital or serving in the forces overseas, cannot vote.

Other assumptions made are:

(2) *Intimidation is not used in any other kind of voting system.* This is not established in the passage. For example, intimidation could be used to make people surrender their voting papers in other kinds of election.

(3) *Postal voting could not be altered to reduce or remove intimidation.*

Passage 6.9

The conclusion is: *It would be better to return to traditional methods of using leaves and roots of plants rather than mass-produced pharmaceuticals.*

The implicit assumptions are:

(1) *Past methods of using plants were as effective as modern medicines.* This may be true or untrue. The passage does not provide evidence to establish this. Modern medicines often use plants in more concentrated forms and combined with other chemicals that are not locally available. This may make them more, or less, effective.

(2) *Modern medicines are being used to cure the same range and types of illnesses as in the past.*

(3) *The range and amounts of plants would be available and accessible to people in the amounts needed.* When one considers the very large amounts of concentrated chemicals stored on chemists' shelves, it is difficult to imagine that the plant equivalent could be made available so readily.

Passage 6.10

The conclusion is: *We should continue to improve sanitation and diet in order to further increase our life expectancy.*

The implicit assumption being used as a reason is that *life expectancy increased in the past because of sanitation and diet.* This may be the case but it hasn't been established as true in the passage. It might be argued, for example, that many people had good diets but not enough food, and died early as a result of famine. Others died as a result of epidemics and many men died through wars, without these necessarily being affected by poor diet or sanitation.

The passage also contains the implicit assumption that diet, sanitation and life expectancy could be improved further, and that continuing to increase life expectancy is a good thing. Not everyone might agree with this.

Passage 6.11

The conclusion is: *Therefore, in order to keep their businesses afloat, new restaurant owners should delay installing new kitchens until the restaurant is established.*

The implicit assumption which is used as a reason is that *new kitchens are an unnecessary expense when a restaurant is new, contributing to the lack of funds at the end of the year.* This is a reasonable assumption to make but it does not follow from what has been said so far in the passage. This kind of conclusion is also an example of a non-sequitur (see p. 76), as the conclusion seems to jump out of nowhere, rather than following the previous sequence of the reasoning.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 6 (continued)

Passage 6.12

The conclusion is: *More should be done to reduce the world's population so that food supplies can go round.*

The implicit assumption used as a reason is that *the size of the world's population is the cause of under-nourishment*. The passage also assumes that there is not enough food to go round. This may or may not be the case: the passage does not present evidence to support this. However, under-nourishment can be caused by eating the wrong foods rather than simply not having food to eat. Some countries consume much more food than their populations actually require so other people might argue that better food distribution is more important than population control.

False premises (p. 80)

Passage 6.13

Sound premises. Petrol prices would be likely to rise for the reasons given.

Passage 6.14

False premises. The argument is based on the false premise that getting wet in the rain gives you a cold. There is no direct link between getting wet and catching a cold. Most of the time when people get wet, they do not later have a cold.

Passage 6.15

False premises. The false premise is that people would and could continue to marry at the same rate each year, which is unlikely. The passage does not take into consideration that some of the population, such as children, would not be eligible to marry, and that others would not choose to.

Passage 6.16

False premises. It is a false premise that a good menu will lead to a new restaurant being filled every night. Most new restaurants struggle to survive and established restaurants local to the one in the passage do not fill to capacity. Good cooking, low prices or a better location might have been reasons for expecting a full restaurant.

Passage 6.17

False premise. The false premise is that the more choice there is, the better the quality of the programmes. This has not been established – and many people would argue to the contrary.

Passage 6.18

Sound premises. The Indian film industry is growing in its worldwide appeal for the reasons given: it is gaining international acclaim, attracts non-Indian audiences and is shown in more countries than in the past.

Passage 6.19

False premise. Even if it were true that people's nationality could be read from their behaviour, the argument would be based on the false premise that similarities are genetically based. Nations such as the English and the French are not genetically homogeneous but descend from a very wide variety of ancestors. The behaviours described are more likely to be the result of cultural than genetic reasons.

Passage 6.20

False premise. The false premise is that the air in the countryside is free of pollution. There are many pollutants, such as agricultural pesticides, that can affect people living in rural areas.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 6 (continued)

Implicit arguments (p. 82)

Passage 6.21

The implicit argument is that if employees do not do as is expected of them, they are likely to lose their jobs or suffer a similar serious penalty such as lack of promotion. This is not stated explicitly but is an implicit threat.

Passage 6.22

The implicit argument is that the opposing candidate lied about fighting for the country and stealing from the nation and won't keep electoral promises about taxes. This is not stated explicitly but is implied.

Passage 6.23

The implicit argument is that Julian and Ian stole the pipes. A series of statements are presented which, if there was a recognisable structure for an argument, would form a series of reasons. The two workers 'worked late', so we are left to assume this means after other people had all gone home; they can drive the lorries so it is implied that they did drive them; they have given no alibi so we are left to assume both that they have no alibi and that this means they must have committed the theft.

Passage 6.24

The implicit argument is that people who emigrate from other countries are more likely to be dishonest. No evidence is presented to support this argument.

Passage 6.25

The implicit argument is that as most people want the death penalty, it should be introduced. This is not stated explicitly.

Ideological assumptions (p. 82)

Passage 6.26

In this passage, people 20 years old are still considered children. The age at which one becomes an adult has varied at different historical times and according to the society.

Passage 6.27

The passage assumes these are good working conditions. It considers it to be acceptable for children to work rather than attend school, that a 12-hour working day is reasonable, and that workers don't have extended holidays. In this passage, work is considered a form of morality and not working is regarded as sinful. Novels of the early nineteenth century describe working conditions such as these, which were not unusual at that time.

Passage 6.28

In this case, the ideological assumption is that women cannot inherit estates. This was the case in Britain for several hundred years, and died out mainly in the twentieth century.

Passage 6.29

This passage assumes that women are too emotional to report news about serious issues. For many years, women were not allowed to read the news in Britain, and arguments such as these were commonplace. It was assumed that women would burst into tears at difficult news. It was also argued that if a woman read the news, it would automatically sound trivial because women were associated only with trivial matters.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 6 (continued)

Word associations (p. 85)

1 = F
2 = A
3 = E
4 = B
5 = G
6 = C
7 = D

Stereotypes (p. 85)

1 This reinforces the stereotype that all girls like pink.
2 This reinforces the stereotype that being a pilot is a job for males and being a steward is a job for females.
3 This reinforces the stereotype that British people only eat roast beef and won't eat food from other countries.
4 This reinforces the stereotype that all red-headed people are hot-tempered.
5 This reinforces the stereotypes that people from the Caribbean all like Reggae and only want to listen to that music, and that people from Spain all like flamenco music and only want to listen to that.
6 This reinforces the stereotype that all football fans are trouble-makers.
7 This reinforces the stereotype that students are lazy and can't fend for themselves. It reinforces the idea of students as younger people with parents who live near enough to visit. It doesn't include the concept of students who do not have parents, are older, from overseas, or brought up in care.
8 This reinforces the stereotype that people are not interested in fashion or computers once they reach a certain age.

© Stellla Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 7

Assuming a causal link (p. 90)

Passage 7.1

The assumed causal link: obesity leads to longer life expectancy. The link does not follow logically from the reasons given: it hasn't been shown that those who are obese live longer, nor why obesity should lead to longer life.

Passage 7.2

The assumed causal link: that it was the roof-top protest that led to the prisoners' release, rather than, for example, them having been found innocent, the evidence against them being found to be flawed, or them having completed their sentences. Something which has happened only twice does not establish a solid trend.

Passage 7.3

The assumed causal links are that the man was murdered, that somebody broke in to do this, and that the knife was the murder weapon. However, in actuality, no murder may have occurred.

Identify the nature of the link (p. 92)

Passage 7.4

A The reasons support the conclusion through causal links: children eat sugar; sugar decays teeth; the children's teeth decay.

Passage 7.5

B The conclusion requires the assumption that students are more likely to be found plagiarising if they work electronically. It assumes that there is something about working electronically which enables this to occur, such as, for example, specialist software, to identify students who copy items found on the internet.

Passage 7.6

C There may appear to be a link between being a great scientist and having long hair but this would be easy to disprove: all that would be needed would be examples of great scientists with short hair, of which there are many. The argument is illogical as it assumes that long hair is a constant, whereas hair can vary in length over relatively short times. To prove the case, the author would have to establish a link between a decrease in scientific ability when hair was cut, and an increase when it grew back.

Passage 7.7

B The conclusion requires the assumption that increases in footballers' wages are paid for primarily by match tickets rather than any other means that clubs have for raising money, such as selling players, advertising, prize money and television payments.

Passage 7.8

B The conclusion requires the assumption that fast food outlets use ice cream containers that hedgehogs can break into. If not, the conclusion would not be supported. It also assumes that nothing or nobody else could have created the wastage except for the hedgehogs, and that there were hedgehogs in the area.

Passage 7.9

A The reasons support the conclusion through causal links: Dubai provides opportunities for jobs and houses to foreign nationals; foreign nationals have settled; the population is rising.

False analogies (p. 97)

Passage 7.10

This compares the earth's atmosphere to a blanket. In this case, the comparison is valid as both are thin coverings that provide protection and warmth.

Passage 7.11

This passage compares a small political party to the biblical character David, and larger political parties to his opponent, Goliath. David was successful against

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

an apparently greater opponent so the comparison is an effective one in arguing that the new party has a chance of success. The validity of the comparison would be demonstrated at the elections if the smaller party did better than the bigger parties.

Passage 7.12

This compares the premises of an argument to the foundations of a building. This comparison is valid as both provide an underlying structure for what is added later. In both cases, if the basis is not solid, later additions may be unstable.

Passage 7.13

This compares emotions to a pressure cooker. This comparison is made in order to argue that emotions cannot be controlled. However, the comparison isn't valid as it isn't comparing like with like: human emotions are not like steam under pressure. The underlying argument is based on false premises: that emotions cannot be controlled and that pressure cookers inevitably explode at boiling point. However, there are methods for managing emotions. An explosion isn't inevitable, either, when the contents of a pressure cooker reach boiling point, as there are control mechanisms to let out the steam. The comparison does not help us to understand why the defendant couldn't control his emotions.

Passage 7.14

This compares failures in the stock market to health and safety matters for the human body. The passage is based on the assumption that it is reasonable to expect compensation for accidents and ill-health, but, in reality, that varies depending on circumstances such as the country and insurance policies. The author is attempting to make the argument for financial compensation seem more plausible by comparing financial loss to other major events for which compensation seems reasonable. The comparison is not valid because:

- Ill-health and accidents do not automatically bring compensation.
- Even if compensation for major health issues were automatic, the comparison still would not be valid. Health and finance are not comparable in terms of the kinds of choices people have, their control over the risks, and the advance action they can take to avert the consequences.

Necessary and sufficient conditions (p. 95)

	Proposition	Necessary?	Sufficient?
Ex.	*Example: Birds have wings. The item has wings. Therefore it is a bird.*	*Yes. Wings are a necessary condition for the item being a bird.*	*No. The reasons given to support the argument that the item is a bird are not sufficient to satisfy the definition of a bird. This would include 'usually flies', is animate, lays eggs, has two legs, has feathers. The information given is not sufficient to rule out an aeroplane or a toy.*
1	The report makes reference to branches. It must be about a tree.	No. It is not a necessary condition: a report could be about a tree without referring to branches.	No. The reasons given to support the argument that the report is about a tree are not sufficient to prove the case. The report could be referring to branches of an organisation such as a bank.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 7 (continued)

	Proposition	Necessary?	Sufficient?
2	The boxer doesn't eat meat or fish. He does eat dairy products and vegetables. The boxer is a vegetarian.	Yes. It is a necessary condition of being a vegetarian that you don't eat meat or fish but do eat vegetables.	Yes. The reasons given for identifying the boxer as a vegetarian are sufficient to satisfy the definition of a vegetarian.
3	Amir is under the age of 20. Teenagers are less than 20 years old. Amir must be a teenager.	Yes. Being less than 20 is a necessary condition of being a teenager.	No. The reasons given to support the argument that Amir is a teenager are not sufficient to meet the definition of a teenager. Amir must also be over the age of 12 to qualify as a teenager.
4	Claire does not play any musical instrument. Therefore, she is not a musician.	No. Playing an instrument is not a necessary condition of being a musician. A composer or conductor might not play an instrument.	No. The reason given to support the argument that Claire is not a musician is not sufficient to prove the case. We would need to know other information such as that Claire was not a composer or a conductor and did not meet any other definition of 'musician'.
5	The bishop arrived on a vehicle with two wheels, one in front of the other. The bishop must have been on a bicycle.	Yes. It is necessary that the vehicle had two wheels, one in front of the other, in order for the bishop to have arrived on a bicycle.	No. The details given about the vehicle are not sufficient to establish that it was a bicycle. Therefore, the details do not support the conclusion that the bishop arrived on a bicycle. It might have been a scooter or motorbike.
6	A television usually costs more than a radio. This one costs less than a radio, so it must be a bargain.	No. It isn't always a necessary condition for a television to cost less than a radio for it to be a bargain.	No. We do not know whether the radio is priced at its normal rate. If the radio is more expensive than usual, then the TV could also be more expensive and still cost less than the radio. For the television to be a bargain, we would need to know that there was not a reason for the lower price, such as it being damaged in some way.
7	Li Yeung had the benefit of an exceptionally happy childhood. She must be a very happy adult.	No. Having an exceptionally happy childhood is not a necessary condition of being a happy adult. A person could have had a miserable childhood but their circumstances might change in later life.	No. Even an exceptionally happy childhood is not a sufficient condition for being a happy adult: many events may have intervened to make a person's circumstances unhappy.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 8

Authenticity (p. 112)

1 Probably authentic: such documents originated in cathedrals and could have become lost in library stacks over the years. Cathedrals are unlikely to forge a document as this would be exposed and would reflect badly on a religious organisation. However, checks would need to be made to validate the manuscript's age and origins, or *provenance*.
2 Probably not authentic: such items are rare and usually found in libraries, museums, private collections or religious institutions.
3 A collection of 1000 autographs by Elvis Presley could be authentic but such a collection would be valuable and it is unlikely that it would be bought without a viewing. It is more likely that an authentic collection would be sold at auction.
4 Probably not authentic: it is unlikely, though not impossible, that such an unpublished diary would fall into the possession of a student.
5 Probably authentic: such letters are found in collections in major libraries.
6 Probably not authentic: such valuable pictures are found occasionally in attics of old houses or behind other paintings, but not usually in modern garages and not in such large numbers.
7 Probably authentic: it could be carbon dated to check its age so would be difficult to fake.
8 Probably authentic: such items might well be kept at a prison and the governor could have overall responsibility for their care.

Controlling for variables (p. 122)

Passage 8.4 (p. 118)

The experiment requires a control group to compare changes in night vision between those who ate the capsules of carrot extracts and those who didn't. Some variables that would need to be controlled are: diet, which could affect the results; activities which might tire the eyes; previous levels of vision and visual problems; whether participants already had diets high in carrots, allowing no further room for improvement.

Passage 8.5 (p. 118)

The research should take into account such variables as whether participants liked any kind of perfumed soap at all, and whether the scents were equally strong. If not, then participants might have chosen on the basis of the strength of the perfume rather than its scent.

Passage 8.6 (p. 118)

There are many variables that could affect the research outcomes here. The researchers need to check such details as: how closely related the participants were to the bereaved; the frequency and kind of contact and interaction between the people in the sample and the deceased before the bereavement; whether participants attended the funeral; the kinds of work that participants are involved in; for how much time they were usually absent from work before the bereavement; whether they had any illnesses or other conditions likely to make them miss work. Each group would need to have roughly equal numbers of people from each circumstance. However, it could be that a particular combination of these variables has an effect on time off work and it would be hard to control for that in the first set of research.

Triangulation (p. 125)

(1) You would probably want to contact the venue to find out if there really were cheap tickets available on the night.
(2) This could be triangulated with reports from other manufacturers about how their brakes were tested and the results, as well as reports in trade magazines. There may also be general information in consumer magazines about different braking systems. If you knew anybody who had bought a car with the new

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

brakes, you could ask their opinion. If you can drive, you would want to try out the braking system for yourself.

(3) If the book provides references, you can check the original sources to see if they were reported accurately. You would expect to see references to specific 'poor laws' on begging, and the dates of these. You can also check other books to see if these contradict or support the chapter in the book. However, several books may refer to the same secondary source, which itself might be incorrect. Where possible, it is useful to check the primary sources, or published versions of these, for yourself.

Evaluating a body of evidence (p. 126 and texts pp. 229–33)

(a) Identifying reputable sources

Very reputable
Text 3: journal article
Text 8: a chapter of an academic book
Text 10: journal article
Text 11: journal article
Text 12: journal article

Fairly trustworthy
Text 2: popular magazines
Text 4: editorial in a small town local newspaper
Text 6: trade magazine
Text 7: columnist in a national paper

Little authority
Text 1: internet chat room
Text 5: letter to a national paper
Text 9: personal website

(b) Vested interests

The authors of the following texts may have a vested interest in the outcome of the argument:

Text 1: as the author downloads from the internet;
Text 2: as the author may be currying favour with his readers, who are likely to share free cuttings;
Text 5 (and possibly Text 9): the argument appears to be a rationalisation for not paying for downloaded copies.

(c) Reliable evidence of internet users' views

The most reliable sources for indicating what internet users believe are those written by internet users themselves, and those indicated by research evidence. In this case, that would be Text 1 and possibly Texts 5 and 9, by internet users. More information would be needed to ensure Texts 5 and 9 were indeed by internet users. Text 12 gives details of the behaviour of internet users, drawn from research, and this behaviour is indicative of their beliefs. However, more investigation would be needed to check on these internet users' motivations for paying.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 9

Identifying theory (p. 132 and texts pp. 229–33)

Only two texts have an explicit theoretical position. These are:

Text 10: that moral and ethical issues should be regarded as 'dilemmas' rather than as simple questions of right and wrong.
Text 12: that behaviour is affected primarily by how easy it is to act in an ethical way.

Categorising arguments (p. 133 and texts pp. 229–33)

Text 1: sophistical, artistic and philanthropic
Text 2: sophistical and philanthropic
Text 3: economic and legal
Text 4: economic
Text 5: philanthropic and sophistical
Text 6: economic
Text 7: legal, economic, ethical
Text 8: sophistical and legal
Text 9: sophistical and legal
Text 10: ethical, legal
Text 11: none of these
Text 12: ethical, economic

Accurate interpretations when reading (p. 134 and texts pp. 229–33)

Passage 9.1

B Misinterpretation. The text doesn't state that the author, personally, is offering a service.

Passage 9.2

A Accurate interpretation.

Passage 9.3

B Misinterpretation. The text is very clear that it regards piracy as stealing, and makes no exceptions. Passage 9.3 waters this down, using words such as 'not usually' and 'most customers', suggesting there may be exceptions.

Passage 9.4

A Accurate interpretation.

Passage 9.5

B Misinterpretation. The author's argument is that gardeners who give away cuttings are cheating the people who breed new species of plant. It is true that the text implies that small gardeners will not be prosecuted in practice, but that isn't the argument.

Passage 9.6

B Misinterpretation. The passage does argue that there hasn't been a democratic process to decide that the law should make ultimate decisions of right and wrong. It also argues that positive changes have occurred when people stand up for what they believe. The text does not make recommendations. The argument is more abstract, pointing out that questions of right and wrong are complex and that there are different ways of looking at ethical issues. The implication is that there should be more public discussion of the concept of justice. However, the text doesn't advocate that people stop obeying the law on those grounds.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 10

Setting the scene for the reader (p. 151)

'Is productionism dead?'

Passage 10.1 provides a good introduction to the subject that an intelligent reader without an in-depth knowledge of the subject could follow. The author defines what is meant by 'productionism' and summarises why the theory was developed. The introduction informs the reader about positive and negative aspects of productionism covered in the essay. The author's position and conclusions are presented clearly to orientate the reader.

Passage 10.2 is written in a flowery or theatrical style, and makes grand sweeping statements. However, the style makes it difficult for a reader who does not know the subject well to work out what productionism is. The author's general position is clear, but the reader is not told how the argument will be developed.

Passage 10.3 launches too quickly into the subject, giving little introduction to orientate the reader. The author presents examples of the effects of productionism without having explained what it is and how it led to these effects.

Passage 10.4 makes too much use of broad generalisations about human society. Some of these may be true, but would be hard to prove and are not directly relevant to the essay. As a result, the essay starts very slowly, and uses a lot of words to say very little of relevance.

Words used to signpost conclusions (p. 157)

If you used different words to signpost the argument than those used in the passages opposite, check the table on page 158 to see if you used suitable alternatives. The signal words are indicated in italic.

Passage 10.5

Deaf people have their own languages, based on signs, body position and facial expressions. *However*, as few hearing people understand these languages, communication between deaf and hearing people is not usually very effective. *Although* deaf people often form strong social and cultural groups, they are often excluded from mainstream culture and their talents are not used effectively within the economy. *Similarly*, hearing people can feel excluded from deaf conversations and uncertain of how to behave around deaf people. *Therefore*, it would be in everyone's interests if sign languages were taught in school so that deaf and hearing children grew up able to communicate effectively with each other.

Passage 10.6

Globalisation appears to be inevitable but there is disagreement about whether this is a positive development. *On the one hand*, there are those who argue that increased contact between countries leads to better understanding and has reduced the likelihood of future wars. *Furthermore*, they see benefits to democracy and human rights from information being widely available electronically, so that different nations can compare conditions in their country with those elsewhere. *On the other hand*, there are those who see globalisation as a destructive force. They argue that it leads to less powerful peoples losing their indigenous languages as the languages of more powerful countries are used internationally for business and politics. *Moreover*, they argue that globalisation often means big business buying up resources and land in poorer countries, thus distorting local economies and draining their resources. *Therefore*, although there are some potential benefits to globalisation, some controls are needed to protect poorer economies from exploitation.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Answers to activities in Chapter 13

Example 1: Serina (p. 222)

Serina would not make a good impression on the employer, for the following reasons.

She asks if it is possible to visit the employer, stating that she is available for a visit on any weekend. To the employer, this would suggest:

1) That Serina does not make use of available information, which has already indicated that visits are possible via an open evening on April 20th. This suggests that Serina doesn't pay good attention to detail, and/or can't follow instructions.
2) That Serina doesn't give thoughtful consideration to the needs of the employer – week-end visits are not likely to be convenient for the employer.

These points would be true for other jobs too, and not just for teachers.

Serina's application does not give a sense of having thought through the needs of the job. As a primary teacher, she may need to be available before term starts to help plan and prepare for the new term. It would be important to the job role that she is present from the start of the term to help settle in children at a critical time of the year. Many jobs recruit new posts such as graduate jobs in line with their annual work cycle. Employers are likely to have good reasons for expecting employees to start work on the dates they have indicated – especially if there are particular tasks, training, or orientation that needs to take place at a given time.

Employers would expect applicants to understand the nature of the role and to have arranged personal matters well in advance, in order to accommodate the requirements of the job. They would not expect that they needed to point out such things. By asking, Serina suggests she either doesn't understand the nature of the role or doesn't care sufficiently about it.

Such enquiries give the employers useful clues about an applicant's attitude and whether they are right for the job.

Example 2: Arno (p. 222)

Arno is likely to be wasting his time for this role. The employer clearly is looking for someone who has already worked in the field and understands the nature of the specialist fields that he or she would be managing. For a deputy director role, this calls for significantly more experience than Arno has to date.

Managing a budget of £500 would not be the equivalent of managing a service budget. Whilst personal experience is often valuable, that is not sufficient without the relevant work experience.

If this area of work interests Arno, he could look for a more junior role in one of the service departments, gain experience and then apply for more senior roles.

Example 3: Kim (p. 222)

Kim would be better advised to spend his time in preparing fewer but better applications. His answers to this section on independent and group working are weakened in the following ways.

He spelt 'Kiaru Holdings' incorrectly as 'Kairu Holdings'.

He hasn't addressed the employer requirement for the ability to work independently so would get no points for that.

Although he provides a sense of where he has been a team member, he doesn't provide any meaningful detail that helps the employer know his strengths as a team player and how these would be relevant to the job.

Saying others find you 'easy to get on with' is vague and doesn't really add anything to the application.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

As a keen member of the Radio Society, Kim could have written something along the lines of this.

> *I am self-reliant and able to work effectively when working independently and as part of a team. I was an active member of the university's Radio Society, and took on personal responsibility for the Enterprise Report that came out each week. This included making proposals to the team on the content, researching the items, writing these up, agreeing the draft with team, and ensuring my report was ready to go out on time.*
>
> *My particular strengths in team working are in communicating inside and outside the team, making sure everyone's voice is heard, running focus groups, and using social networking usefully to achieve team objectives. I believe these skills would be useful when working with the customer base at Kiaru Holdings.*

Example 4: Lizzy (p. 222)

Lizzy would not make a good impression on the employer, for the following reasons.

Her application contains many grammatical errors and missing words, which make it look careless, and which also make it harder to understand.

She has clearly applied previously to a different company, MTZ-Co, but hasn't spotted that she has left in a mention of that company. Whilst employers know that people apply for lots of jobs, they take a dim view of such errors.

She seems interested in the job for reasons that would not be relevant to the business itself – her desire to live in the capital city. Employers are more likely to offer opportunities to candidates whose responses suggest that they have a particular interest in the business, such as its values, mission, training programme, clients, and the nature of the work itself.

She does not seem aware of how her application is wasting the employer's time – her approach is rather 'chatty' and goes into far too much detail about matters that an employer doesn't need to know about her life and interests.

The person specification requires flexibility about travel and working hours. Lizzy discusses flexibility only in the context of moving to the capital: that isn't what the employer would mean by flexibility to travel.

Lizzy has attached her CV, despite the job advertisement asking candidates, specifically, not to do so. Employers might find this irritating at best, and could use this as evidence of not being able to follow instructions or to pay attention to details.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Bibliography

Anon. (1803) *Such is Buonaparte* (London: J. Ginger).

Arnheim, R. (1954, 1974) *Art and Visual Perception: The Psychology of the Creative Eye* (Berkeley: University of California Press).

Ashcroft, M.Y. (1977) To Escape a Monster's Clutches: Notes and Documents Illustrating Preparations in North Yorkshire to Repel the Invasion. North Yorkshire, CRO Public No. 15.

Bailey, I. and Compston, H. (2012) *Feeling the Heat: The Politics of Climate Change in Rapidly Industrializing Countries* (Basingstoke: Palgrave Macmillan).

Barrell, J. (1980) *The Dark Side of the Landscape: The Rural Poor in English Painting, 1730–1840* (Cambridge: Cambridge University Press).

Bodner, G.M. (1988) 'Consumer Chemistry: Critical Thinking at the Concrete Level'. *Journal of Chemistry Education*, **65** (3), 212–13.

Borton, T. (1970) *Reach, Touch and Teach* (London: Hutchinson).

Boud, D., Keogh, R. and Walker, D. (eds) (1985) *Reflection: Turning Experience into Learning* (London: Routledge).

Bowlby, J. (1980) *Attachment and Loss*, Vol. 3: *Loss, Sadness and Depression* (New York: Basic Books).

Boyle, F. (1997) *The Guardian Careers Guide: Law* (London: Fourth Estate).

British Council (2013) *Skills Development in South Asia*. Economist Intelligence Unit Report commissioned by the British Council, cited surveys by India's National Skill Development Corporation (NSDC) (https://www.britishcouncil.org/sites/default/files/south-asia-skills-report-summary.pdf).

British Council (2014) *India Employability Survey Report*. Foreword by Rob Lynes (www.kcl.ac.uk).

Campbell, A. (1984) *The Girls in the Gang* (Oxford: Basil Blackwell).

Carwell, H. (1977) *Blacks in Science: Astrophysicist to Zoologist* (Hicksville, NY: Exposition Press).

CBI and UUK (2009) *Future Fit: Preparing Graduates for the World of Work* (www.cbi.org.uk).

Chan, J., Goh, J. and Prest, K. (eds) (2015) *Soft Skills, Hard Challenges. Understanding the Nature of China's Skills Gap* (British Council) (www.britishcouncil.org/education/ihe).

Cholmeley, C. (1803) Letter of Catherine Cholmeley to Francis Cholmeley, 16 August 1803. In M.Y. Ashcroft (1977) *To Escape a Monster's Clutches: Notes and Documents Illustrating Preparations in North Yorkshire to Repel the Invasion*. North Yorkshire, CRO Public No. 15.

Colley, L. (2003) Captives: Britain, Empire and the World 1600–1850 (London: Pimlico).

Collins, P. (1998) 'Negotiating Selves: Reflections on "Unstructured" Interviewing'. *Sociological Research Online*, **3** (3) (www.socresonline.org.uk/3/3/2.html, January 2001).

Coren, A. (1997) *A Psychodynamic Approach to Education* (London: Sheldon Press).

Cottrell, S. (2010) *Skills for Success: The Personal Development Planning Handbook*, 2nd edn (Basingstoke: Palgrave Macmillan).

Cottrell, S. (2015) *Skills for Success: Personal Development and Employability*, 3rd edn (London: Palgrave).

Cowell, B., Keeley, S., Shemberg, M. and Zinnbauer, M. (1995) 'Coping with Student Resistance to Critical Thinking: What the Psychotherapy Literature Can Tell Us'. *College Teaching*, **43** (4).

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Crane, T. (2001) *Elements of Mind: An Introduction to the Philosophy of Mind* (Oxford: Oxford University Press).

Csikszentmihalyi, M. (1992) *Flow: The Psychology of Happiness* (London: Random House).

Diamond, A., Walkley, L., Forbes, P., Hughes, T. and Sheen, J. (2011) *Global Graduates into Global Leaders.* Report of the CIHE, AGR and CFE (www.cfe.org.uk).

Donaldson, M. (1978) *Children's Minds* (London: Fontana).

Driscoll, J. (1994) 'Reflective Practice for Practice'. *Senior Nurse*, **13** (7), 47–50.

Dunbar, R. (1996) *Grooming, Gossip and the Evolution of Language* (London: Faber & Faber).

Eco, U. (1998) *Serendipities: Language and Lunacy* (London: Weidenfeld & Nicolson).

Elliott, J.H. (1972) *The Old World and the New, 1492–1650* (Cambridge: Cambridge University Press).

Elliot, L. (2015) 'Can the World Economy Survive without Fossil Fuels?' The Guardian 8/4/15, http://www.theguardian.com/news/2015/apr/08/can-world-economy-survive-without-fossil-fuels, downloaded 14/4/16.

Encyclopaedia Britannica (1797) 3rd edition, Edinburgh.

Ennis, R.H. (1987) 'A Taxonomy of Critical Thinking Dispositions and Abilities'. In J. Baron and R. Sternberg (eds), *Teaching Thinking Skills: Theory and Practice* (New York: W.H. Freeman).

Farndon, J. (1994) *Dictionary of the Earth* (London: Dorling Kindersley).

Farrar, S. (2004a) 'It's Very Evolved of Us to Ape a Yawn'. *Times Higher Educational Supplement*, 12 March 2004, p. 13.

Farrar, S. (2004b) 'It's Brit Art, but Not as we Know it'. *Times Higher Educational Supplement*, 9 July 2004, p. 8.

Farrar, S. (2004c) 'Old Sea Chart is So Current'. *Times Higher Educational Supplement*, 16 July 2004, p. 5.

Fillion, L. and Arazi, S. (2002) 'Does Organic Food Taste Better? A Claim Substantiation Approach'. *Nutrition & Food Science*, **32** (4), 153–7.

Fischer, K. (2012) 'Job Preparation and International Collaboration Are Themes of Indian Higher-Education Conference'. Chronicle of Higher Education, 6 November 2012 (http://chronicle.com/article/Job-Preparation-and/135584/ accessed 8/8/16).

Fisher, A. (2004) *The Logic of Real Arguments*, 2nd edn (Cambridge: Cambridge University Press).

Fisher, D. and Hanstock, T. (1998) *Citing References* (Oxford: Blackwell).

Foster, R. (2004) *Rhythms of Life* (London: Profile Books).

Gates, L. (ed) (1984), *Black Literature and Literary Theory* (pp. 263–83) (New York: Methuen).

Garnham, A. and Oakhill, J. (1994) *Thinking and Reasoning* (Oxford: Blackwell).

Gibbs, G. (1988) *Learning by Doing: A Guide to Reading and Learning Methods* (Oxford: Further Education Unit, Oxford Polytechnic).

Gilligan, C. (1977) 'In a Different Voice: Women's Conceptions of Self and Morality'. *Harvard Educational Review*, **47**, 418–517.

Grassroots Global Justice Alliance (2015) Call to Action: The COP21 *Failed Humanity*, http://ggjalliance.org/ParisFailure2015, downloaded 16/4/16.

Green, E.P. and Short, F.T. (2004) *World Atlas of Sea Grasses* (Berkeley: University of California Press).

Greenfield, S. (1997) *The Human Brain: A Guided Tour* (London: Phoenix).

Hammacher, A.M. (1986) *Magritte* (London: Thames & Hudson).

Hart Research Associates (2013) *It Takes More than a Major: Employer Priorities for College Learning and Student Success. An online survey conducted on behalf of the Association of American Colleges and Universities, (AACU)* (Washington DC: Hart Research Associates).

Hogan, C. (2004) 'Giving Lawyers the Slip'. *The Times*, 24 August 2004, p. 26.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

ISEC (n.d.) *International Scholarly Exchange Curriculum Program* (Bejing: ISEC, Chinese Ministry of Education).

Jacobs, P.A., Brunton, M., Melville, M.M., Brittain, R.P. and McClermont, W.F. (1965) 'Aggressive Behaviour, Mental Subnormality and the XYY Male'. *Nature*, **208**, 1351–2.

Jamet, M., *What do Green NGOs make of the COP21 Climate Deal?*, http://www.euronews.com/2015/12/14/what-do-green-ngosmake-of-cop21-climate-deal, downloaded 16/4/16.

Johns, C. and Freshwater, D. (1998) *Transforming Nursing through Reflective Practice* (Oxford: Blackwell Scientific).

Kahneman, D. (2011) *Thinking, Fast and Slow* (London: Penguin).

Kohlberg, L. (1981) *Essays on Moral Development*, vol. 1 (New York: Harper & Row).

Korn, M. (2014) 'Bosses Seek "Critical Thinking" but What Is That?' *Wall Street Journal*, 21 October 2014.

Lane, H. (1984) *When the Mind Hears: A History of Deaf People and Their Language* (Cambridge: Cambridge University Press).

Lang, T. and Heasman, M.A. (2004) *Food Wars: The Global Battle for Mouths, Minds and Markets* (London; Sterling, VA: Earthscan).

Lavater, J.K. (1797) *Essays on Physiognomy*, translated by Rev. C. Moore and illustrated after Lavater by Barlow (London: London Publishers).

Le Page, M. (2016) *Developing Nations Urged to Boycott Paris Agreement Signing*, Climate Home, http://www.climatechangenews.com/2016/03/29/developing-nations-urged-to-boycott-parisagreement-signing, downloaded 15/4/16.

Loftus, E.F. (1979) *Eyewitness Testimony* (Cambridge, MA: Harvard University Press).

Lowden, K., Hall, S., Elliot, D. and Lewin, J. (2011) *Employers' Perceptions of the Employability Skills of New Graduates* (London: Edge Foundation).

McMurray, L. (1981) *George Washington Carver* (New York: Oxford University Press).

McPeck, J.H. (1981) *Critical Thinking and Education* (New York: St Martin's Press).

Miles, S. (1988) *British Sign Language: A Beginner's Guide* (London: BBC Books).

Morris, S. (2004) *Life's Solution: Inevitable Humans in a Lonely Universe* (Cambridge: Cambridge University Press).

National Committee of Inquiry into Higher Education (1997) *Higher Education in the Learning Society* (London: HMSO).

Pagel, M. (2004) 'No Banana-eating Snakes or Flying Donkeys are to be Found Here'. *Times Higher Educational Supplement*, 16 July 2004.

Palmer, T. (2004) *Perilous Planet Earth: Catastrophes and Catastrophism Through the Ages* (Cambridge: Cambridge University Press).

Papers in the Bodleian Library. Curzon Collection, vol. 22, ff. 89–90. Letter from Henry Peter Lord Brougham to C.H. Parry, 3 September 1803.

Pears, R. and Shields, G. (2016) *Cite them Right: The Essential Referencing Guide*, 10th edn (London: Palgrave).

Pendleton, G. (1976) 'English Conservative Propaganda During the French Revolution, 1780–1802', Ph.D. (unpub.), Emory University.

Peters, R.S. (1974) 'Moral Development: a Plea for Pluralism'. In R.S. Peters (ed.), *Psychology and Ethical Development* (London: Allen & Unwin).

Piliavin, J.A., Dovidio, J.F., Gaertner, S.L. and Clark, R.D. (1981) *Emergency Intervention* (New York: Academic Press).

Platek, S.M., Critton, S.R., Myers, T.E. and Gallup, G.G. Jr (2003) 'Contagious Yawning: the Role of Self-awareness and Mental State Attribution'. *Cognitive Brain Research*, **17** (2), 223–7.

Postgate, J. (1994) *The Outer Reaches of Life* (Cambridge: Cambridge University Press).

'Pratt' (1803) *Pratt's Address to His Countrymen or the True Born Englishman's Castle* (London: J. Asperne).

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Raman, M. (2016) *The Signing Ceremony of the Paris Agreement in New York on 22nd April – Why There is no Need to 'Rush' into Signing*, Third World Network, https://www.scribd.com/doc/306273316/ Note-on-the-Signing-Ceremony-in-New-York, downloaded 17/4/16.

Rose, S. (2004) *The New Brain Sciences: Perils and Prospects* (Milton Keynes: Open University Press).

Rowbotham, M. (2000) *Goodbye America! Globalisation, Debt and the Dollar Empire* (New York: John Carpenter).

Sachs, O. (1985) *The Man who Mistook his Wife for a Hat* (London: Picador).

Salzberger-Wittenberg, I., Williams, G. and Osborne, E. (1983) *The Emotional Experience of Learning and Teaching* (London: Karnac Books).

Sattin, A. (2004) *The Gates of Africa: Death, Discovery and the Search for Timbuktu* (London: HarperCollins).

Shulman, L. (1986) 'Those who Understand: Knowledge Growth in Teaching'. *Educational Researcher*, **15** (2), 4–14.

Smith, L. (1992) 'Ethical Issues in Interviewing'. *Journal of Advanced Nursing*, **17**, 98–103.

Stein, C. (1997) *Lying: Achieving Emotional Literacy* (London: Bloomsbury).

Tajfel, H. (1981) *Human Groups and Social Categories* (Cambridge: Cambridge University Press).

Thompson, E.P. (1963) *The Making of the English Working Class* (Harmondsworth, Middlesex: Penguin).

Trevathan, W., McKenna, J. and Smith, E.O. (1999) *Evolutionary Medicine* (Oxford: Oxford University Press).

United Nations Treaty Collection (2015) *The Paris Agreement*, https://treaties.un.org/pages/ViewDetails.aspx?src=TREATY&mtdsg_no=XXVII-7-d&chapter=27&lang=en, downloaded 11/4/16.

Vidal, J. and Vaughan, A. (2015) 'Paris Climate Agreement "May Signal End of Fossil Fuel Era"'. The Guardian 13/12/15, http://www.theguardian.com/environment/2015/dec/13/paris-climateagreement-signal-end-of-fossil-fuel-era, downloaded 15/4/16.

White, C. (1779) *An Account of the Infinite Gradations in Man, and in Different Animals and Vegetables; and from the Former to the Latter.* Read to the Literary and Philosophical Society of Manchester at Different Meetings (Manchester: Literary and Philosophical Society).

Williams, G. (1997) *Internal Landscapes and Foreign Bodies: Eating Disorders and Other Pathologies* (London: Tavistock Clinic Series, Duckworth).

Willis, S. (1994) 'Eruptions of Funk: Historicizing Toni Morrison'. In L. Gates, Jr (ed.), *Black Literature and Literary Theory* (New York: Methuen).

Wilson, J.Q. and Hernstein, R.J. (1985) *Crime and Human Nature* (New York: Simon).

Wollstonecraft, M. (1792) *Vindication of the Rights of Women.* (Republished in 1975 by Penguin, Harmondsworth, Middlesex.)

Worwood, V.A. (1999) *The Fragrant Heavens: The Spiritual Dimension of Fragrance and Aromatherapy* (London: Bantam Books).

www.emeagwali.com for the scientist Emeagwali.

www.princeton.edu/~mcbrown/display/carver.html George Washington Carver, Jr: Chemurgist; 6/8/2004.

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

Index

abstracts, using, 110
academic writing, 48, 107, 109, 131,144, 147, 148–9, 152
  see also essays; writing critically
accuracy, 5, 6, 9, 48, 134, 299, 152
  see also precision
agreement, 46
ambiguity, 8, 36
analogies, 96–7, 294, 201, 272
analysis, critical, 1, 4, 8, 45, 48, 52, 54, 101, 137, 148
  use analysis rather than description, 52, 185
  when writing, 147–62, 164–6
  see also argument, identifying; comparison; categorising; selection; evidence, evaluating; selection
argument, xiv, 34, 46
  arguments as reasons, 34
  contributing arguments, 34
  counter arguments, 53–4, 59, 101, 149, 155–6, 163, 175, 181, 182, 183, 240, 241, 258, 260
  features, 33, 34, 36, 37, 43, 57, 236
  flawed arguments, 89, 90–105, 176
  identifying arguments, 33, 37–9, 43, 45–55
  implicit arguments, 73, 81, 292
  and non-argument 45, 48–55, 282
  overall argument, xiv, 34
  and theoretical perspective, 131, 132
  see also structuring an argument
  'winning' an argument, 10
assertions, xiv
assumptions, 73–8, 87, 249, 250
  ideological, 81, 292
  implicit assumptions, 76–7
  unsupported, 81, 83, 90, 91, 96, 100, 270, 271
  used as reasons, 77–8, 289–90
attention, focusing, 1, 17, 23–6, 29, 45, 277
  to detail, 5, 12, 13–15, 17
audience, x, 148
authenticity, 112, 297
author, xii
author's position, 34–5, 36, 280, 46, 57, 58, 59, 72, 283–4, 96, 163, 175, 181, 182, 183, 240
  dealing with evidence against, 163

background information, necessary and unnecessary, 177, 240, 258, 259, 269
  see also description
barriers to critical thinking, 1, 10–12, 16
benefits of critical thinking skills, 4
career planning and critical thinking, 211–8
castle of cards, 100, 271
categorising, 17, 19, 27–8, 133
  grouping points, 149
  reasons and arguments, 299, 182
  theories, 123, 132, 133, 299
cause and effect, 90–2, 105, 294, 271
citations 167
  see also references
clarity in critical thinking, 58, 59, 148
close reading, 20, 29–31, 129
comparisons, making, 18, 21, 24, 27, 96
complicity, with the reader, 98, 250, 251, 272
conclusions, xiv, 37, 42, 43
  as deduction, 42, 68, 175
  evidence-based, 115–6, 175, 177, 181, 184, 158–61
  interim, *see* intermediate
  intermediate, xiv, 57, 65–6, 181, 182, 240, 241, 242, 259– 261
  location of, 39, 41–3
  logical conclusions, 68, 242, 261
  see also deduction
  as reasons, *see* conclusions, intermediate
  as summaries, 42, 43, 54, 68, 286, 241, 242, 261
  supported by reasons, 240
  tentative conclusions 159
  writing conclusions, 156–7, 160, 162, 166, 167, 176
connoted meanings, 83–4, 86
consistency
  checking your own writing for, 185
  internal, xii, xiv, 59–60, 283–4, 175, 177, 270–2
  logical, xii, xiv, 61–2
contributing arguments, *see* argument
correlation, 91–2
counter arguments, *see* argument
credible sources, 176
critical reflection, *see* reflection
critical thinking
  in academic contexts, 7–9, 11, 12
  as cognitive activity, 1
  as process, 2, 16
  what is critical thinking, viii, 1, 2, 3, 5, 6, 8, 9
  why it matters, ix, 4
criticism
  of peers, 8–9
  what is criticism, 2
currency, 113

deduction, 42, 43, 68
  and unwarranted leaps, 100
deflection, of the reader, 98, 251, 270

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

denoted meanings, 83–4, 86
description, 48, 54, 55, 175, 240, 259
  identifying background information, 38–9, 45, 52, 53, 54
difference, identifying, 24
disagreement, 46, 47
distortions to argument, 175, 181, 185

emotion, and critical thinking, 1, 5, 11
emotive language, 101, 176, 248, 250, 270
employability and critical thinking, 211–28
  when looking for a job, 217–8
  applying for jobs, 219–22, 226
  understanding employers' needs, 219–23, 226
  demonstrating critical thinking to employers, 225
  critical self-evaluation of job applications, 226–8
  using the person specification, 219, 226
  where job applicants go wrong, 222, 302–3
essays, 3, 152, 161–7, 209
  see also academic writing; writing critically
evidence, 107–27, 297
  evaluating the evidence, 3, 8, 9, 107, 109, 110, 126, 127, 299
  interpreting evidence, 6
  reputable sources, 111
  selecting evidence, 114–16
  using supporting evidence, 175, 176–7, 181, 182, 185, 240, 249, 250
  see also primary sources; secondary sources; triangulation
exclusion, 98
explanation, 49, 53, 54, 55
extraneous material, 52
eye-witness testimony, 124

facts, 123, 185
false analogies, 96–7, 250, 272, 294
false correlations, 91–92
false premises, 38, 73, 79–80, 270, 291
features of an argument, *see* argument
flawed arguments, *see* argument
following directions, 19
frames of reference, 23

generalisations, 121, 176

identifying arguments, *see* arguments
influences on judgement, 6
in-groups, 98
interim conclusions, *see* conclusions, intermediate
intermediate conclusions, *see* conclusions
introductions, for critical writing, 150, 164
jobs, *see* employability; reflection and critical thinking; career planning
journal articles as evidence, 111
  notes from, 139
  references from, 145
judgements, making critical, 165, 206

key features of an argument, *see* argument, features

latent messages, 84
line of reasoning, *see* reasoning
literature searches, 110
  on-line, 110, 241
  writing up, 152
  see also primary source; secondary source
logical conclusions, 68–9, 286
logical order, 57, 70–1, 72, 149, 175, 177, 181, 283, 287
  see also sequencing

misrepresentation, 103, 249, 270

necessary conditions, meeting, 93–5, 249, 272 295–6
non-sequitors, 76,
note-making, 129, 146
  to support reading, 135
  selecting what to note, 140–143
  structuring notes, 137–9
  why make notes, 135
  see also references, quotations

objectivity, 5
opinion, 123
out-groups, 98

personal attacks, 9, 101, 249, 272
personal strategies, 6, 9, 12
personality and critical thinking, 2
persuasion
  and audience, 43, 46, 96, 147, 148
  through flawed argument, *see* argument, flawed
  through reasons, 36, 43, 46
  through latent methods, 73, 81, 87, 98
  see also presenting an argument
plagiarism, 146, 168
  see also references
position, *see* author's position
precision, 5, 6, 8, 10, 59, 249, 269
predicate, xiv, 38, 79
premises, xiv, 38, 79
  see also false premises
presenting an argument, 2, 3, 4, 9, 14–15, 23, 46, 72
  see also line of reasoning; persuasion; writing critically
primary sources, 107, 108, 124, 241, 259–62, 273
priorities for developing critical thinking, 13–17
probability, levels of, 119, 120, 121, 185
professional life, and critical thinking, 4, 8, 17, 107, 190, 211–16, 218, 223–4
propositions, 37, 38, 39, 43

quotations, choosing, 136
  see also quotations; references

© Stellla Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

reading
and accurate interpretation, 134, 152, 299
close reading, 29–31, 134, 163
critically, 2, 4, 129, 163
efficiently, 33, 45, 57, 129, 135
information for job applications, 217–9, 222
selectively, 133, 136
see also note-making
reasoning, 3
line of reasoning, xiv, 43, 46, 81, 104, 153–4, 158, 177, 184
see also logical order
reasons, xv, 3, 167, 173
and implicit assumptions, 77
independent and joint, 63
intermediate conclusions as reasons, 65–6
supporting the conclusion, xiv, xv, 38–9, 53–54, 63, 65, 289–91, 93–4, 184, 280–1
references, 152, 167–8, 184, 185
using other people's, 109
to other people's work, 114, 136, 144–5, 152, 175, 180, 181
poor referencing, 249, 250, 269, 271, 273
see also quotations
reflection and critical thinking, xiii
good and bad, 207–8
career planning, 211–16, 218
characteristics of, 188–9, 198
experience, 188, 189, 207
transformative role, 188, 189, 201
and theory, 189, 194, 195, 207, 209
and professional practice, 190, 205–6, 211, 223–4
approaches to (methodology), 191–7
audience, 193, 195, 208
models of, 192, 195, 201–4
confidentiality, 195
phases of reflection, 198–200
relevance to the argument, 3, 4, 45, 115, 116, 175, 181, 184, 259
see also selection
see also barriers; self-awareness; self-evaluation
replication, 113
reputable sources, *see* evidence
research skills, 5, 7

salient characteristics, 27, 48
samples
representative, 117–8
significance, 120–2
scepticism, 2, 9
secondary sources, 108, 110, 111, 114, 240, 258, 259, 273
selection, 8, 114, 133, 140–3, 148
see also relevance
self-awareness and critical thinking, 3, 4, 5, 6, 8, 10
see also barriers; reflection and critical thinking
self-evaluation, 4, 6, 10, 13–15, 18–22, 214–5, 226–8
sequencing, 18, 25–6
in critical writing, 149
see also logical order
signal words, 4, 6, 147, 149, 153–8, 181–3, 241, 258–62
signposting, 149
see also line of reasoning; signal words
similarities, recognising, 21
skills associated with critical thinking, 4, 5, 17
see also primary sources
sleight of hand, 100, 184
stereotyping, 84–5, 250, 270, 293,
structure of an argument, 57, 89, 147
using intermediate conclusions, 65–6
when writing, 148–9, 175, 176–7, 181, 184, 258
see also signal words
substantive points, xv
sufficient conditions, meeting, 94–5, 272, 295–6
summary, 53–4
conclusions, 42
similarity to argument, 49
summarising the argument, 177, 184, 258
synthesis, 166, 198

tautology, 104, 251, 269
theoretical perspective, 131
theory and argument, 132, 183
types of theory, 133
using when writing, 185
triangulation, 124, 125–6
two wrongs don't make a right, 104

unwarranted leaps, 100, 249, 251, 270

validity, 112
value judgements, 48
and own prejudices, 6, 13
variables, 112
controlling for, 122, 297
vested interest, 36, 46, 113, 176, 182

writing critically, 147, 168
and audience, 74, 147, 148
characteristics of, 148–9
evaluating critical writing, 171–86, 235–73
evaluating your own writing critically, 184–5, 186
introducing the line of reasoning, 153
setting the scene for the reader, 74, 147, 150–1, 169, 258
signalling the direction of the argument, 154–8, 258
tentative style, 159
see also author's position; consistency; line of reasoning; signposting

© Stella Cottrell (2005, 2011, 2017) *Critical Thinking Skills*, Palgrave

'More critical analysis needed' – Have you ever seen this comment on your work or struggled to apply this in your writing? If so, this book is for you! *Critical Thinking Skills* will help you develop your reflective thinking skills, improve your critical analysis and show you how to construct effective arguments. It breaks down a complex subject into easy-to-follow blocks, beginning with the basics and building step-by-step towards evaluations of longer texts. Clear explanations and plenty of activities help to develop understanding at each stage.

'I think this book is a must have!' – **Deidre Kelleher,** *University College Dublin, Ireland*

'I liked this book – it is fun and new!' – **Julie Engstrom**, *Brigham Young University – Idaho, USA*

'The book that I've loved most in the entire year. Totally life changing.' – **Student**

'The best book I have ever read on critical thinking. I would seriously recommend it.' – **Student**

'This book has really helped me to grasp critical thinking and has helped push my grades up to 1sts and 2:1s.' – **Student**

This third edition has been fully revised and updated, and now features a brand new chapter on using critical thinking for your future career and employability.

Use this book to:

- Critically assess other people's arguments
- Recognise flawed reasoning
- Evaluate the material used to support arguments
- Apply critical thinking when reading, writing and making notes
- Write excellent essays and reports.

The author, Stella Cottrell, is widely acknowledged as the leading study skills expert and her unique approach and lively, accessible writing style quickly established the first edition of *Critical Thinking Skills* as an essential guide for students of all disciplines and levels.

STELLA COTTRELL is an international bestselling author, with over a million sales worldwide. Her other titles include *The Study Skills Handbook, Skills for Success, The Palgrave Student Planner, Dissertations and Project Reports, Mindfulness for Students* and *The Exam Skills Handbook.* Stella was formerly director of Lifelong Learning at the University of Leeds, and Pro Vice-Chancellor for Learning and Teaching at the University of East London.

A companion website, which features additional resources, is available at **www.macmillanihe.com/cts3e**

macmillan international HIGHER EDUCATION

RED GLOBE PRESS

ISBN 978-1-137-55050-7

9 781137 550507 90101

macmillanihe.com